

Pre-training, Fine-tuning, and Prompting



CS 288 Spring 2026
UC Berkeley
cal-cs288.github.io/sp26

Berkeley **BAIR**
EECS

Course Progress So Far

- We've taken a long journey in neural architectures—from perceptrons and MLPs, through RNNs and LSTMs, and finally to Transformers (including modern variants).
- Along the way, we learned how to perform supervised learning.
- Today, we'll learn
 - Pre-training: The biggest breakthrough in NLP
 - And how it is used in practice: fine-tuning and prompting

Today's lecture: Pre-training

- Self-supervised learning techniques that allow us to obtain large-scale labeled data for free from raw text.
- Called pre-training because it is often followed by supervised training, which historically was what we simply referred to as “training.”
- Earliest forms of pre-training: Word embeddings, Contextualized word embeddings
- “Pre-trained Transformers”: First widely-used through BERT (masked LM objective), then GPT (generative LM objective)

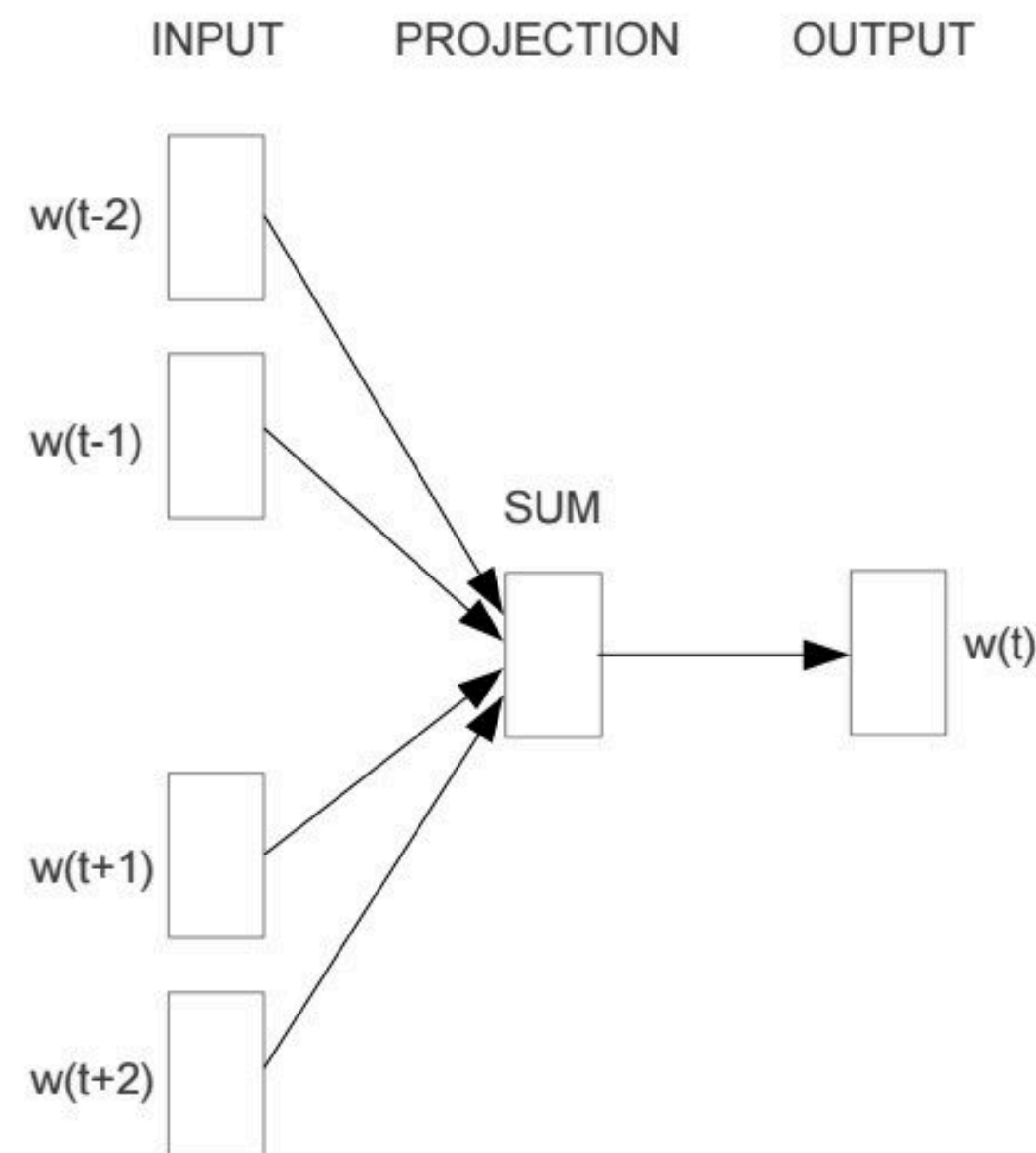
Contextualized Word Embeddings

Recap: word2vec

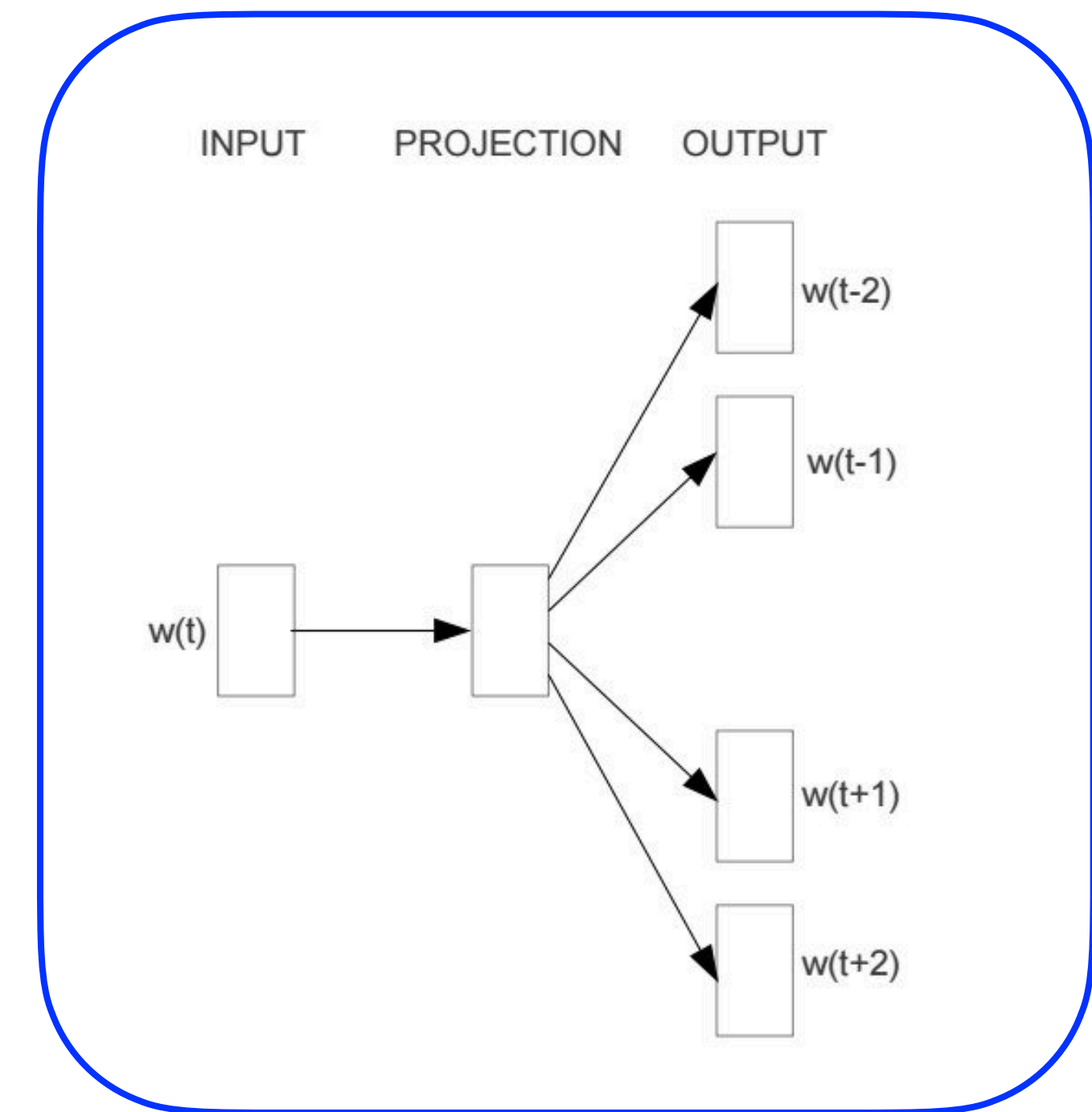
- (Mikolov et al 2013a): Efficient Estimation of Word Representations in Vector Space
- (Mikolov et al 2013b): Distributed Representations of Words and Phrases and their Compositionality



Tomáš Mikolov



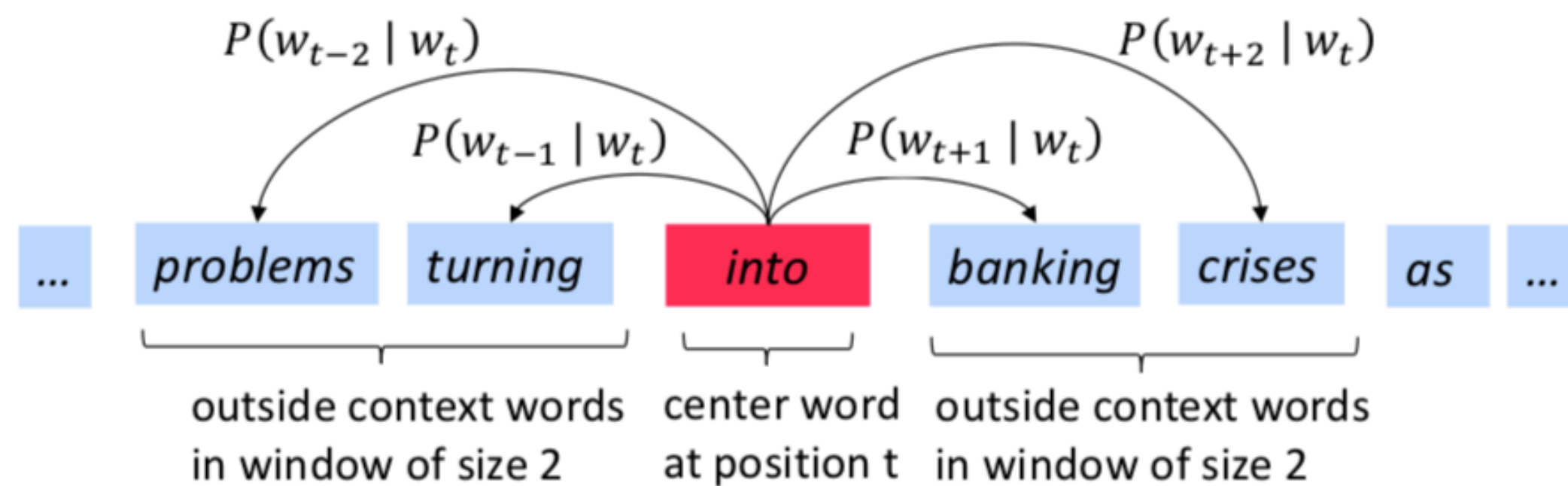
Continuous Bag of Words (CBOW)



Skip-gram

Recap: word2vec

- Key idea: Use each word to predict other words in its context



Our goal is to find parameters that can maximize

$$\underline{P(\text{problems} \mid \text{into}) \times P(\text{turning} \mid \text{into}) \times P(\text{banking} \mid \text{into}) \times P(\text{crises} \mid \text{into})}$$

Recap: word2vec

- Key idea: Use each word to predict other words in its context

Key tricks:

- We turn unlabeled text into supervised learning data
- We train a model on a prediction task, not for the sake of the prediction, but to learn high-quality representations (word embeddings) — *recurring themes in pre-training*

Our goal is to find parameters that can maximize

$$\underline{P(\text{problems} \mid \text{into}) \times P(\text{turning} \mid \text{into}) \times P(\text{banking} \mid \text{into}) \times P(\text{crises} \mid \text{into}) \times}$$

$$\underline{P(\text{turning} \mid \text{banking}) \times P(\text{into} \mid \text{banking}) \times P(\text{crises} \mid \text{banking}) \times P(\text{as} \mid \text{banking}) \dots}$$

Limitations of word2vec

- One vector for each word type
 - (a.k.a. static embeddings)
- Complex characteristics of word use: syntax and semantics
- Polysemous words (e.g., mouse, bank)

$$v(\text{play}) = \begin{pmatrix} -0.224 \\ 0.130 \\ -0.290 \\ 0.276 \end{pmatrix}$$

mouse¹ : a *mouse* controlling a computer system in 1968.

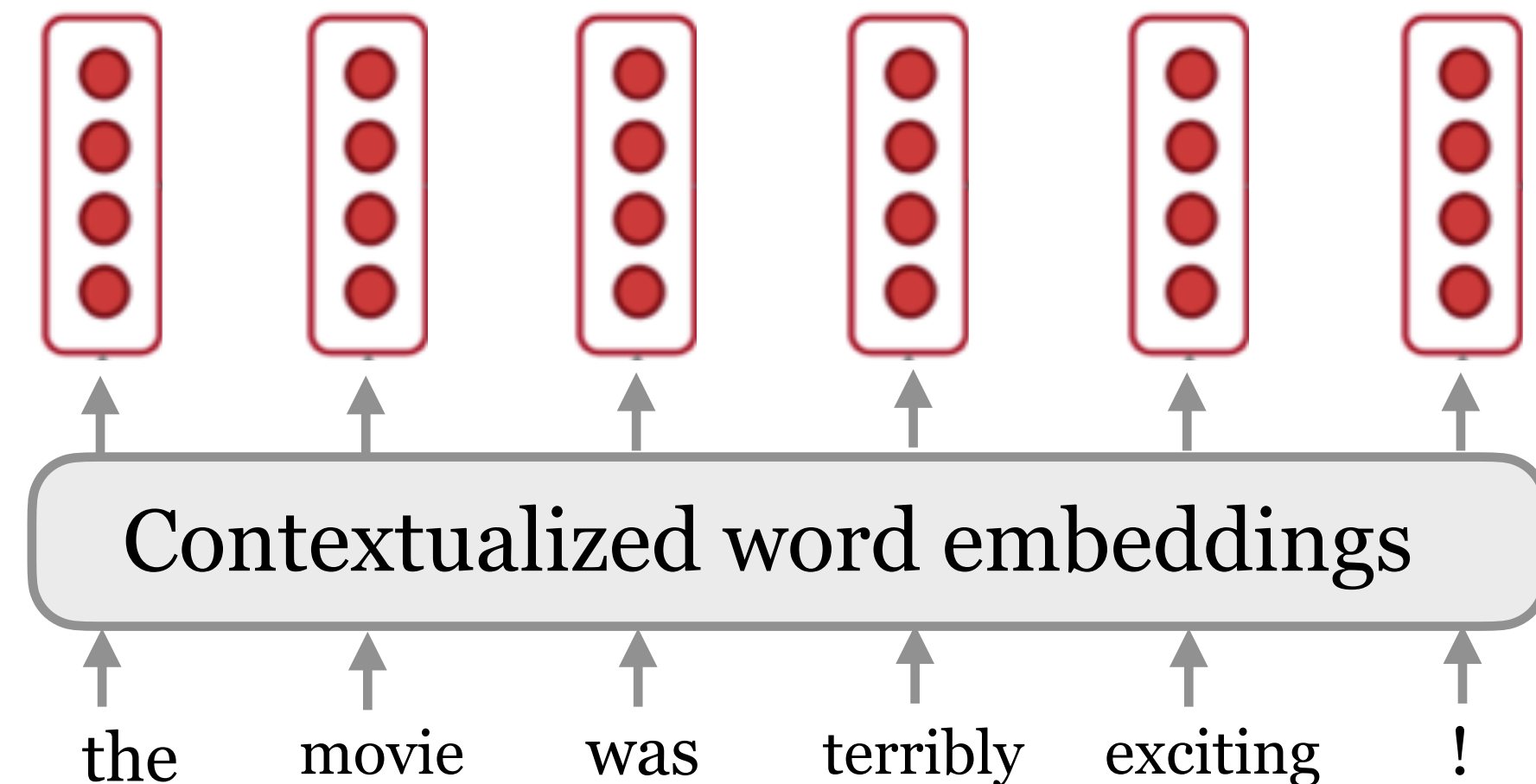
mouse² : a quiet animal like a *mouse*

bank¹ : ...a *bank* can hold the investments in a custodial account ...

bank² : ...as agriculture burgeons on the east *bank*, the river ...

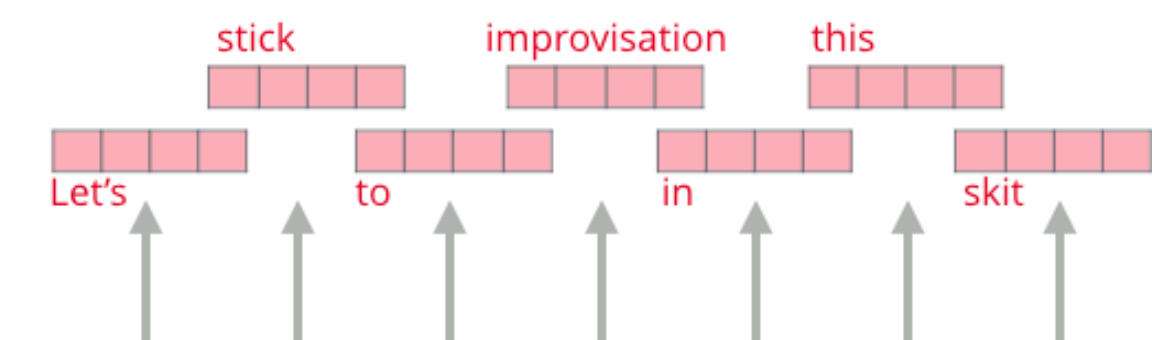
Contextualized word embeddings

Let's build a vector for each word conditioned on its **context**!



$$f: (w_1, w_2, \dots, w_n) \longrightarrow \mathbf{x}_1, \dots, \mathbf{x}_n \in \mathbb{R}^d$$

ELMo
Embeddings



Words to embed



Contextualized word embeddings



Contextualized word embeddings

Sent #1: Chico Ruiz made a spectacular **play** on Alusik's grounder { . . . } $v(\text{play}) = ?$

Sent #2: Olivia De Havilland signed to do a Broadway **play** for Garson { . . . } $v(\text{play}) = ?$

Sent #3: Kieffer was commended for his ability to hit in the clutch , as well as his all-round excellent **play** { . . . } $v(\text{play}) = ?$

Sent #4: { . . . } they were actors who had been handed fat roles in a successful **play** { . . . } $v(\text{play}) = ?$

Sent #5: Concepts **play** an important role in all aspects of cognition { . . . } $v(\text{play}) = ?$

Contextualized word embeddings

Sent #1: Chico Ruiz made a spectacular **play** on Alusik's grounder { . . . }

Which of the following $v(\text{play})$ is expected to have the most similar vector to the first one?

- (A) Olivia De Havilland signed to do a Broadway **play** for Garson { . . . }
- (B) Kieffer was commended for his ability to hit in the clutch , as well as his all-round excellent **play** { . . . }
- (C) { . . . } they were actors who had been handed fat roles in a successful **play** { . . . }
- (D) Concepts **play** an important role in all aspects of cognition { . . . }

(B) is correct.

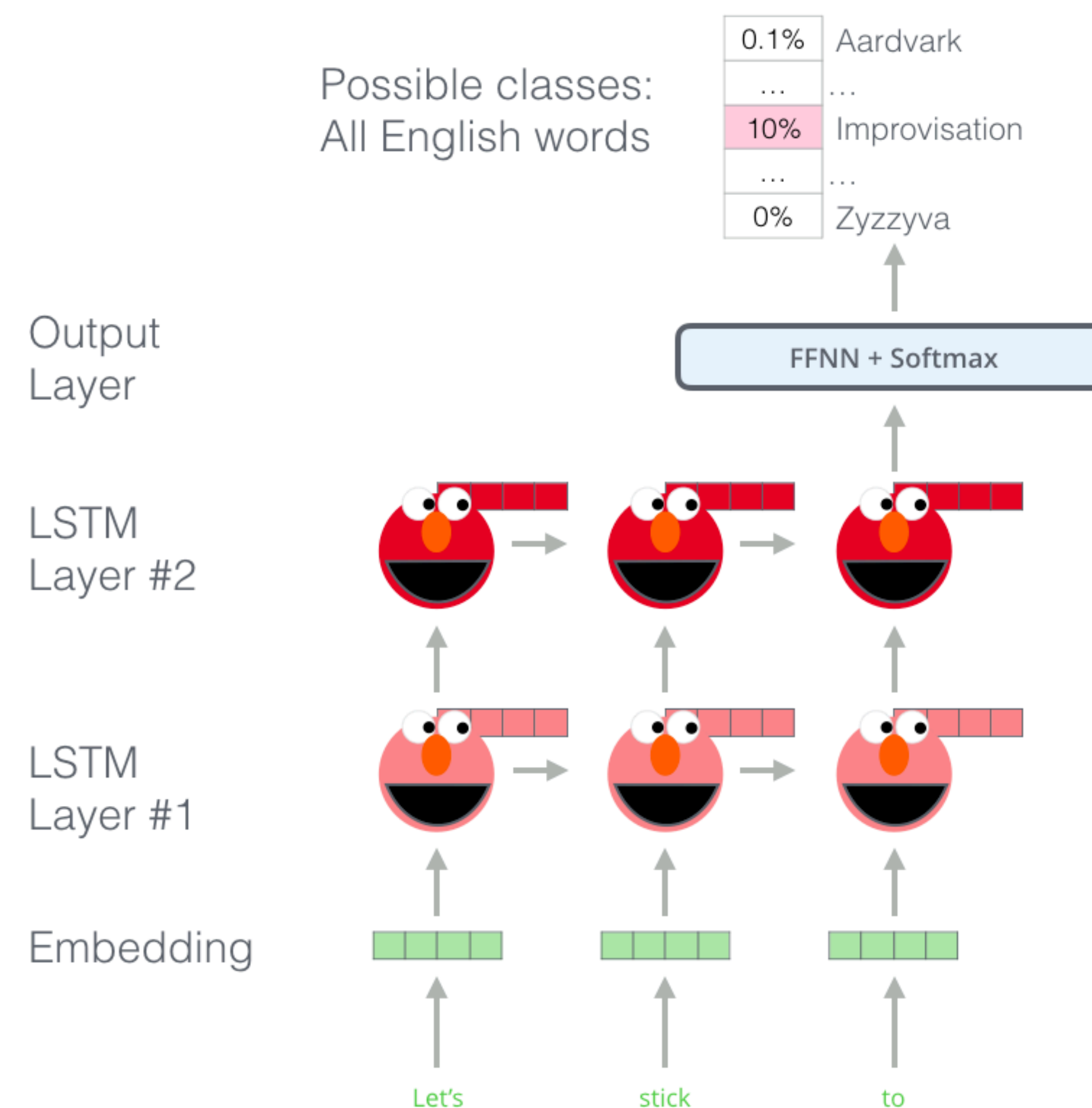
Contextualized word embeddings

Source		Nearest Neighbors
GloVe	play	playing, game, games, played, players, plays, player, Play, football, multiplayer
biLM	Chico Ruiz made a spectacular <u>play</u> on Alusik 's grounder {...}	Kieffer , the only junior in the group , was commended for his ability to hit in the clutch , as well as his all-round excellent <u>play</u> .
	Olivia De Havilland signed to do a Broadway <u>play</u> for Garson {...}	{...} they were actors who had been handed fat roles in a successful <u>play</u> , and had talent enough to fill the roles competently , with nice understatement .

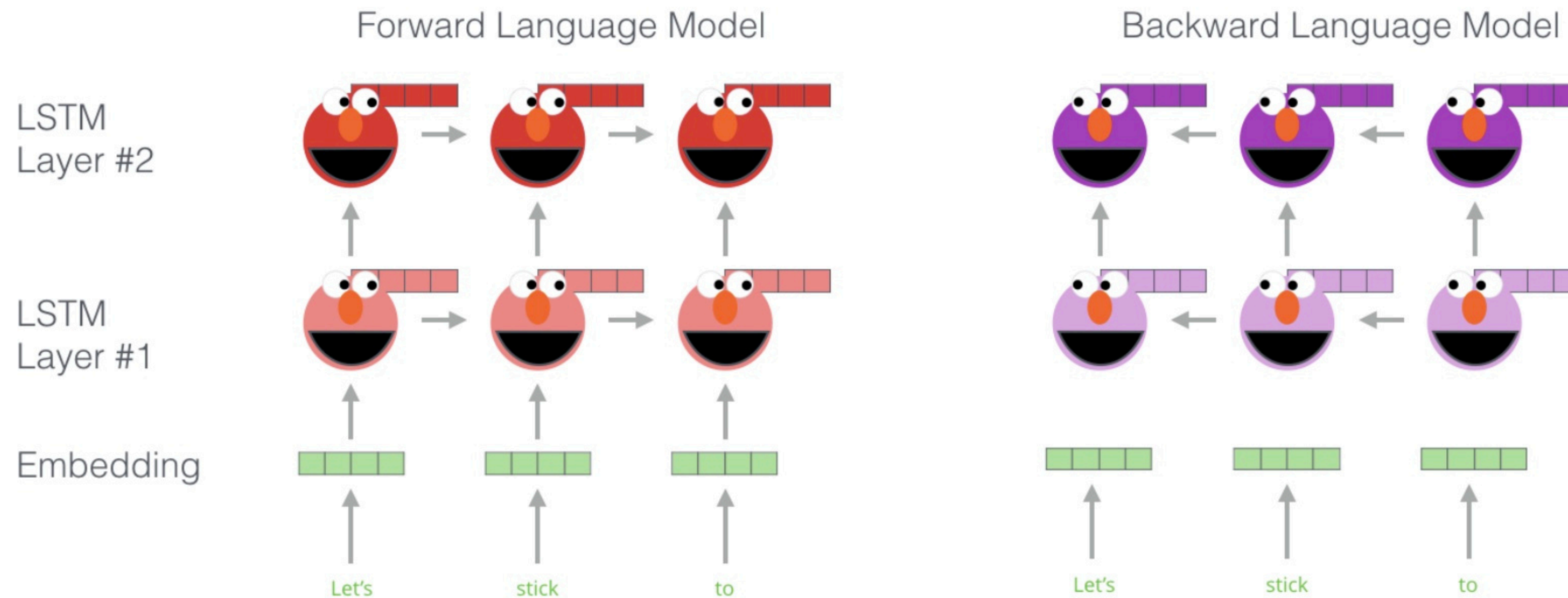
ELMo: Embeddings from Language Models (Feb 2018)

The key idea of ELMo:

- Train *two* stacked LSTM-based language models on a **large** corpus
- Use the **hidden states** of the LSTMs for each token to compute a vector representation of each word



How does ELMo work?



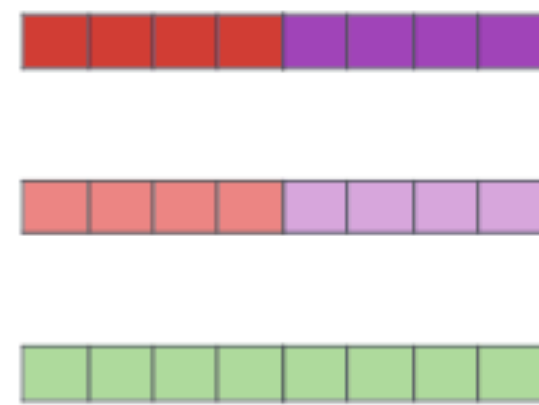
$$\mathbf{ELMo}_k^{task} = E(R_k; \Theta^{task}) = \gamma^{task} \sum_{j=0}^L s_j^{task} \mathbf{h}_{k,j}^{LM}$$

Contextualized word embeddings =
The weighted average of input embeddings + all hidden representations

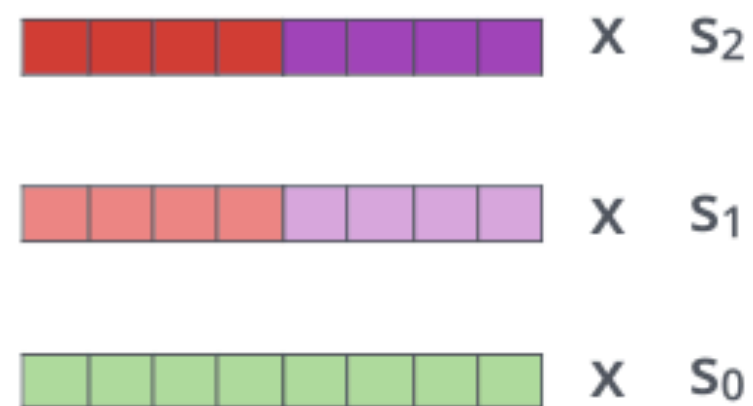
The weights γ^{task} , s_j^{task} are task-dependent and learned

How does ELMo work?

1- Concatenate hidden layers



2- Multiply each vector by a weight based on the task

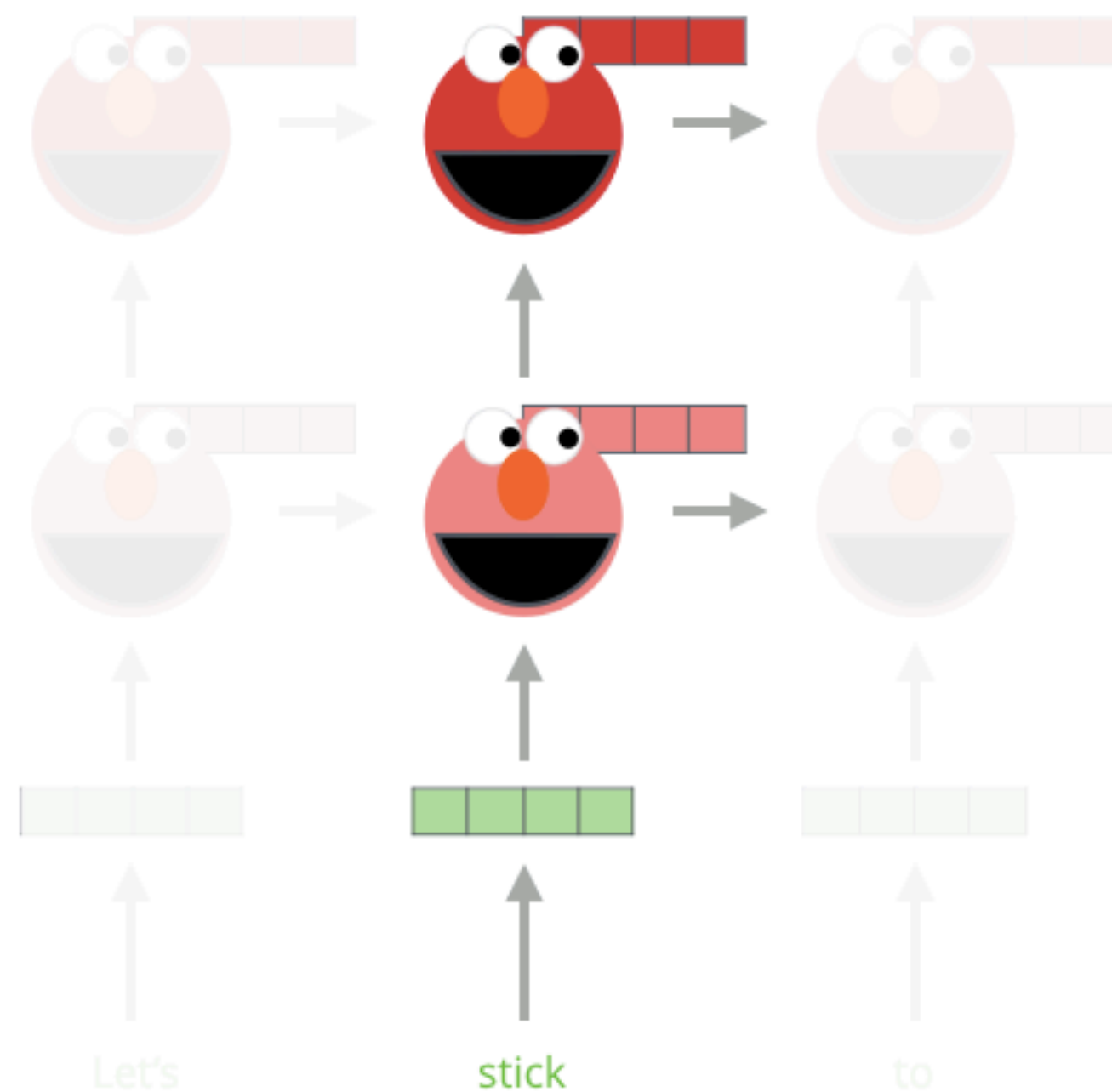


3- Sum the (now weighted) vectors

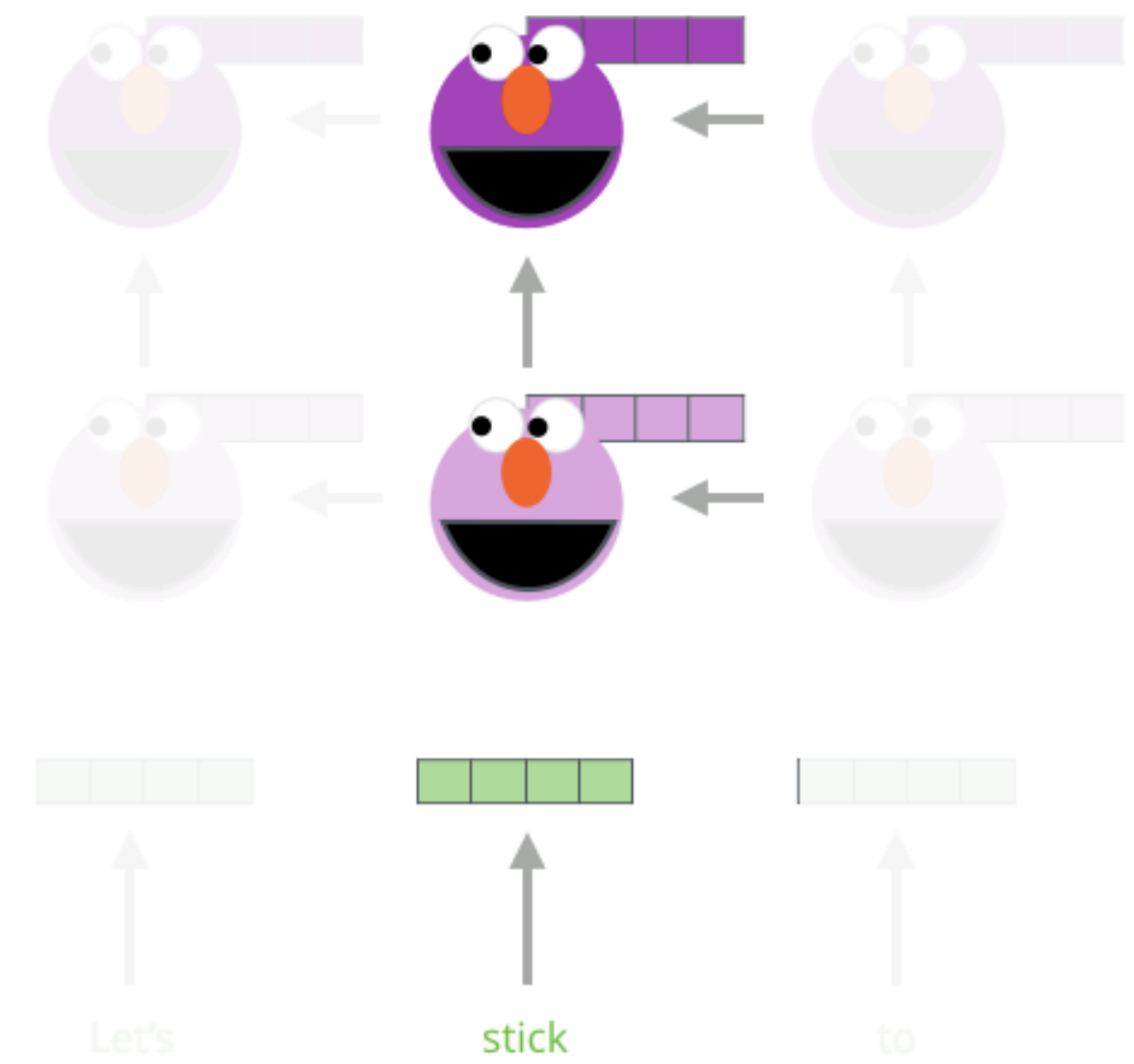


ELMo embedding of "stick" for this task in this context

Forward Language Model



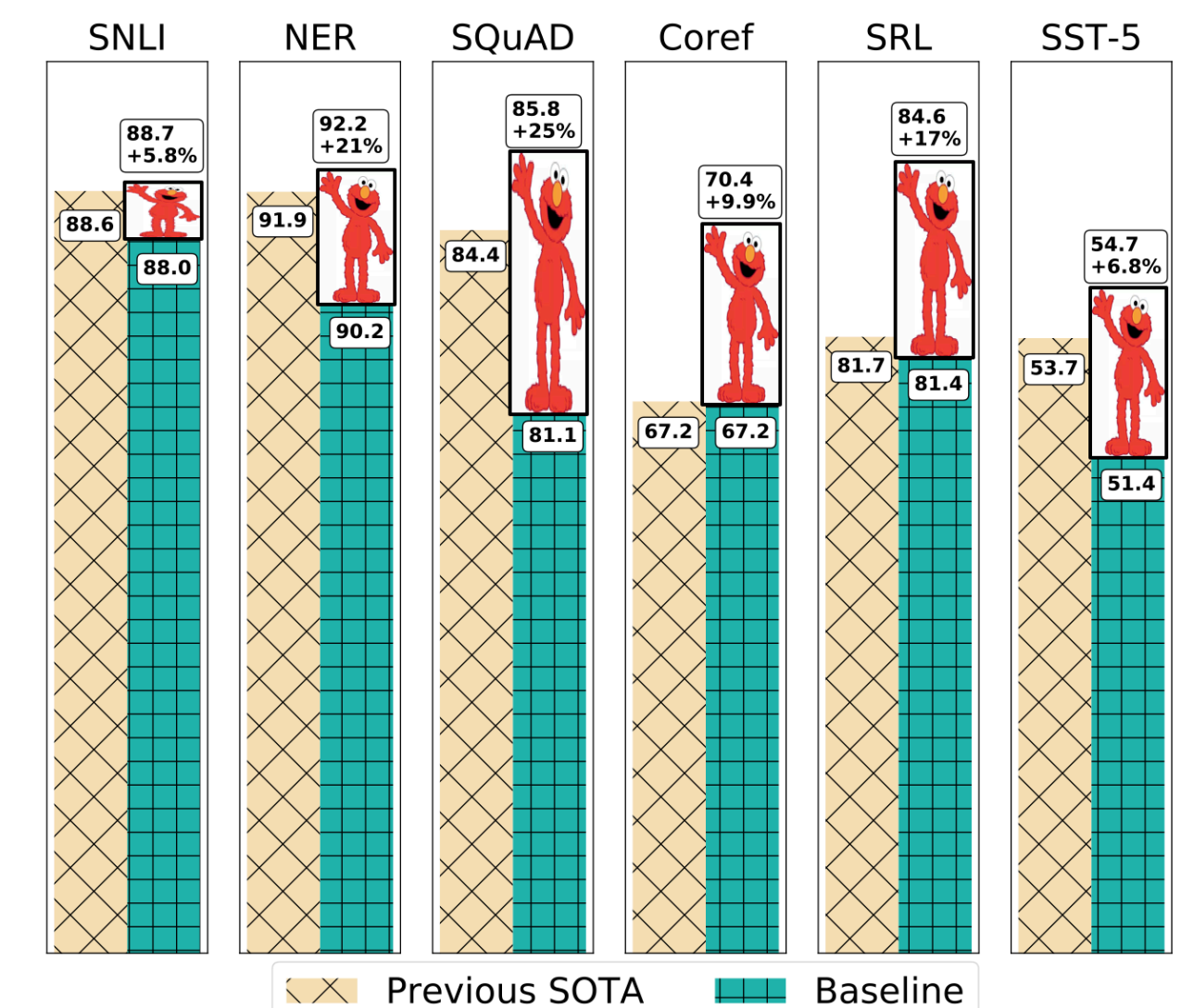
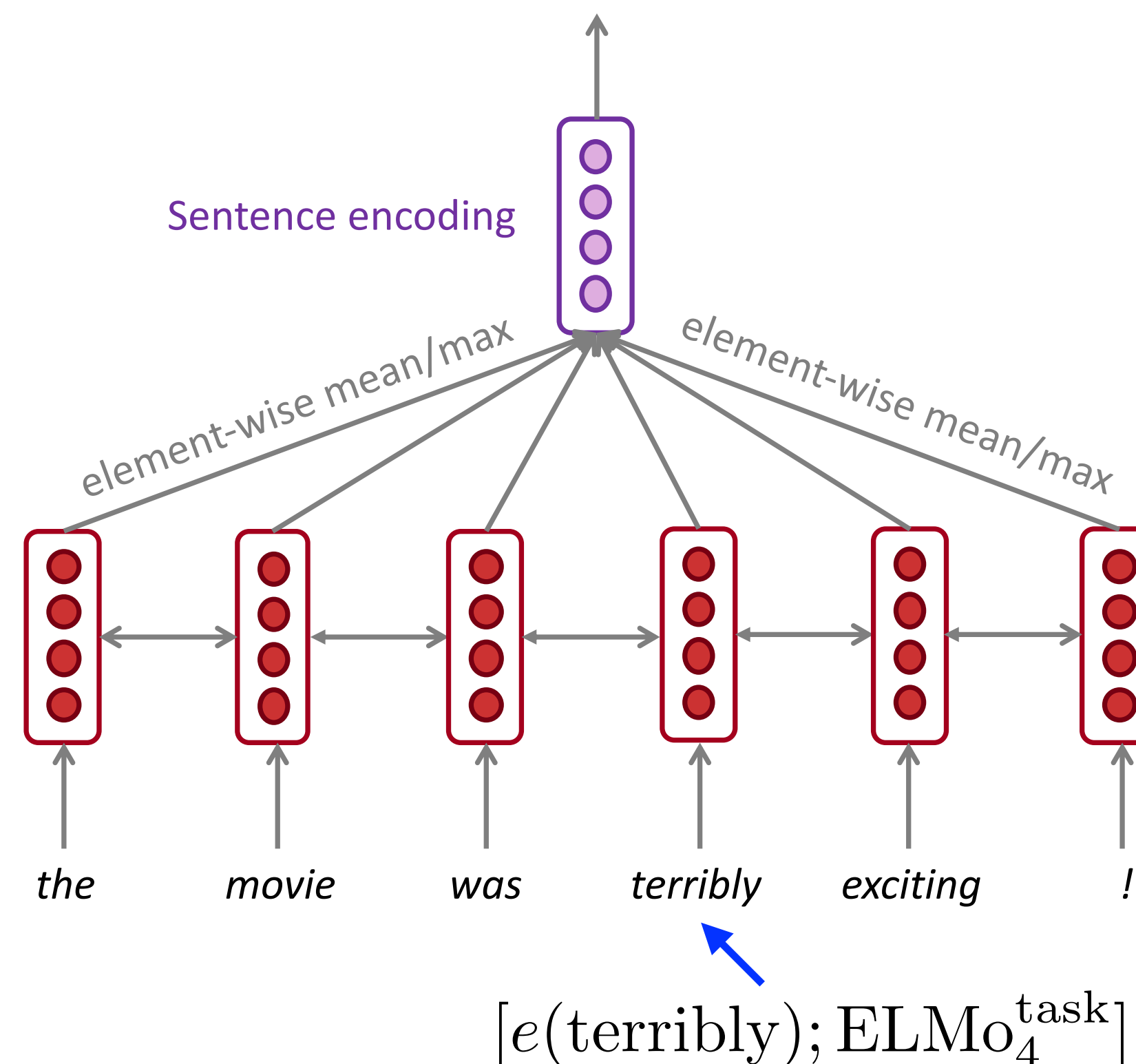
Backward Language Model



ELMo: Pre-training and the use

- Data: 10 epochs on 1B Word Benchmark (trained on **single sentences**)
- Training time: 2 weeks on 3 NVIDIA GTX 1080 GPUs

Example use: A BiLSTM model for sentiment classification



ELMo: Some takeaways

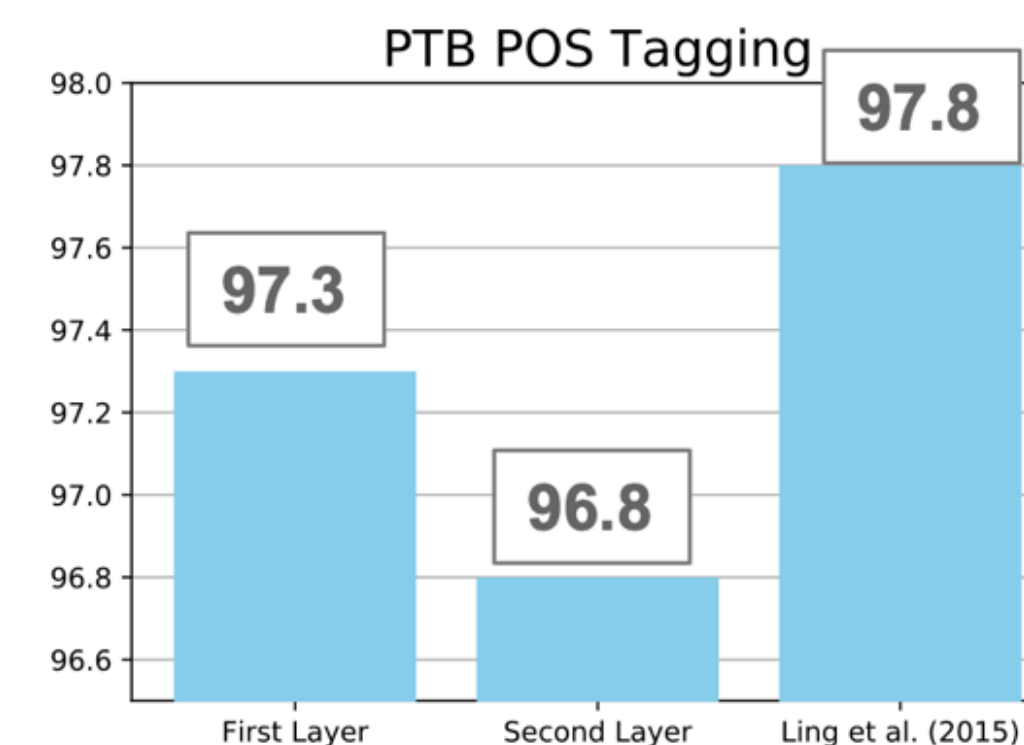
Q: Why use both forward and backward language models?

Because it is important to model both left and right context!

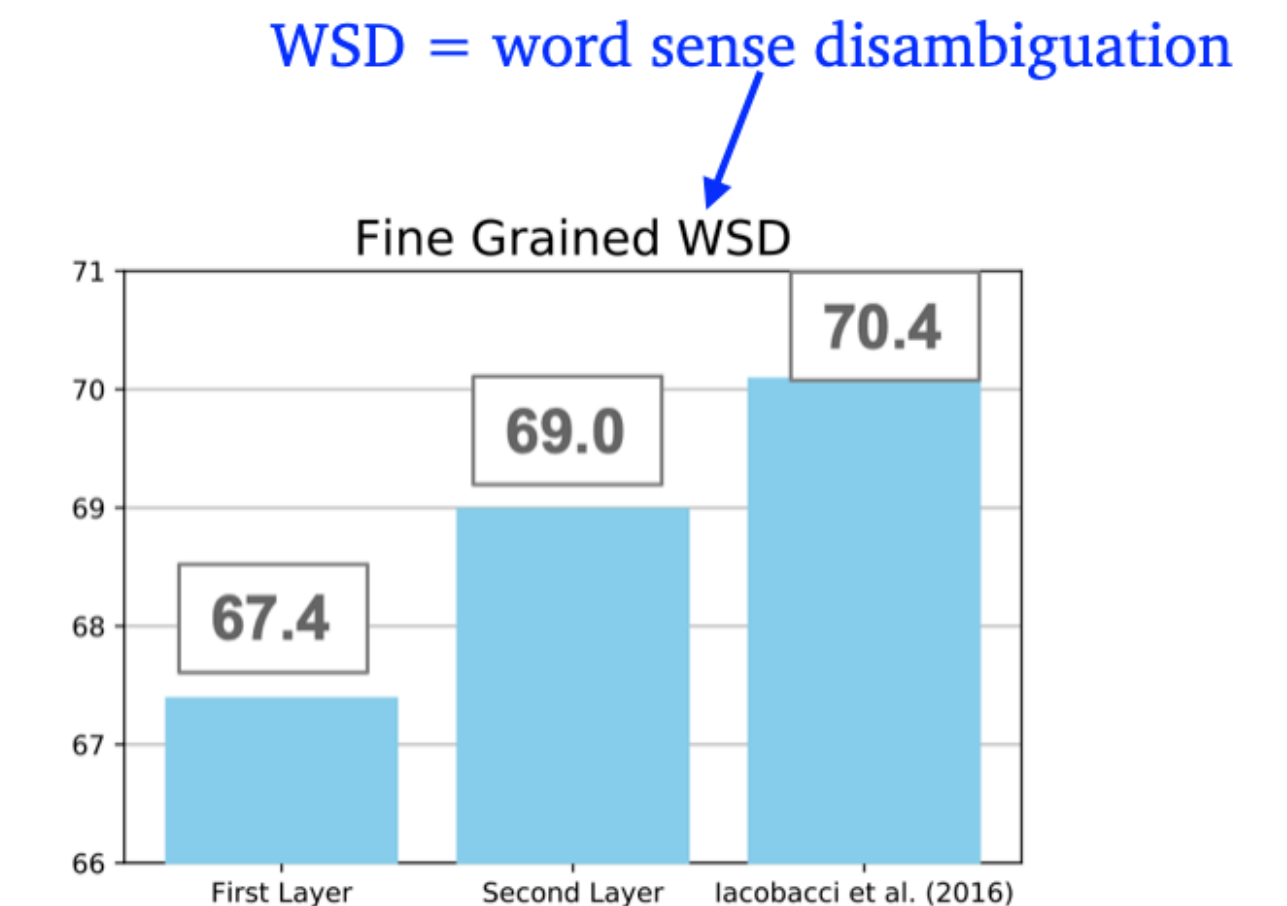
Bidirectionality is very important in language understanding tasks!

Q: Why use the weighted average of different layers instead of just the top layer?

Because different layers are expected to encode different information.



first layer > second layer



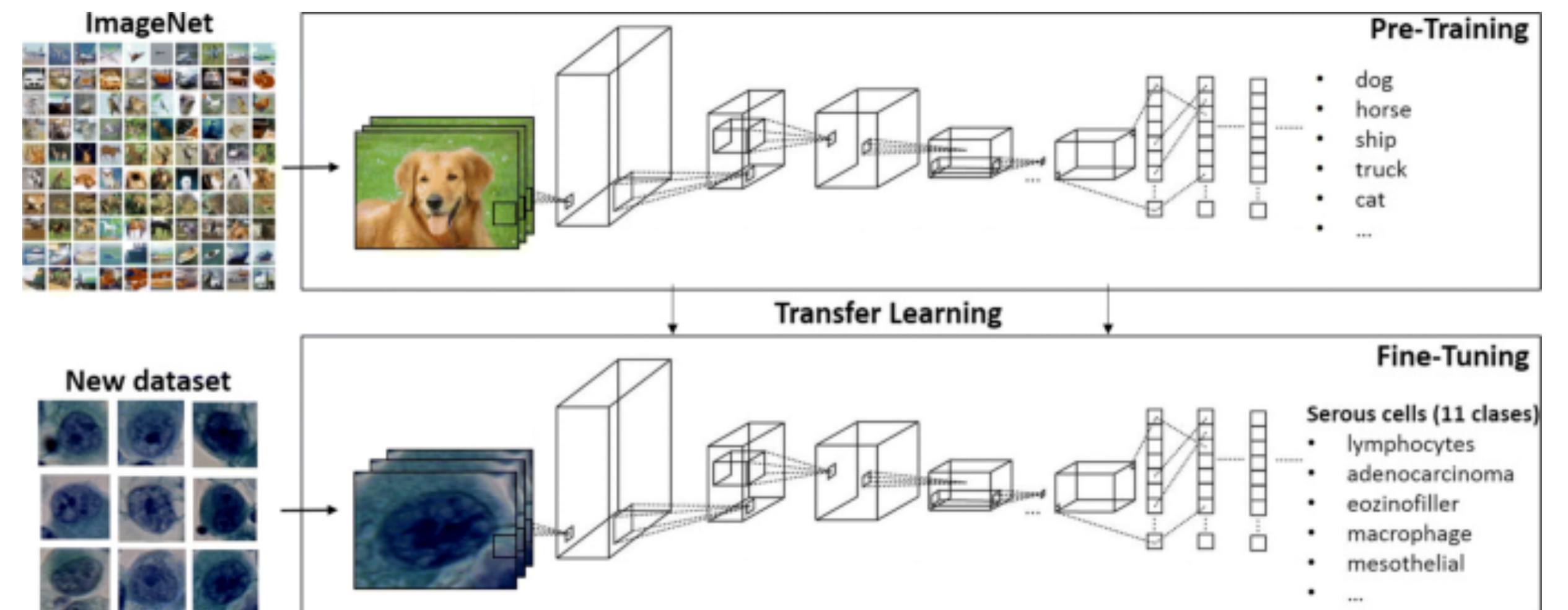
second layer > first layer

Pre-training and Fine-tuning

What is pre-training / fine-tuning?

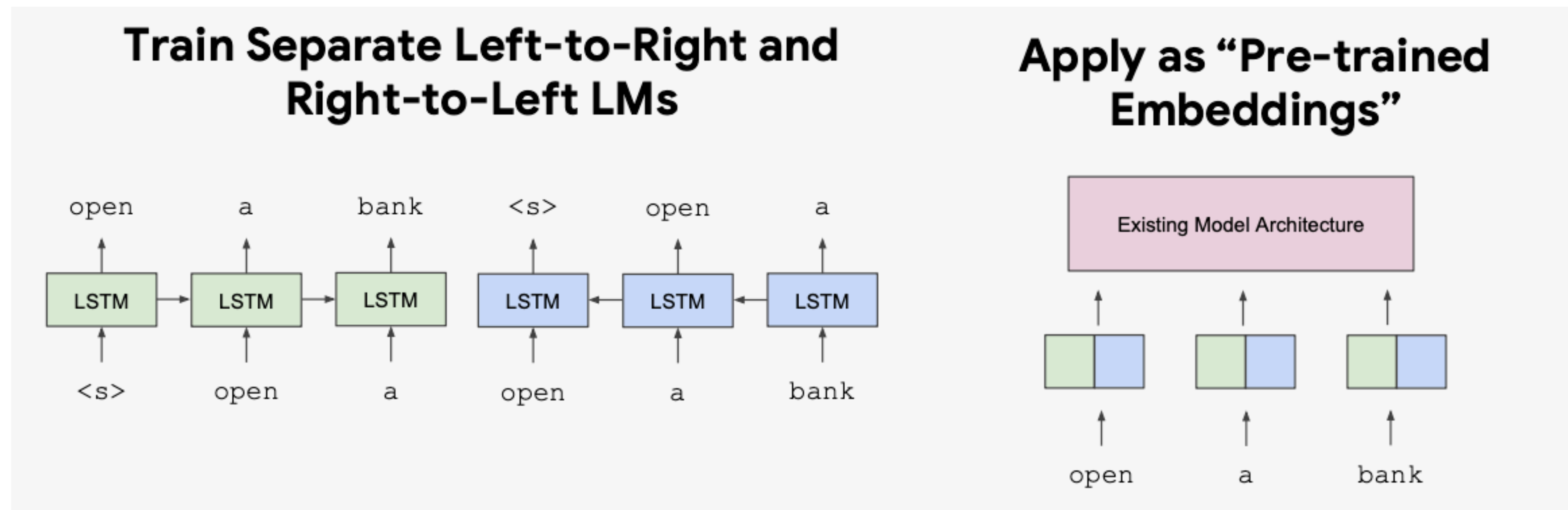
- “Pre-train” a model on a large dataset for task X, then “fine-tune” it on a dataset for task Y
- Key idea: X is somewhat related to Y, so a model that can do X will have some good neural representations for Y as well
- ImageNet pre-training is huge in computer vision: learning generic visual features for recognizing objects

Can we find some task X that can be useful for a wide range of downstream tasks Y?



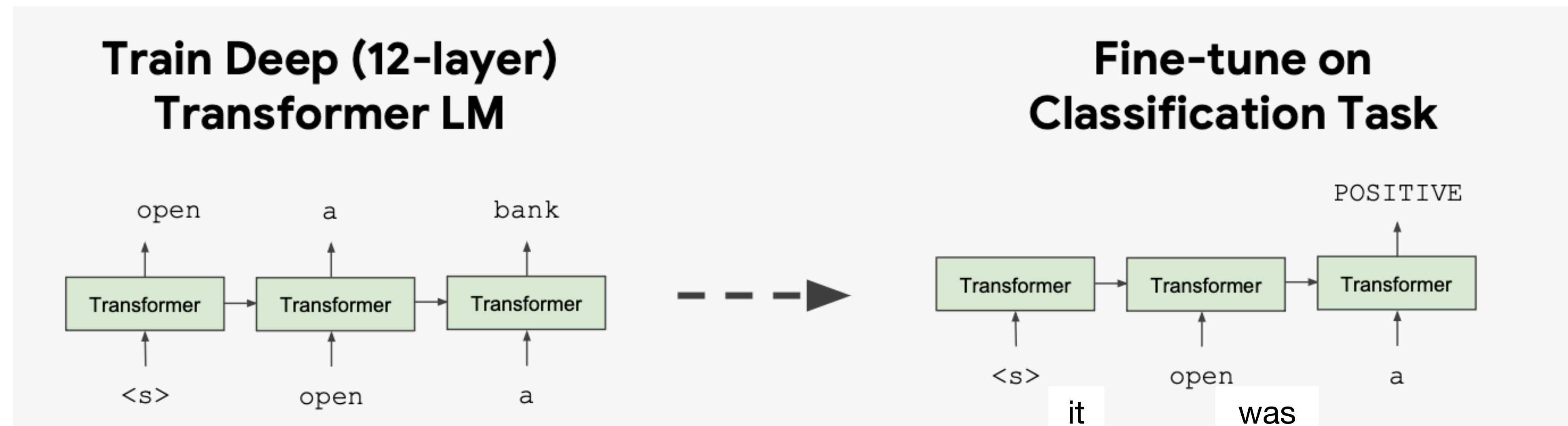
Feature-based vs. fine-tuning approaches

- ELMo is a feature-based approach which only produces word embeddings that can be used as **input representations** of existing neural models



Feature-based vs. fine-tuning approaches

- GPT / BERT (and most of following models) are **fine-tuning approaches**
 - Almost all model weights will be **re-used**, and only a small number of task-specific will be added for downstream tasks



Most of pre-training is reconstructing the input

You learn a lot from reconstructing the input!

- I went to the ocean and saw fish, turtles, _____ and seals.
 - General semantics
- I put ____ fork down on the table
 - Syntactic constraints
- The woman walked across the street checking for traffic over _____ shoulder.
 - Co-reference, relations between different entities within the sentence
- Overall, the value I got from the two hours watching it was the sum total of the popcorn and the drink. The movie was _____.
 - Sentiment
- UC Berkeley is located in _____.
 - Word knowledge

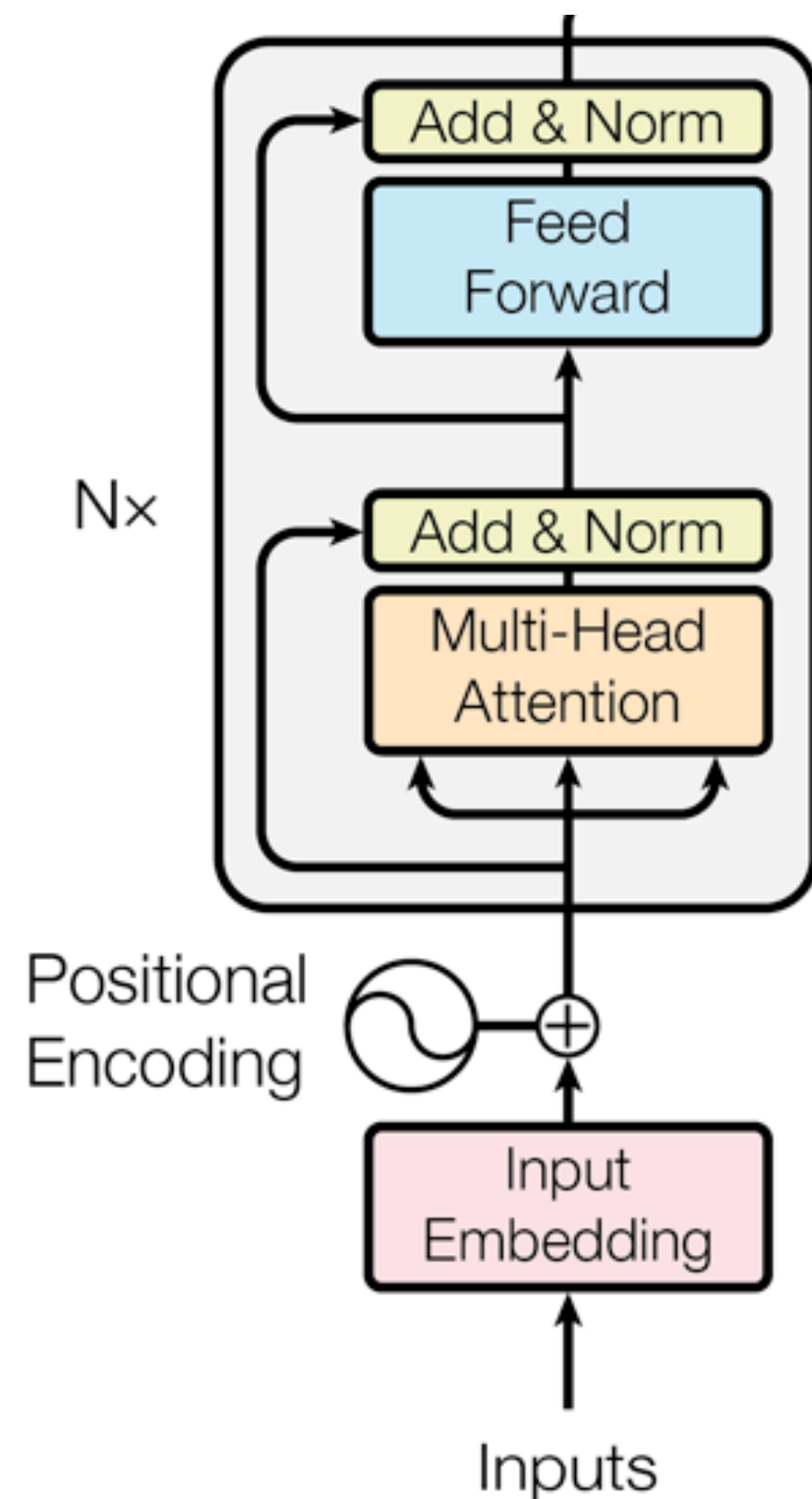
Pre-training for three types of architectures

Architecture	Pretraining objective	Examples
Transformer Encoder	Masked language models	BERT, RoBERTa, ELECTRA
Transformers Encoder-Decoder	Span Corruption	T5, BART
Transformers Decoder	Autoregressive language models (Causal language models)	GPT-2, GPT-3, Llama

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BERT: Bidirectional Encoder Representations from Transformers (October 2018)



- It is a fine-tuning approach based on a deep **bidirectional encoder-only Transformer encoder**
- The key: learn representations based on **bidirectional contexts**

Example #1: we went to the river bank.

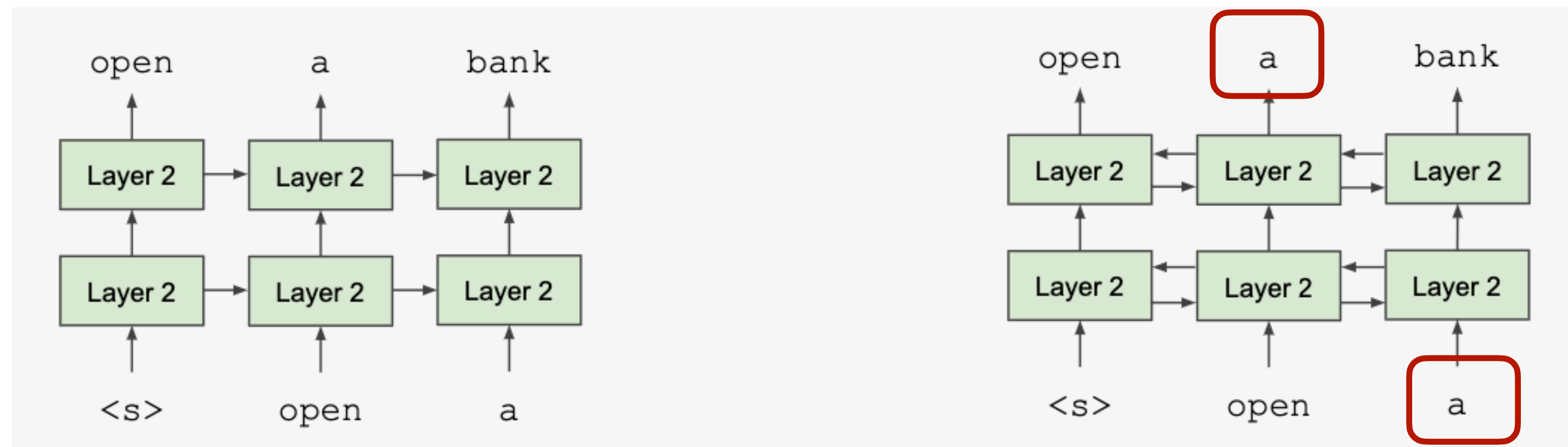
Example #2: I need to go to bank to make a deposit.

- Two new pre-training objectives:
 - **Masked language modeling (MLM)**
 - Next sentence prediction (NSP) - Later work shows that NSP hurts performance though..



Masked Language Modeling (MLM)

- Q: Why we can't do language modeling with bidirectional models?



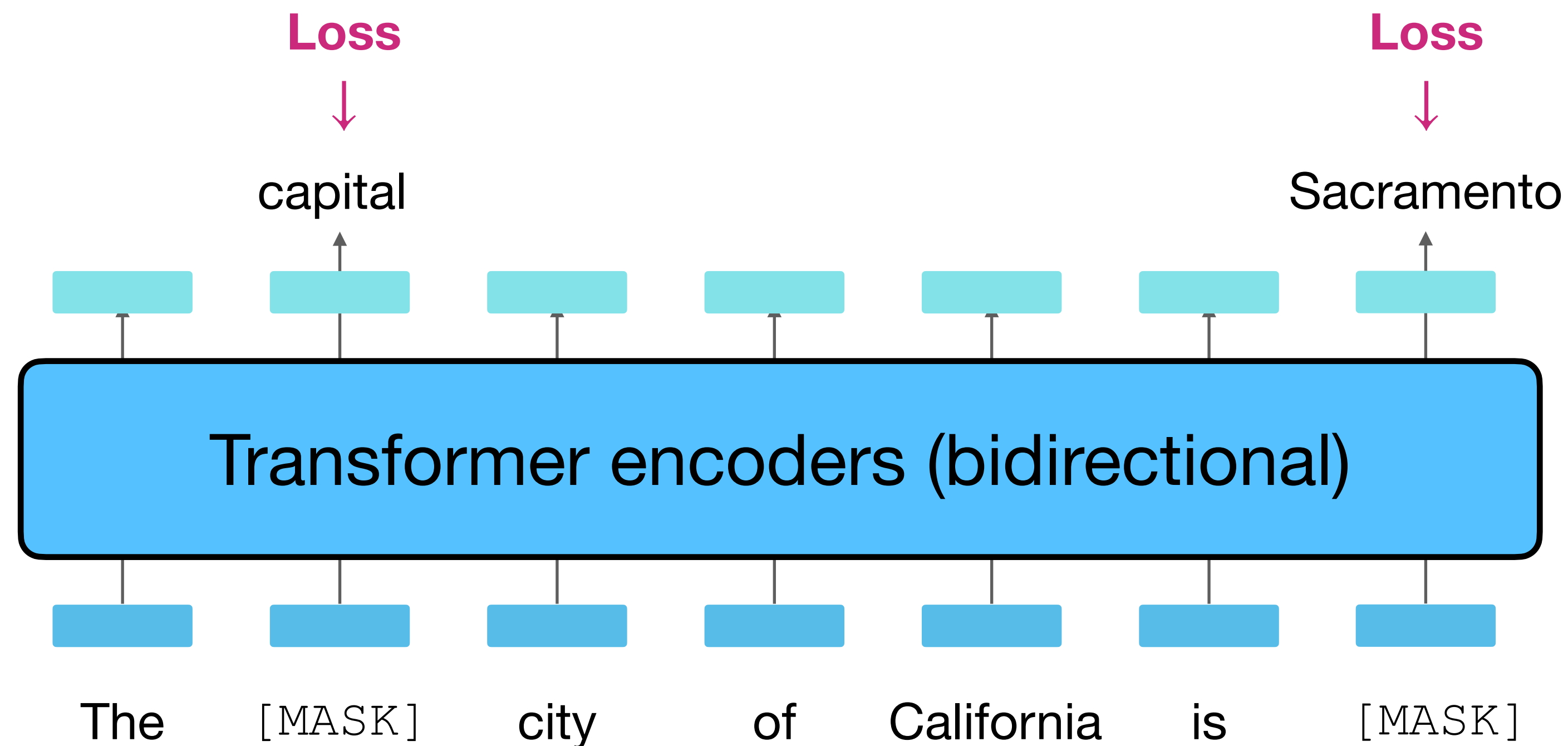
- Solution: Mask out $k\%$ of the input words, and then predict the masked words

store gallon
↑ ↑
the man went to [MASK] to buy a [MASK] of milk

$k = 15\%$ in practice

Masked Language Modeling (MLM)

1. Prepare raw text: "The capital city of California is Sacramento."
2. Mask 15% of the tokens: "The [MASK] city of California is [MASK]."
3. Feed into the Transformers encoder and train with a masked language modeling objective.



MLM: 80-10-10 corruption

For the 15% predicted words,

- 80% of the time, they replace it with [MASK] token

went to the store → went to the [MASK]

- 10% of the time, they replace it with a random word in the vocabulary

went to the store → went to the running

- 10% of the time, they keep it unchanged

went to the store → went to the store

Why? Because [MASK] tokens are never seen during fine-tuning

(See Table 8 of the paper for an ablation study)

Next Sentence Prediction (NSP)

- Motivation: many NLP downstream tasks require understanding the relationship between two sentences (natural language inference, paraphrase detection, QA)
- NSP is designed to reduce the gap between pre-training and fine-tuning

[CLS]: a special token
always at the beginning

[SEP]: a special token used
to separate two segments

Input = [CLS] the man went to [MASK] store [SEP]

he bought a gallon [MASK] milk [SEP]

Label = IsNext

Input = [CLS] the man [MASK] to the store [SEP]

penguin [MASK] are flight ##less birds [SEP]

Label = NotNext

They sample two contiguous segments for 50% of the time and another random segment from the corpus for 50% of the time

(Later MLM work found it less useful and removed it, e.g., RoBERTa)

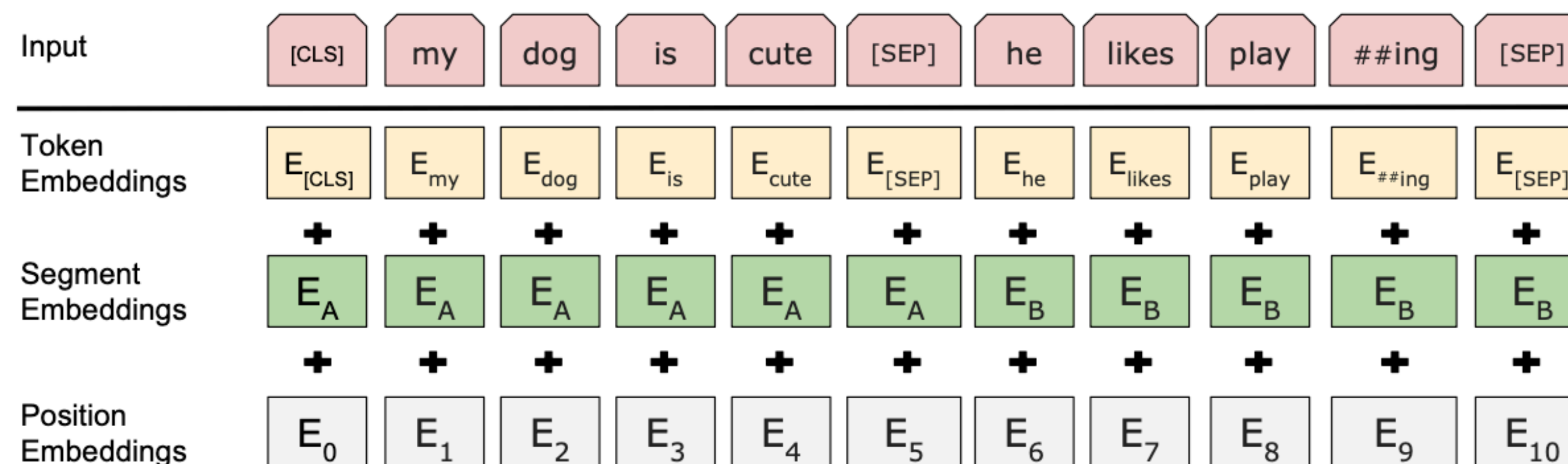
BERT architecture (1/2)

- Vocabulary size: 30,000 **wordpieces** (Wu et al., 2016)

	word		vocab mapping	embedding
Common words	hat	→	hat	
	learn	→	learn	
Variations	taaaaasty	→	taa## aaa## sty	
misspellings	laern	→	la## ern	
novel items	Transformerify	→	Transformer## ify	

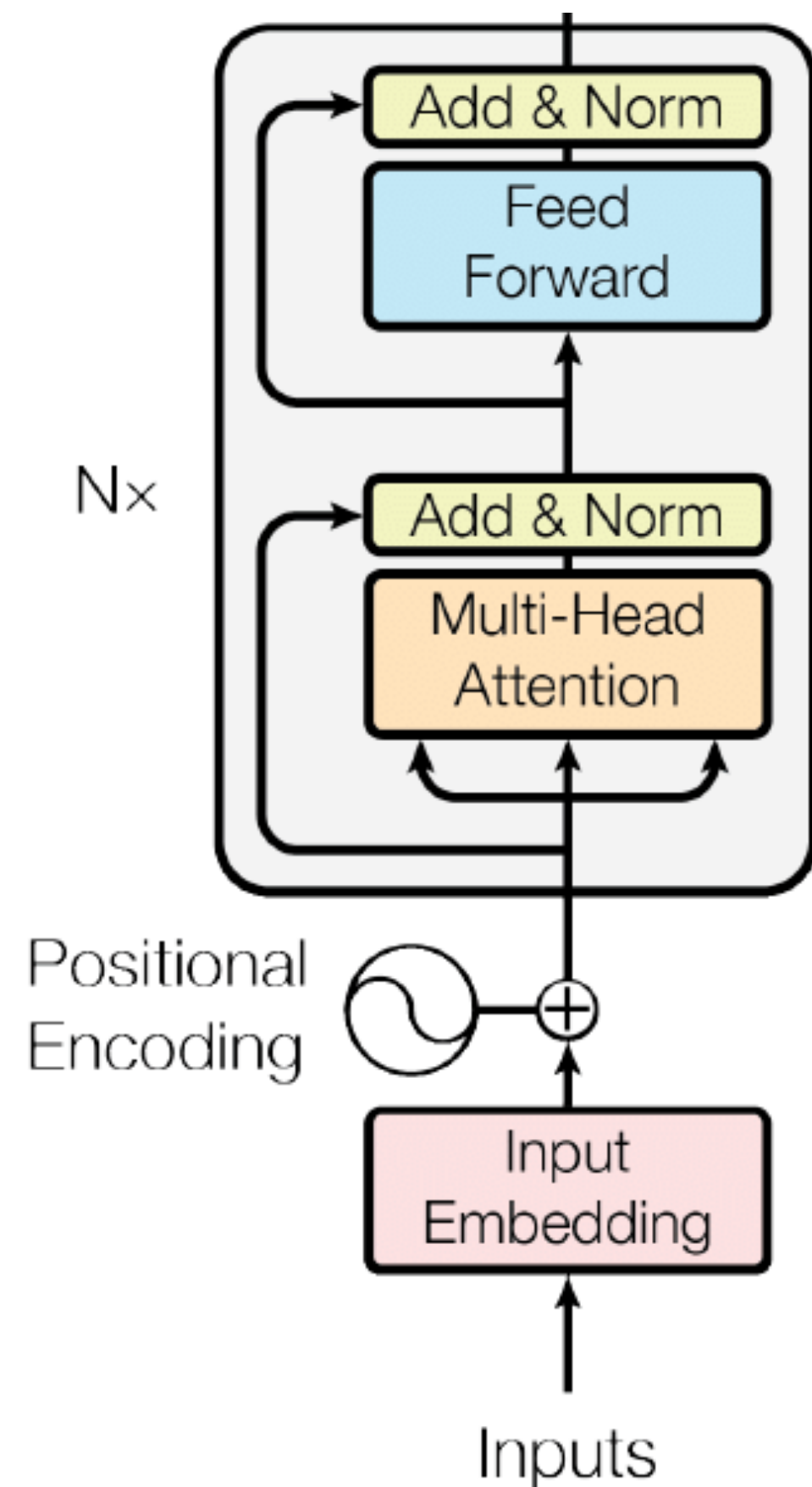
(Image: Stanford CS224N)

- Input embeddings:

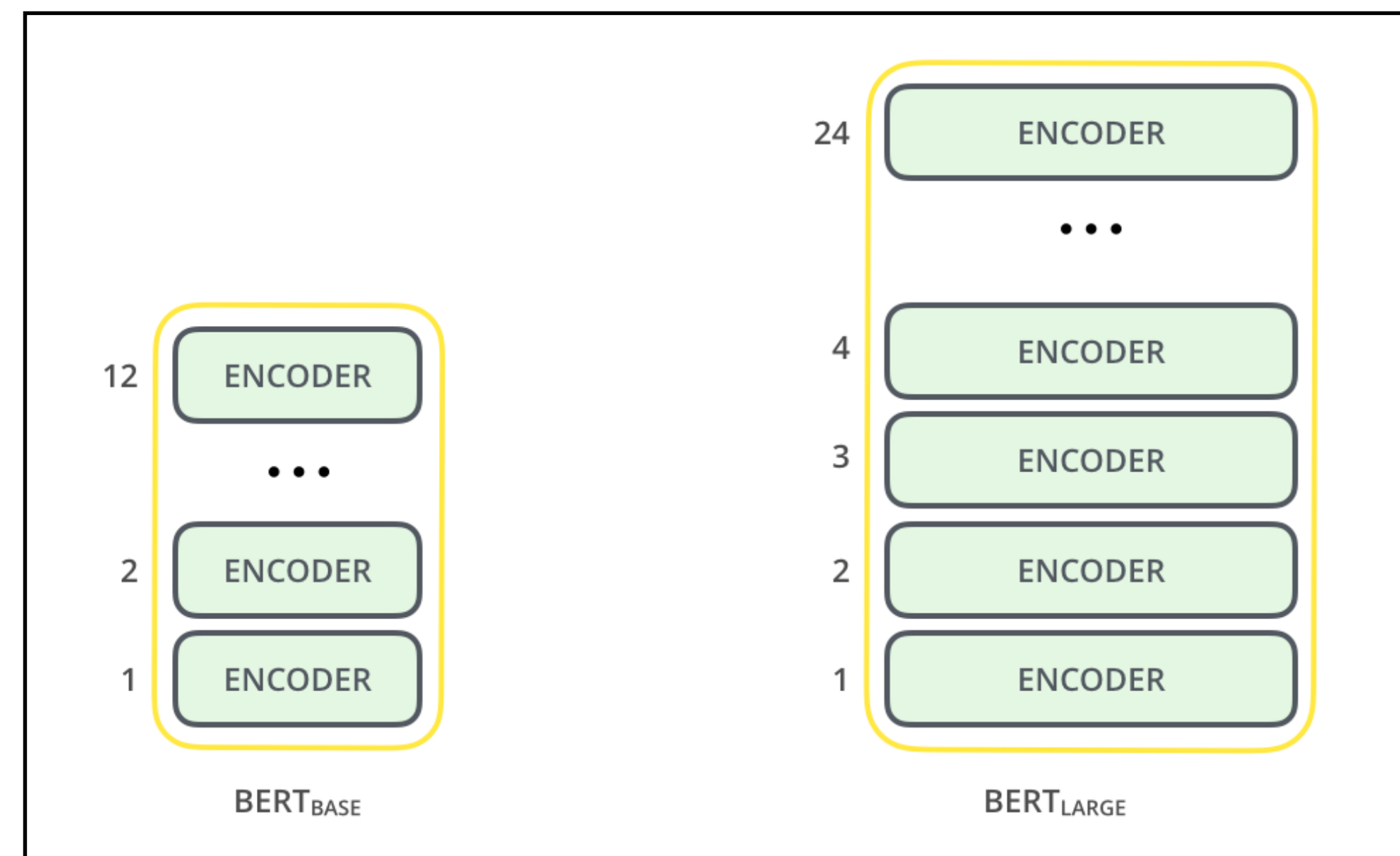


Separate two segments

BERT architecture (2/2)



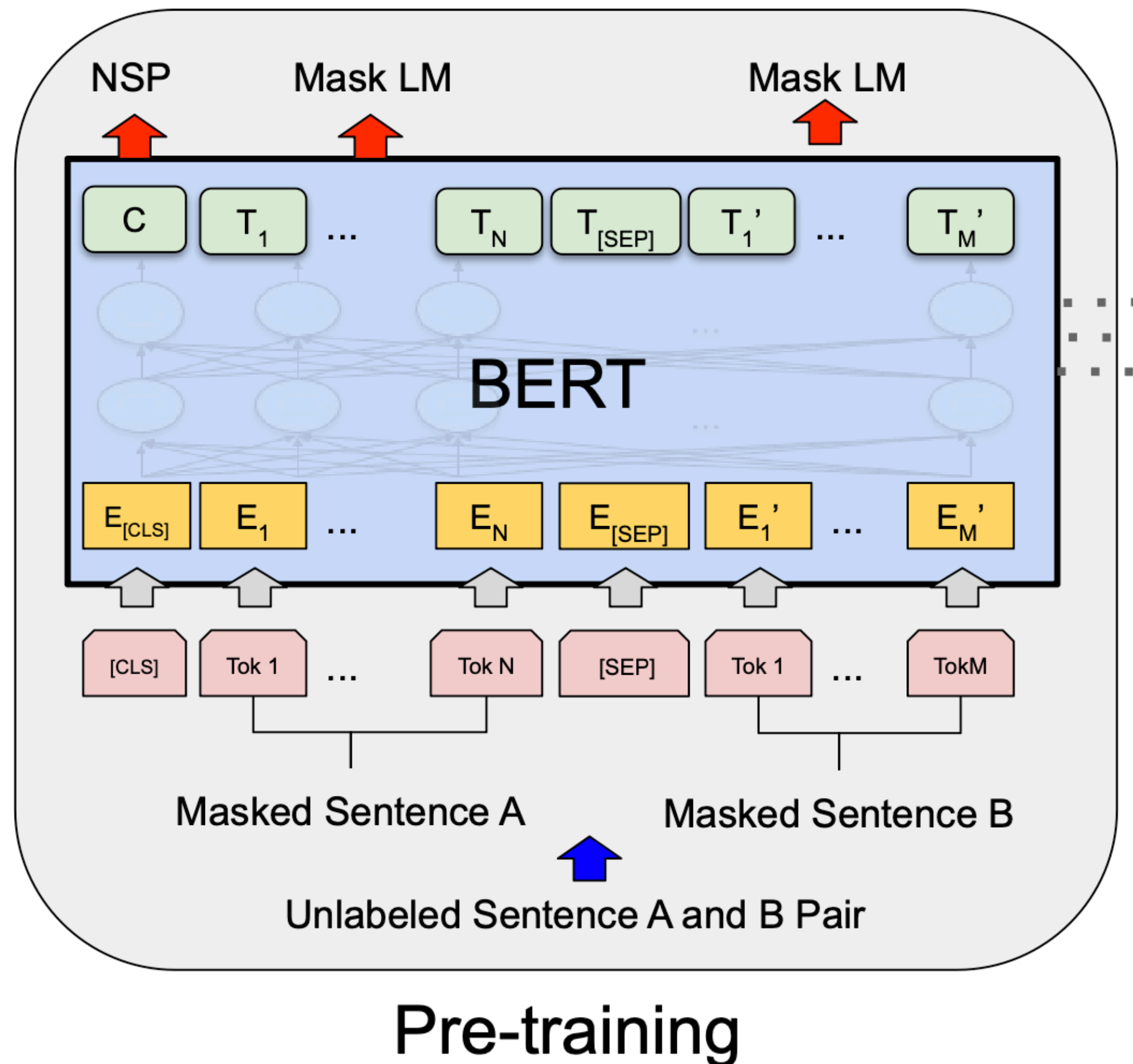
- BERT-base: 12 layers, 768 hidden size, 12 attention heads, 110M parameters
- BERT-large: 24 layers, 1024 hidden size, 16 attention heads, 340M parameters



BERT pre-training (1/2)

- Training corpus: Wikipedia (2.5B) + BooksCorpus (0.8B)
- Max sequence size: 512 wordpiece tokens (roughly 256 and 256 for two non-contiguous sequences)
- Trained for 1M steps, batch size 128k

BERT pre-training (2/2)

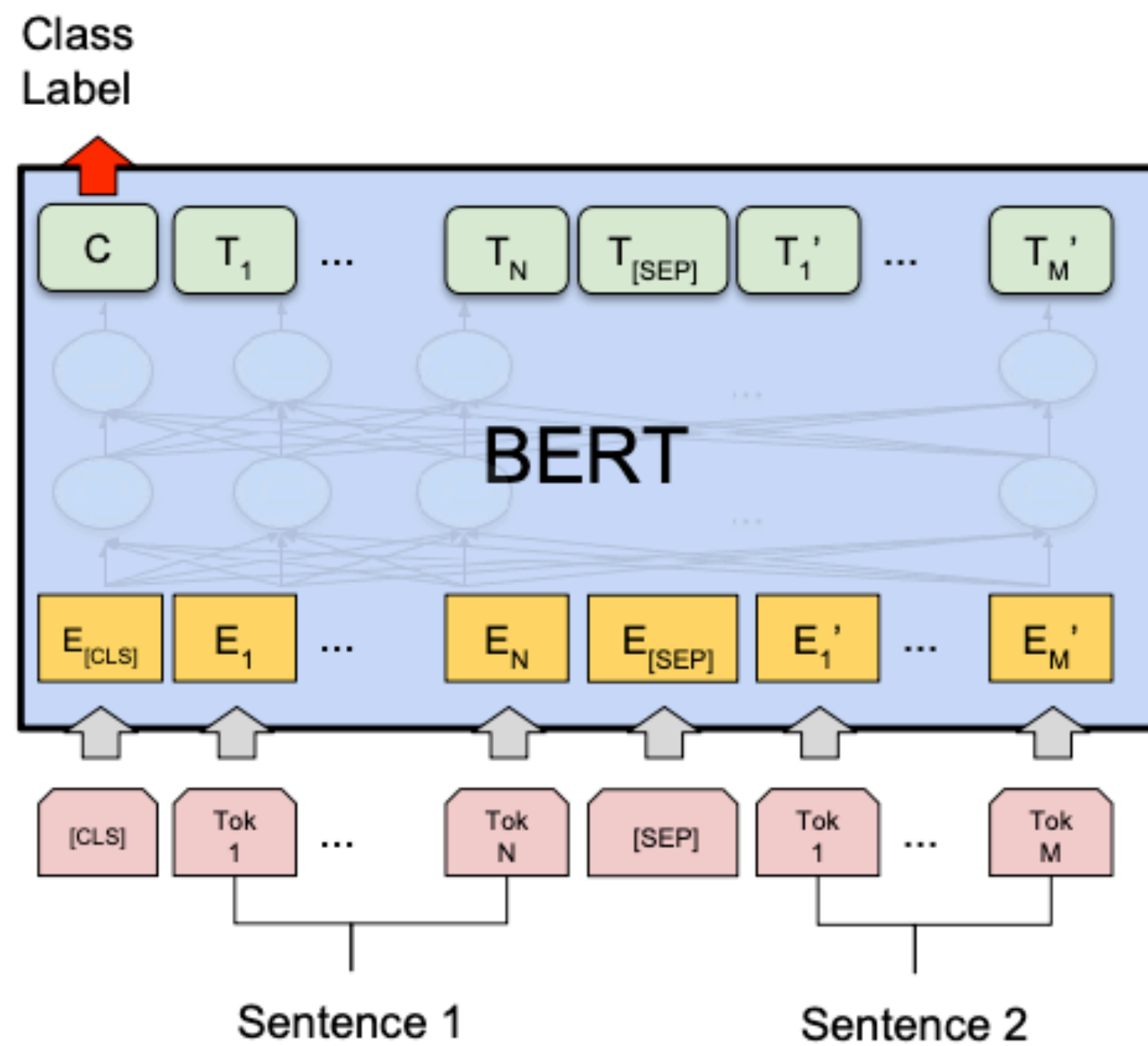


- MLM and NSP are trained together
- [CLS] is pre-trained for NSP
- Other token representations are trained for MLM

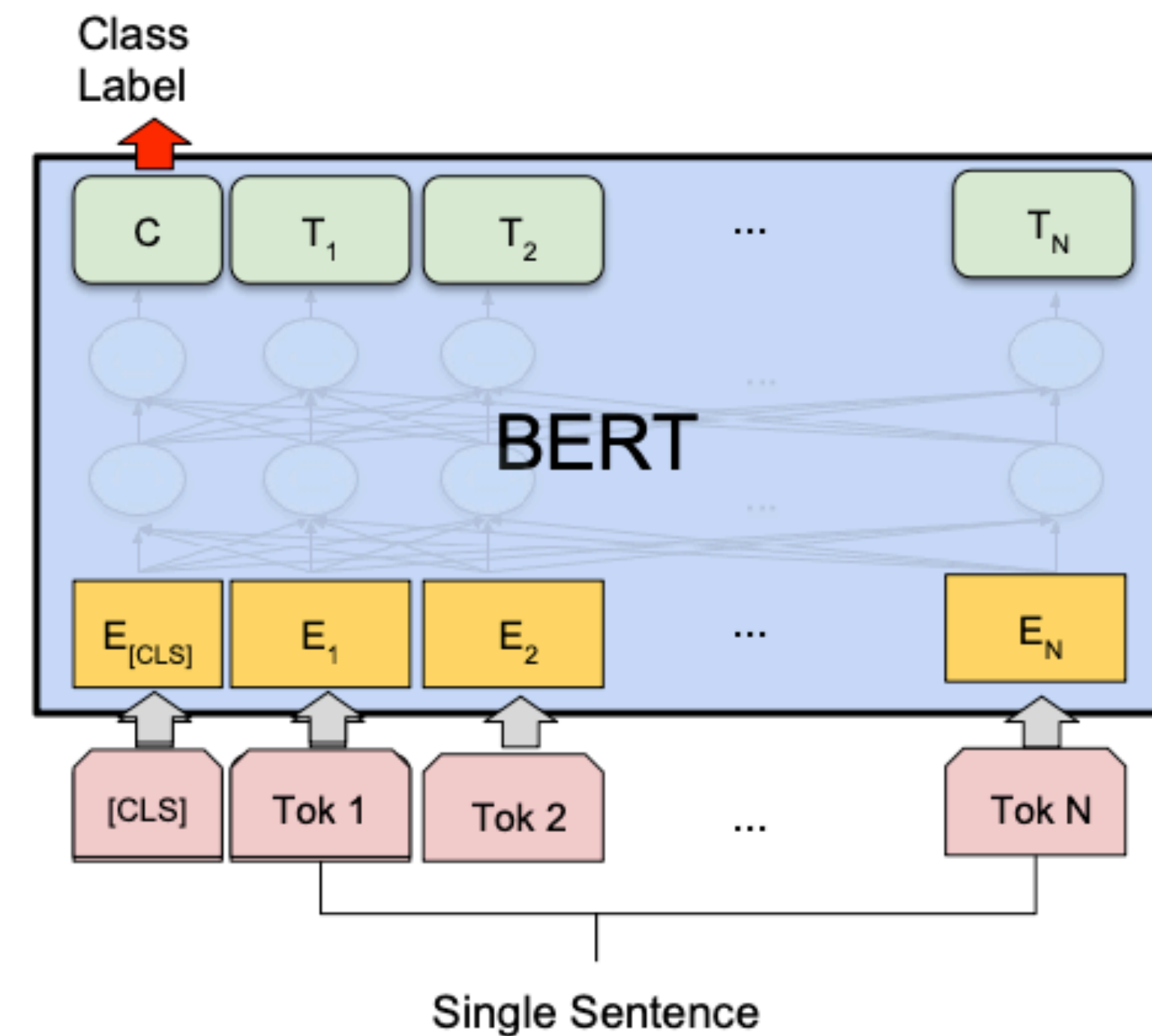
BERT fine-tuning

“Pre-train once, finetune many times.”

sentence-level tasks



(a) Sentence Pair Classification Tasks:
MNLI, QQP, QNLI, STS-B, MRPC,
RTE, SWAG

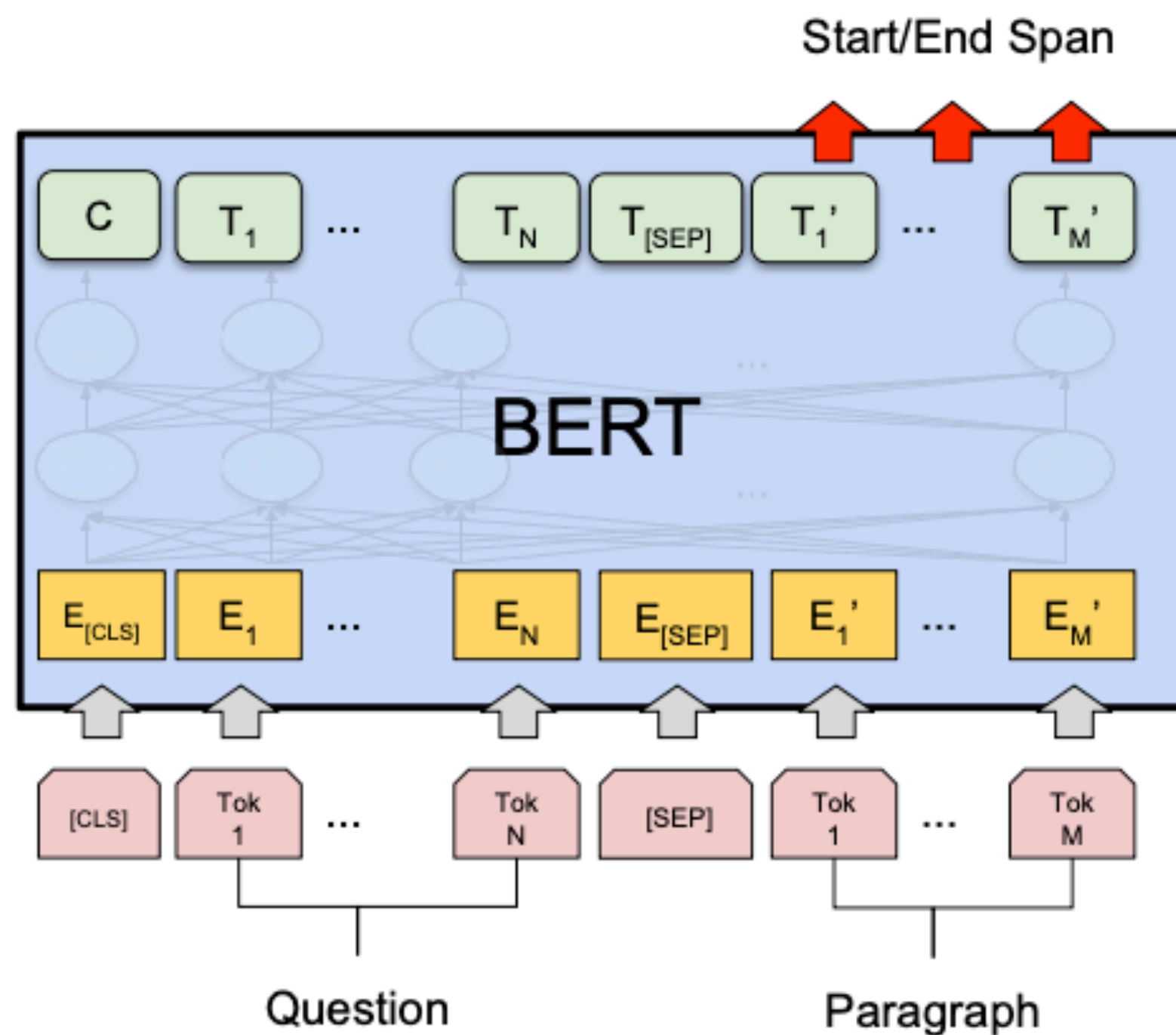


(b) Single Sentence Classification Tasks:
SST-2, CoLA

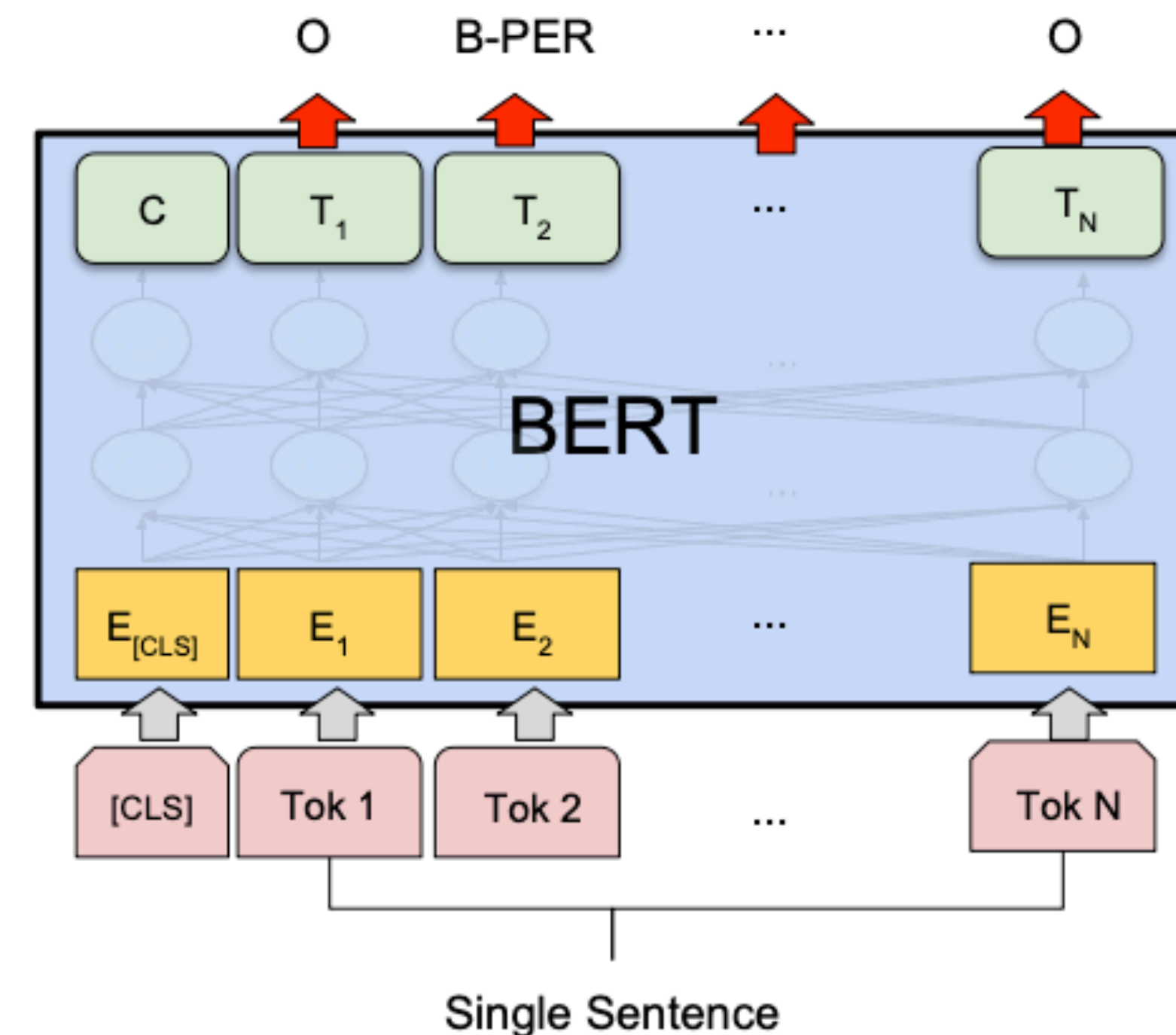
BERT fine-tuning

“Pre-train once, finetune many times.”

token-level tasks

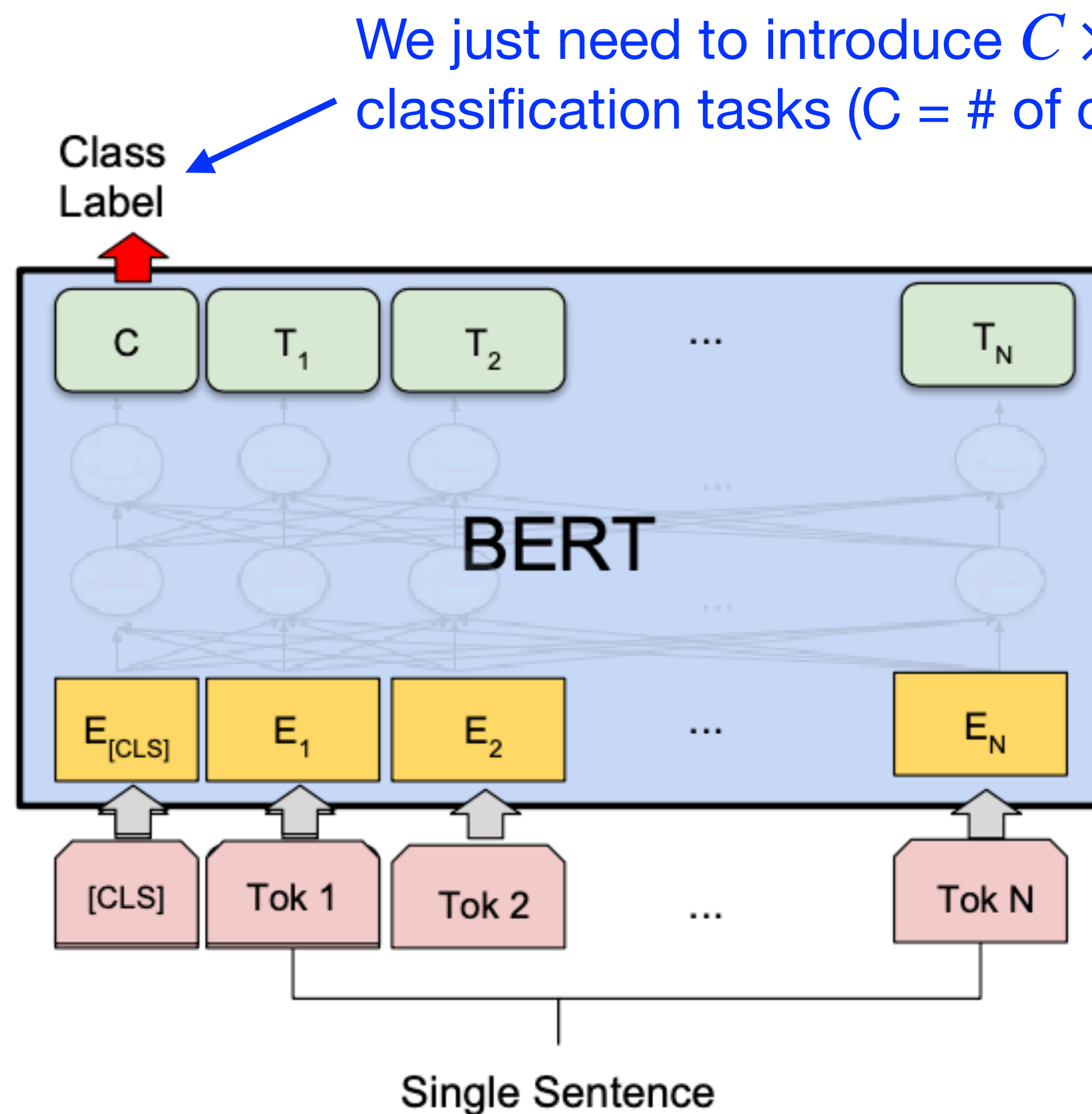


(c) Question Answering Tasks:
SQuAD v1.1



(d) Single Sentence Tagging Tasks:
CoNLL-2003 NER

Example: Sentiment classification



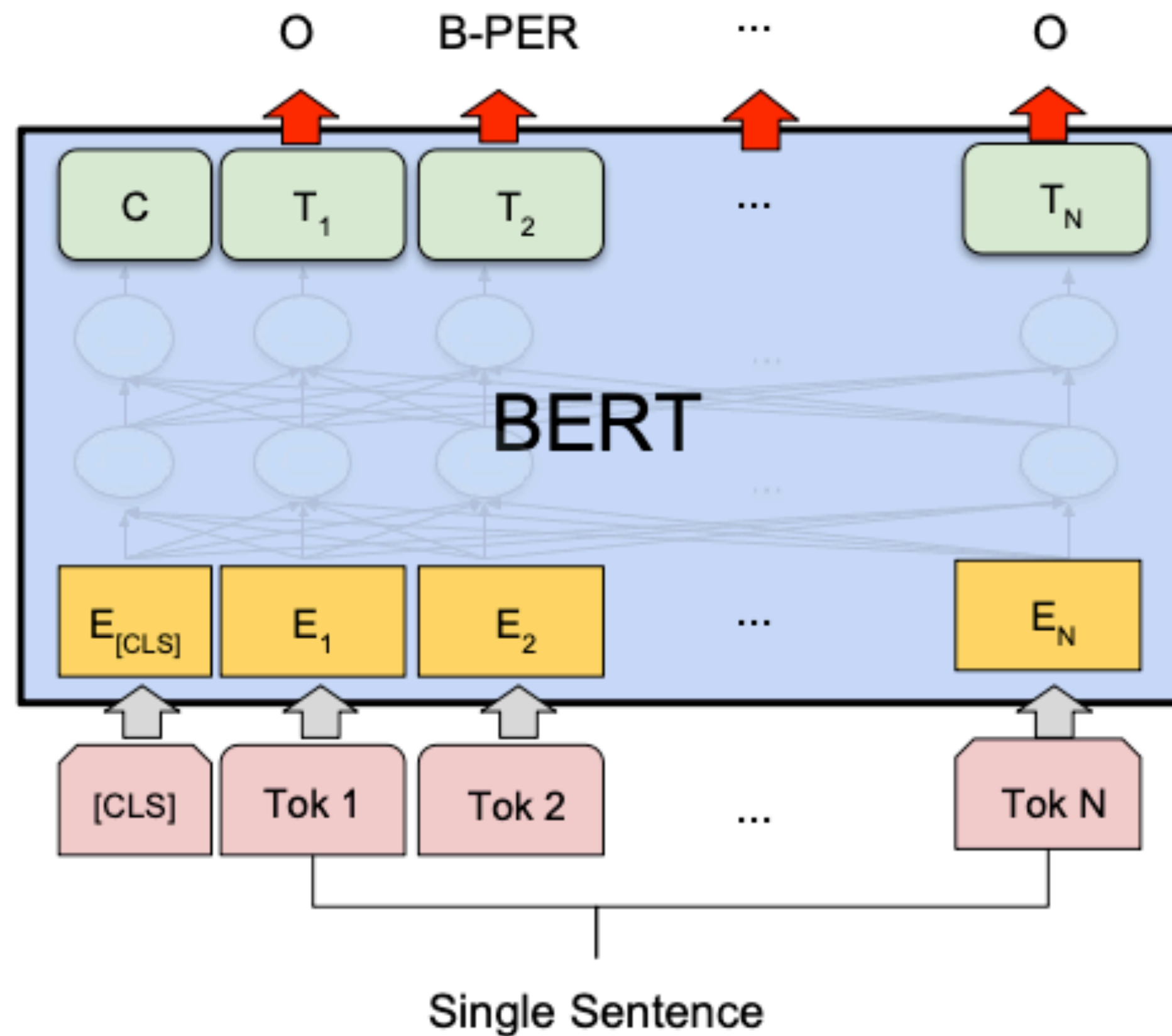
$$P(y = k) = \text{softmax}_k(\mathbf{W}_o \mathbf{h}_{[CLS]})$$

$$\mathbf{W}_o \in \mathbb{R}^{C \times h}$$

All the parameters will be learned together (original BERT parameters + new classifier parameters)

Example: Named entity recognition (NER)

We just need to introduce $C \times h$ parameters for classification tasks (C = # of classes, h = hidden size)!



$$P(y_i = k) = \text{softmax}_k(\mathbf{W}_o \mathbf{h}_i)$$

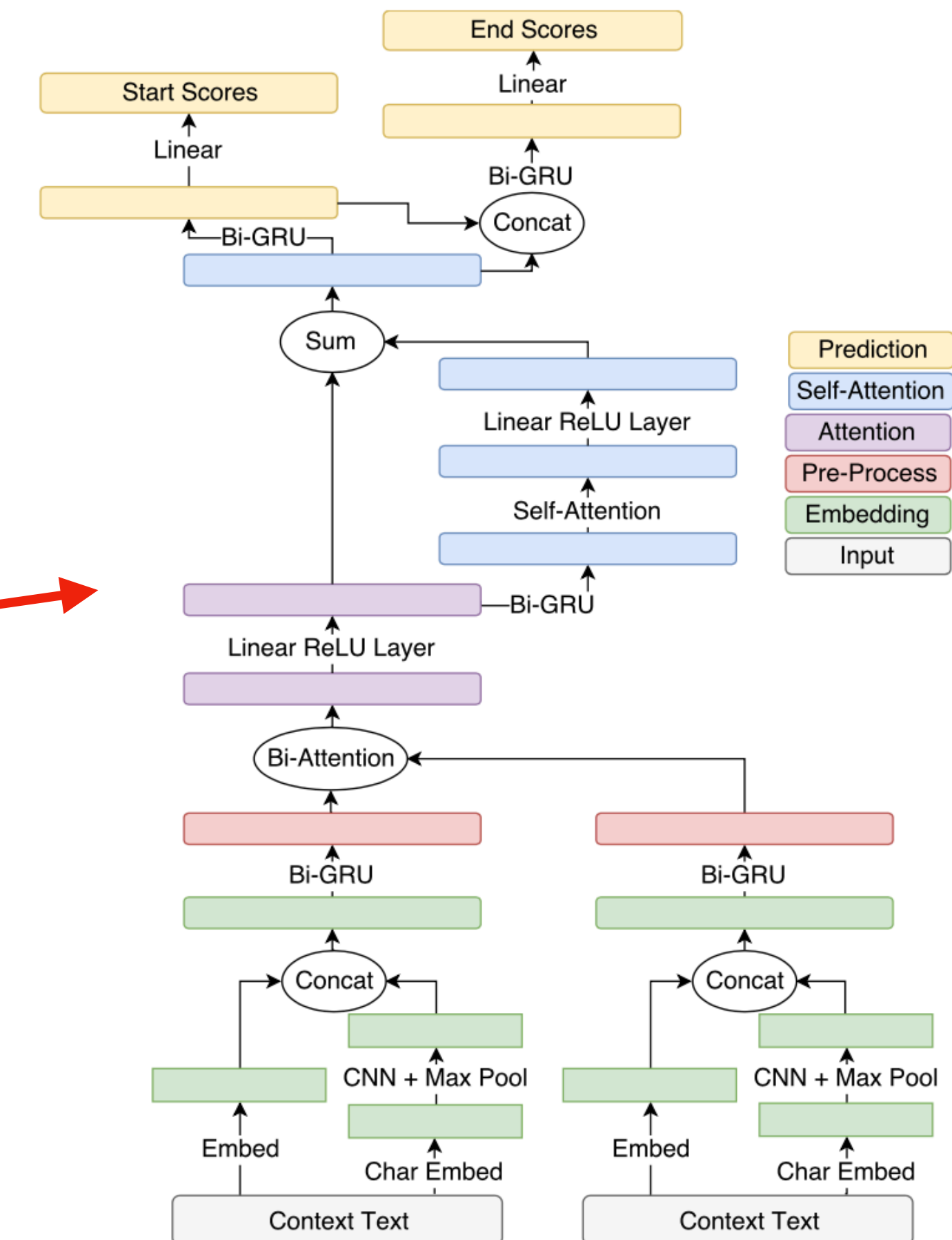
$$\mathbf{W}_o \in \mathbb{R}^{C \times h}$$

Experimental results: GLUE

System	MNLI-(m/mm) 392k	QQP 363k	QNLI 108k	SST-2 67k	CoLA 8.5k	STS-B 5.7k	MRPC 3.5k	RTE 2.5k	Average -
Pre-OpenAI SOTA	80.6/80.1	66.1	82.3	93.2	35.0	81.0	86.0	61.7	74.0
BiLSTM+ELMo+Attn	76.4/76.1	64.8	79.8	90.4	36.0	73.3	84.9	56.8	71.0
OpenAI GPT	82.1/81.4	70.3	87.4	91.3	45.4	80.0	82.3	56.0	75.1
BERT _{BASE}	84.6/83.4	71.2	90.5	93.5	52.1	85.8	88.9	66.4	79.6
BERT _{LARGE}	86.7/85.9	72.1	92.7	94.9	60.5	86.5	89.3	70.1	82.1

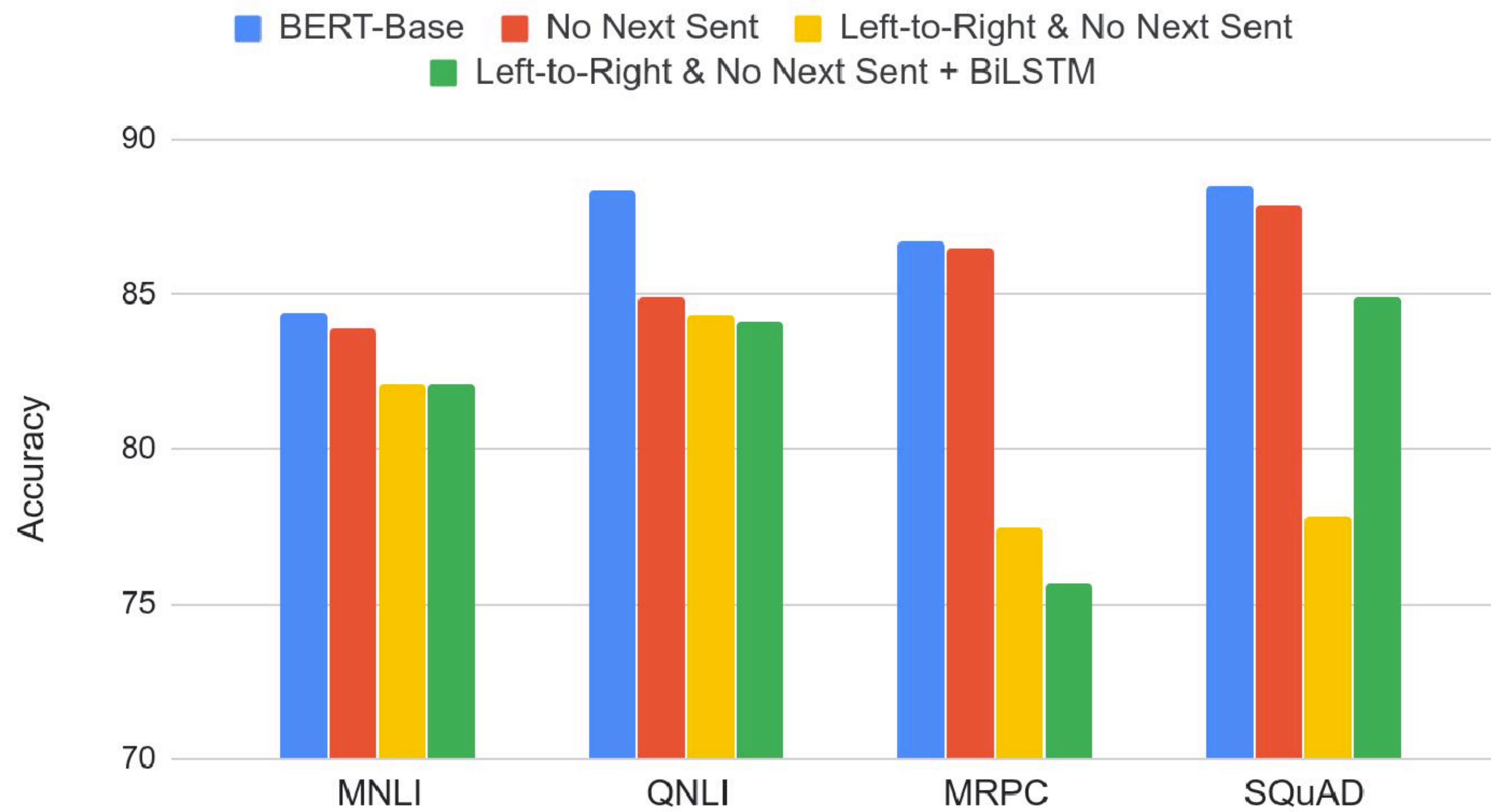
Experimental results: SQuAD

System	Dev		Test	
	EM	F1	EM	F1
Top Leaderboard Systems (Dec 10th, 2018)				
Human	-	-	82.3	91.2
#1 Ensemble - nlnet	-	-	86.0	91.7
#2 Ensemble - QANet	-	-	84.5	90.5
Published				
BiDAF+ELMo (Single)	-	85.6	-	85.8
R.M. Reader (Ensemble)	81.2	87.9	82.3	88.5
Ours				
BERT_{BASE} (Single)	80.8	88.5	-	-
BERT_{LARGE} (Single)	84.1	90.9	-	-
BERT _{LARGE} (Ensemble)	85.8	91.8	-	-
BERT _{LARGE} (Sgl.+TriviaQA)	84.2	91.1	85.1	91.8
BERT _{LARGE} (Ens.+TriviaQA)	86.2	92.2	87.4	93.2



Ablation study: Pre-training tasks

Effect of Pre-training Task



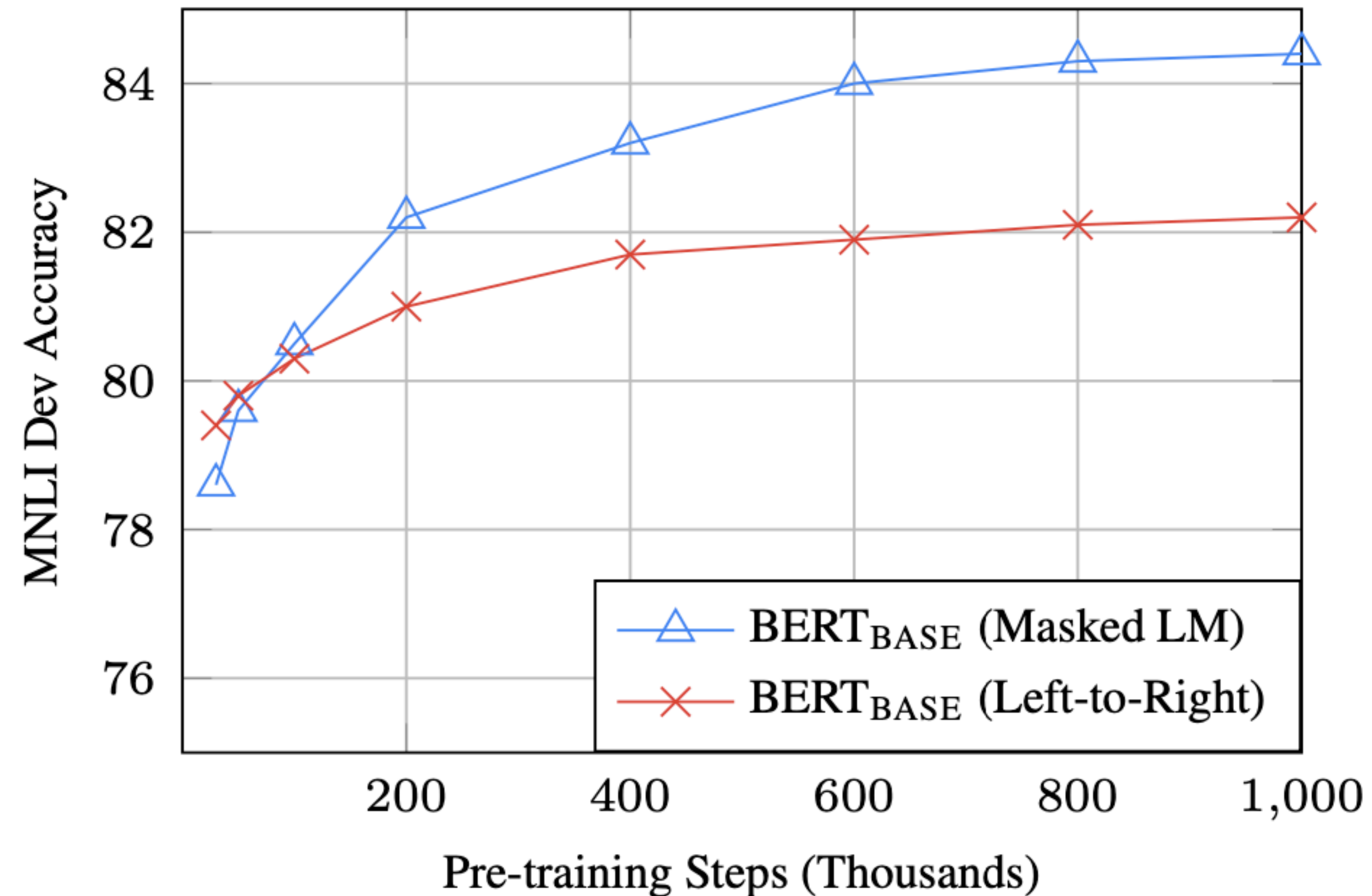
- MLM >> left-to-right LMs
- NSP improves on some tasks
- Note: later work (Joshi et al., 2020; Liu et al., 2019) argued that NSP is not useful

Ablation study: Model sizes

Hyperparams			Dev Set Accuracy			
#L	#H	#A	LM (ppl)	MNLI-m	MRPC	SST-2
3	768	12	5.84	77.9	79.8	88.4
6	768	3	5.24	80.6	82.2	90.7
6	768	12	4.68	81.9	84.8	91.3
12	768	12	3.99	84.4	86.7	92.9
12	1024	16	3.54	85.7	86.9	93.3
24	1024	16	3.23	86.6	87.8	93.7

The bigger, the better!

Ablation study: Training efficiency



MLM takes slightly longer to converge because it only predicts 15% of tokens

RoBERTa (Liu et al. 2019)

- BERT is still under-trained!
- Removed the next sentence prediction training - it adds more noise than benefits!
- Trained longer with 10x data & bigger batch sizes
- Pre-trained on 1,024 V100 GPUs for one day in 2019

Model	data	bsz	steps	SQuAD (v1.1/2.0)	MNLI-m	SST-2
RoBERTa						
with BOOKS + WIKI	16GB	8K	100K	93.6/87.3	89.0	95.3
+ additional data (§3.2)	160GB	8K	100K	94.0/87.7	89.3	95.6
+ pretrain longer	160GB	8K	300K	94.4/88.7	90.0	96.1
+ pretrain even longer	160GB	8K	500K	94.6/89.4	90.2	96.4
BERT _{LARGE}						
with BOOKS + WIKI	13GB	256	1M	90.9/81.8	86.6	93.7



Limitation of pre-trained encoders

- If your task involves generating sequences, BERT and other pre-trained encoders don't naturally lead to nice autoregressive (1-word-at-a-time) generation methods.
- Might want to use a pre-trained decoder

Pre-training for three types of architectures

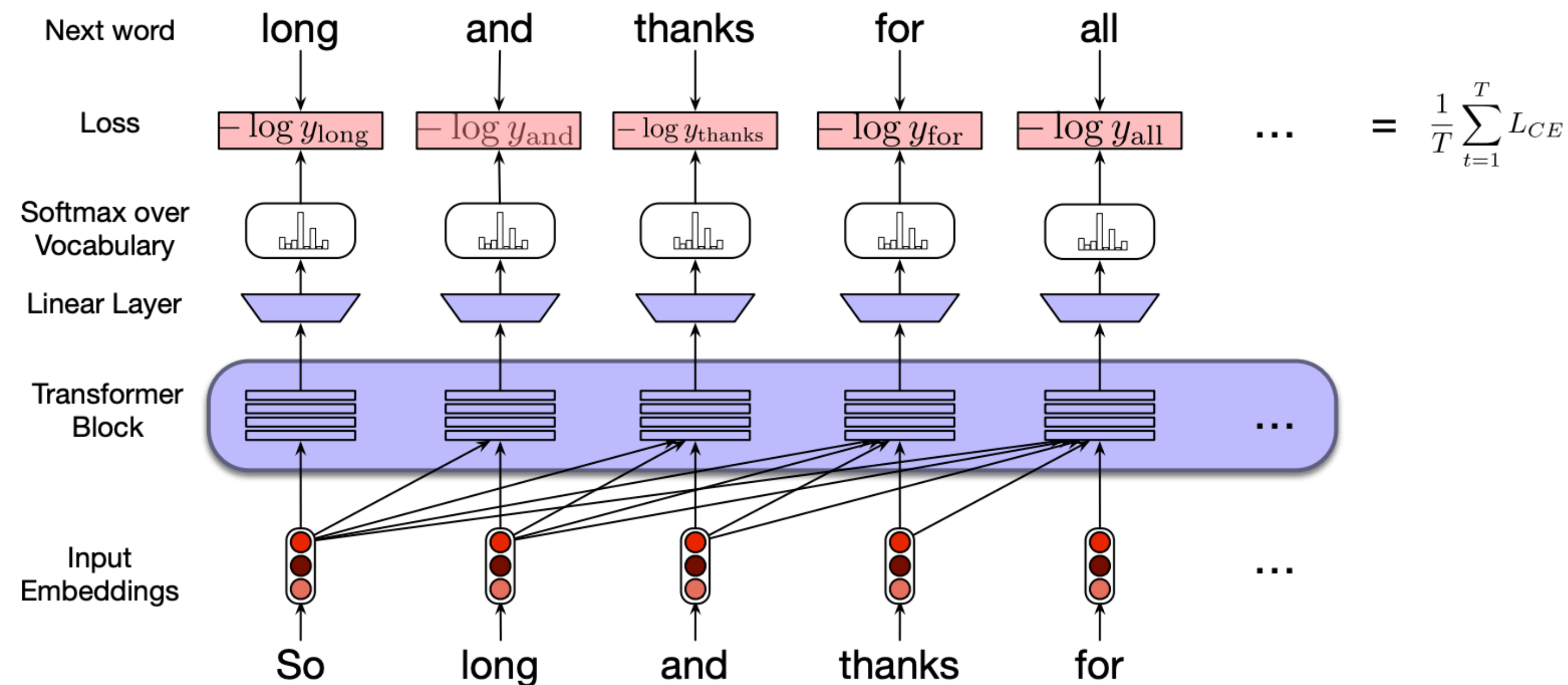
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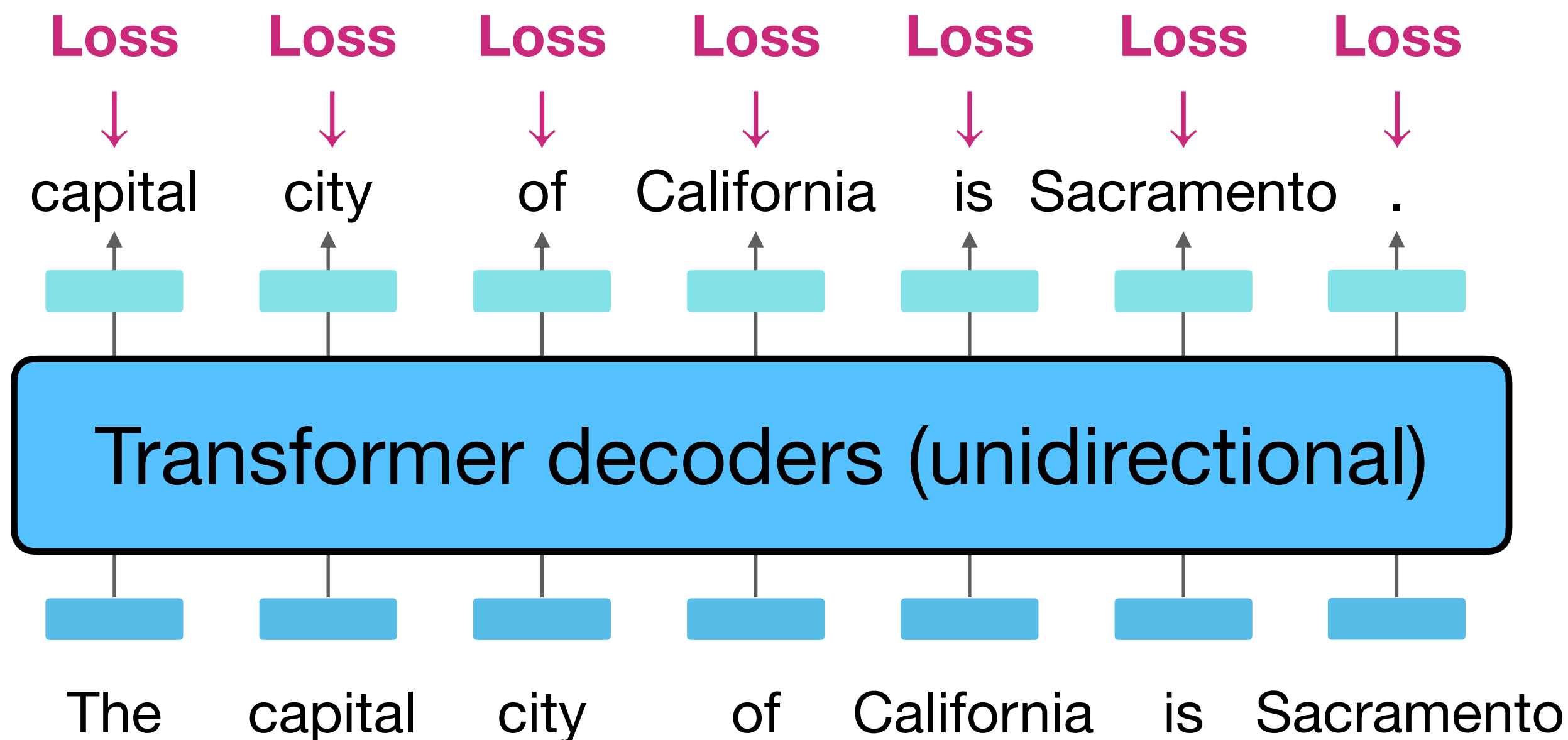
Generative Pre-Training (GPT) (June 2018)

- Use a **Transformer decoder** (unidirectional; left-to-right) instead of LSTMs
- Use **language modeling** as a pre-training objective
- Trained on longer segments of text (**512 BPE tokens**), not just single sentences



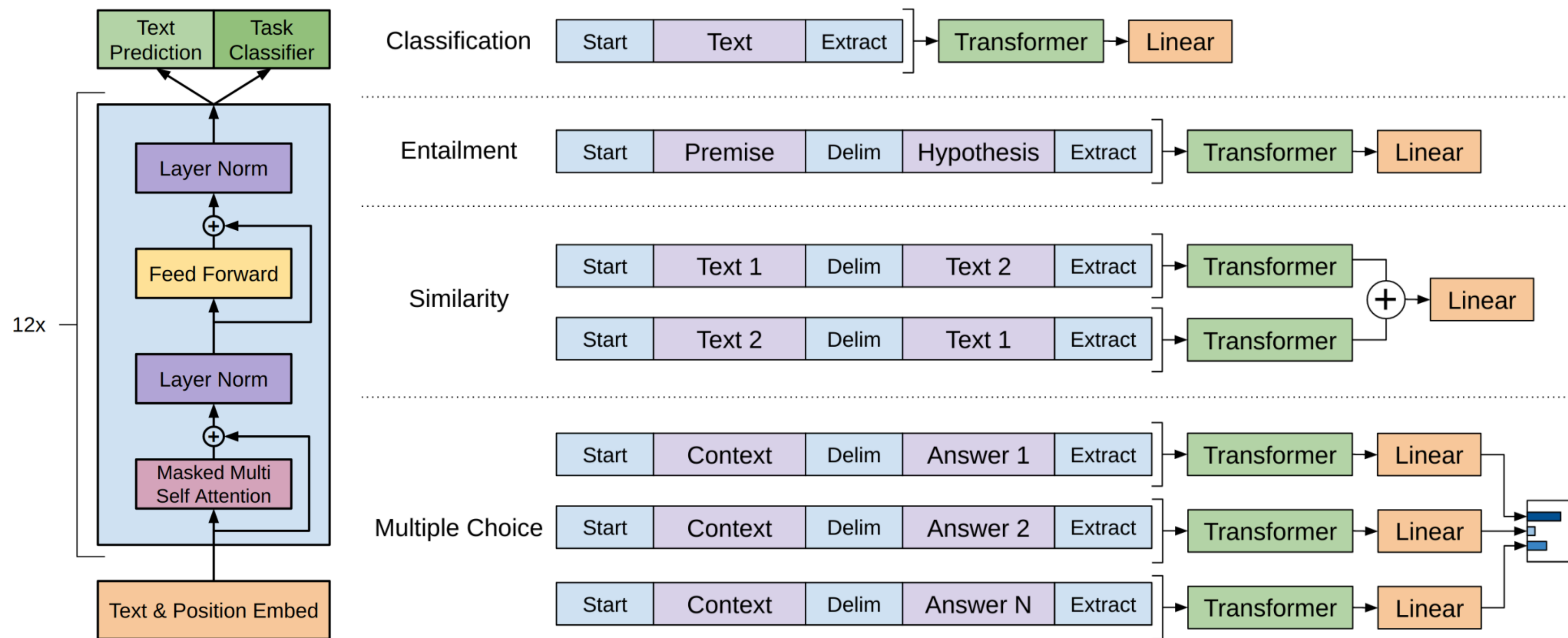
Generative Pre-Training (GPT) (June 2018)

1. Prepare raw text: “The capital city of California is Sacramento.”
2. Feed into the Transformers encoder and train with a (causal) language modeling objective.

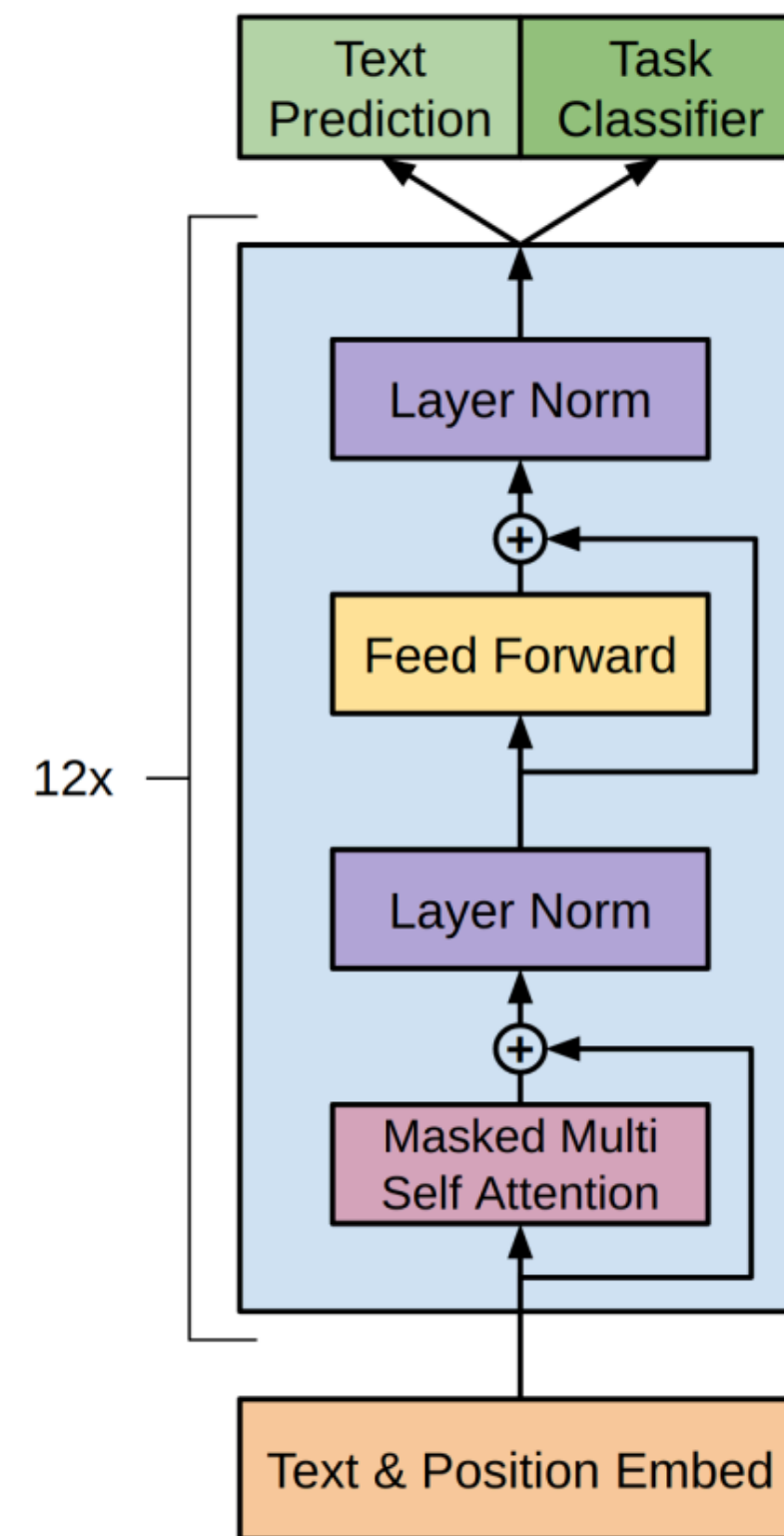


Generative Pre-Training (GPT) (June 2018)

- “Fine-tune” the entire set of model parameters on various downstream tasks



GPT: More details



- 12 layers, 768 hidden size, 12 attention heads, 110M parameters

Same as BERT-base

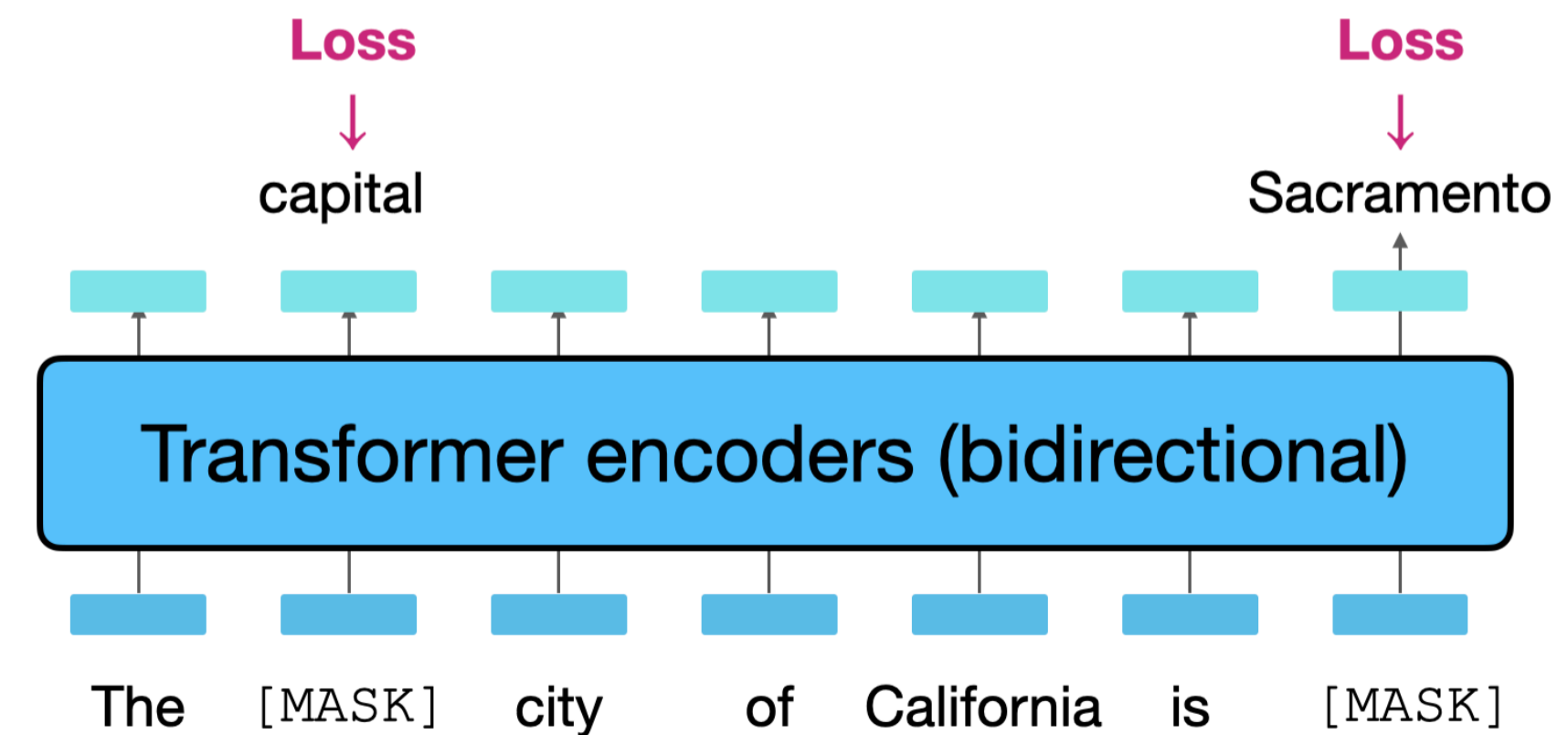
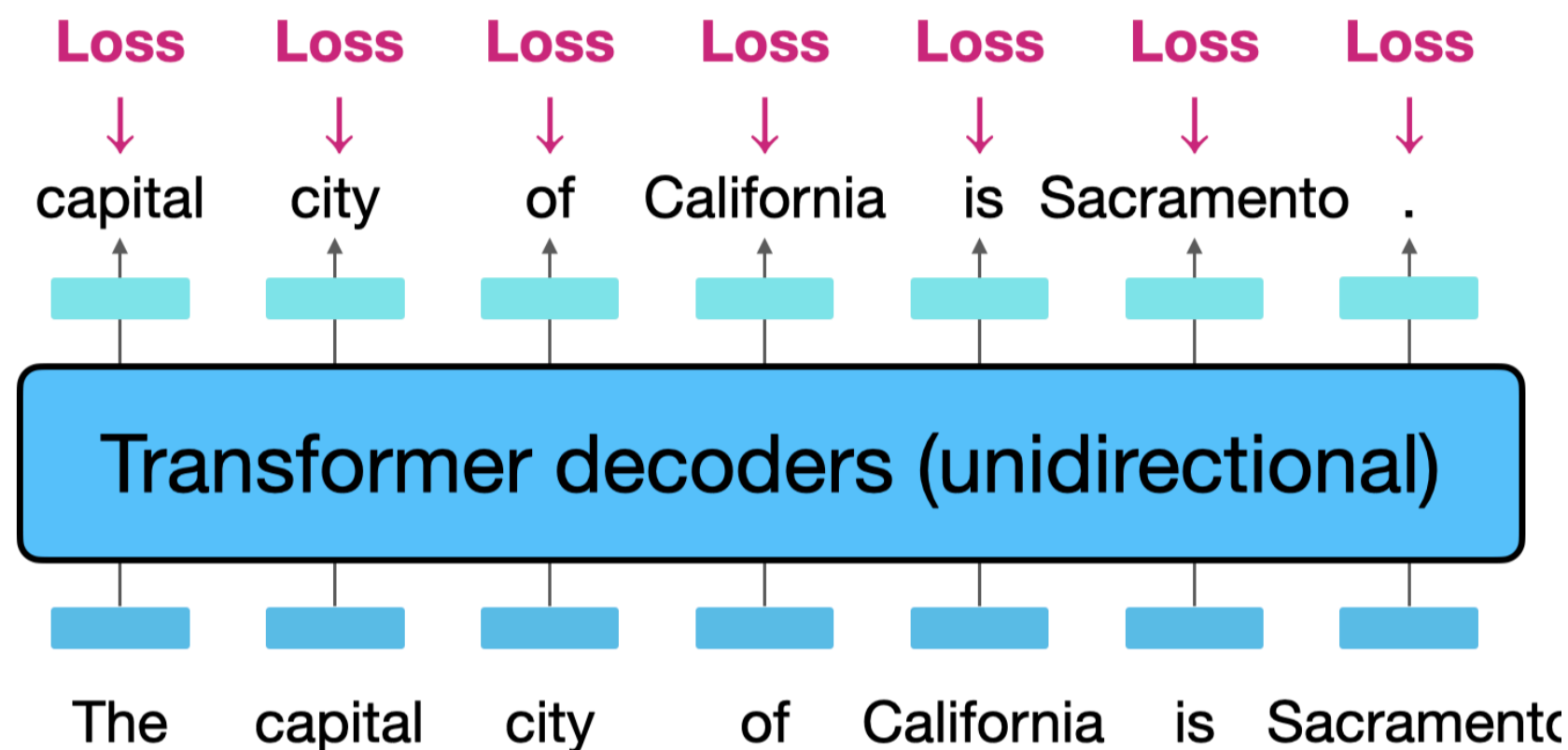
Recall: BERT was trained on this + Wikipedia!

- Training corpus: BooksCorpus (0.8B)
- Max sequence size: 512 wordpiece tokens
- Trained for 100 epochs, batch size 64

Experimental results: GLUE

System	MNLI-(m/mm) 392k	QQP 363k	QNLI 108k	SST-2 67k	CoLA 8.5k	STS-B 5.7k	MRPC 3.5k	RTE 2.5k	Average -
Pre-OpenAI SOTA	80.6/80.1	66.1	82.3	93.2	35.0	81.0	86.0	61.7	74.0
BiLSTM+ELMo+Attn	76.4/76.1	64.8	79.8	90.4	36.0	73.3	84.9	56.8	71.0
OpenAI GPT	82.1/81.4	70.3	87.4	91.3	45.4	80.0	82.3	56.0	75.1
BERT _{BASE}	84.6/83.4	71.2	90.5	93.5	52.1	85.8	88.9	66.4	79.6
BERT _{LARGE}	86.7/85.9	72.1	92.7	94.9	60.5	86.5	89.3	70.1	82.1

Causal (Generative) LMs vs. Masked LMs



- **Causal LMs** receive more training signals per forward pass than **masked LMs**.
- **Causal LMs** learn from noisier signals, since each token is predicted using only preceding context.
- **Causal LMs** can be used for text generation, whereas **masked LMs** cannot.
- **Causal LMs** can be used directly without fine-tuning, whereas **masked LMs** typically require fine-tuning. (Be covered soon.)

Quick quiz

Which paper first used the term “GPT”?

- (A) Radford et al. 2018. [GPT-1 paper, OpenAI]
- (B) Radford et al., 2019. [GPT-2 paper, OpenAI]
- (C) Brown et al., 2020. [GPT-3 paper, OpenAI]
- (D) Devlin et al. 2019. [BERT paper, Google]

(D) is correct.

tional features. The fine-tuning approach, such as the Generative Pre-trained Transformer (OpenAI GPT) (Radford et al., 2018), introduces minimal task-specific parameters, and is trained on the

ELMo vs. GPT vs. BERT

Which of the following statements is INCORRECT?

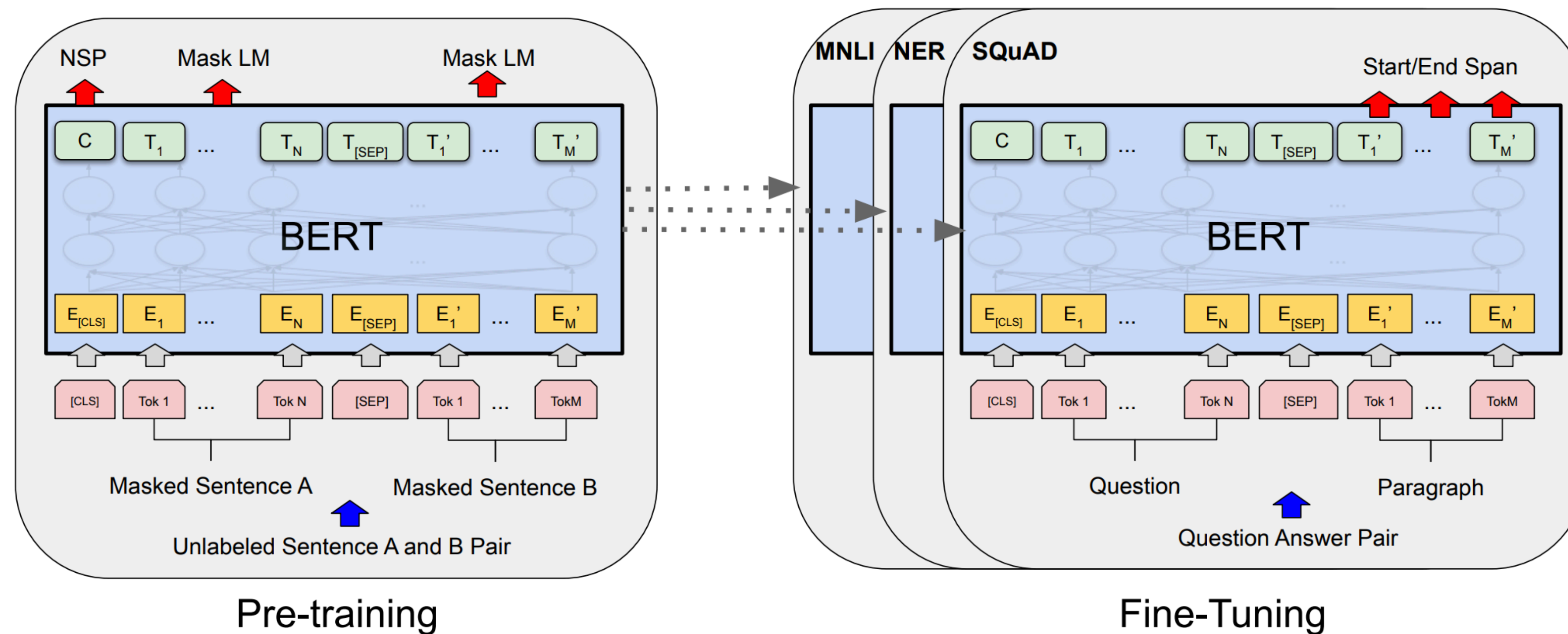
- (A) BERT was trained on more data than ELMo
- (B) BERT builds on Transformer encoder, and GPT builds on Transformer decoder
- (C) ELMo requires different model architectures for different tasks
- (D) BERT was trained on data with longer contexts compared to GPT

(D) is incorrect.

Recap

Recap: Pretraining / finetuning

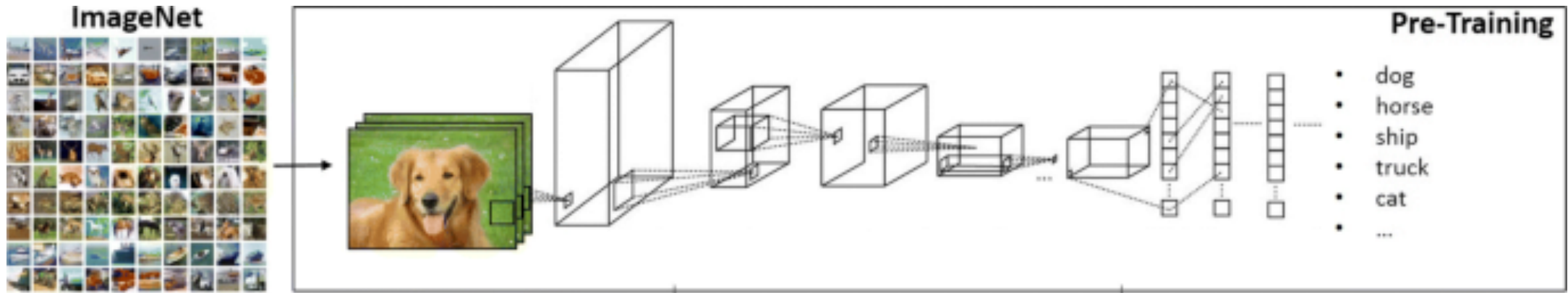
“Pre-train” a model on a large dataset for task X, then “fine-tune” it on a dataset for task Y



“**Fine-tuning** is the process of **taking the network learned by these pre-trained models**, and **further training the model**, often via an added neural net classifier that takes the top layer of the network as input, to perform some downstream task.”

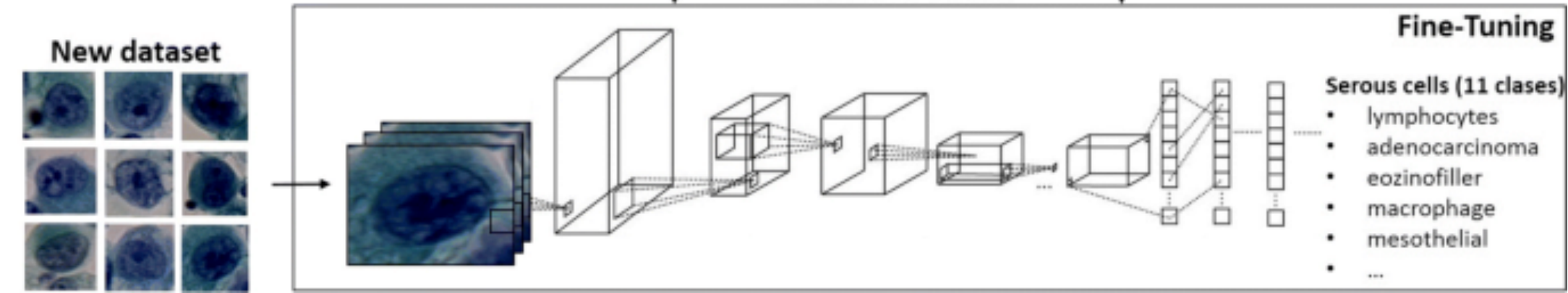
Recap: Pretraining / finetuning

Pre-training



1.28M images, 1000 classes

Fine-tuning



3652 images, 11 classes

Pre-training

Natural language [MASK] (NLP) is a subfield of linguistics, computer science, and artificial [MASK] concerned with the interactions [MASK] computers and human [MASK] ...



processing,
Intelligence,
between,
language

**3.3B tokens
(512 tokens
per segment)**

Fine-tuning

contains no wit, only labored gags
the greatest musicians
very good viewing alternative



Negative
Positive
Positive

**67k examples,
2 classes**

Recap: Most Pre-training = Reconstructing the Input

You learn a lot from reconstructing the input!

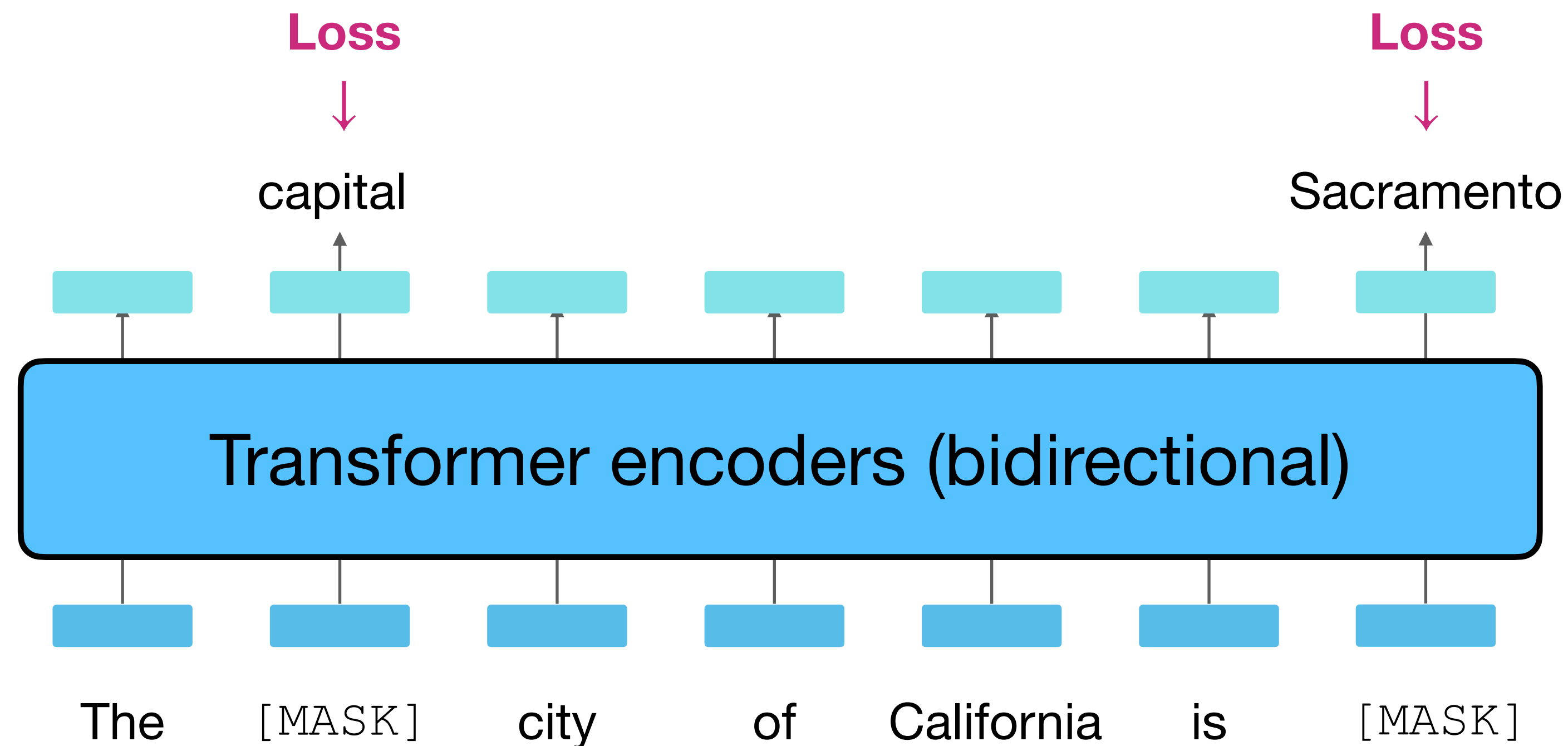
- I went to the ocean and saw fish, turtles, _____ and seals.
 - General semantics
- I put ____ fork down on the table
 - Syntactic constraints
- The woman walked across the street checking for traffic over _____ shoulder.
 - Co-reference, relations between different entities within the sentence
- Overall, the value I got from the two hours watching it was the sum total of the popcorn and the drink. The movie was _____.
 - Sentiment
- UC Berkeley is located in _____.
 - Word knowledge

Pre-training for three types of architectures

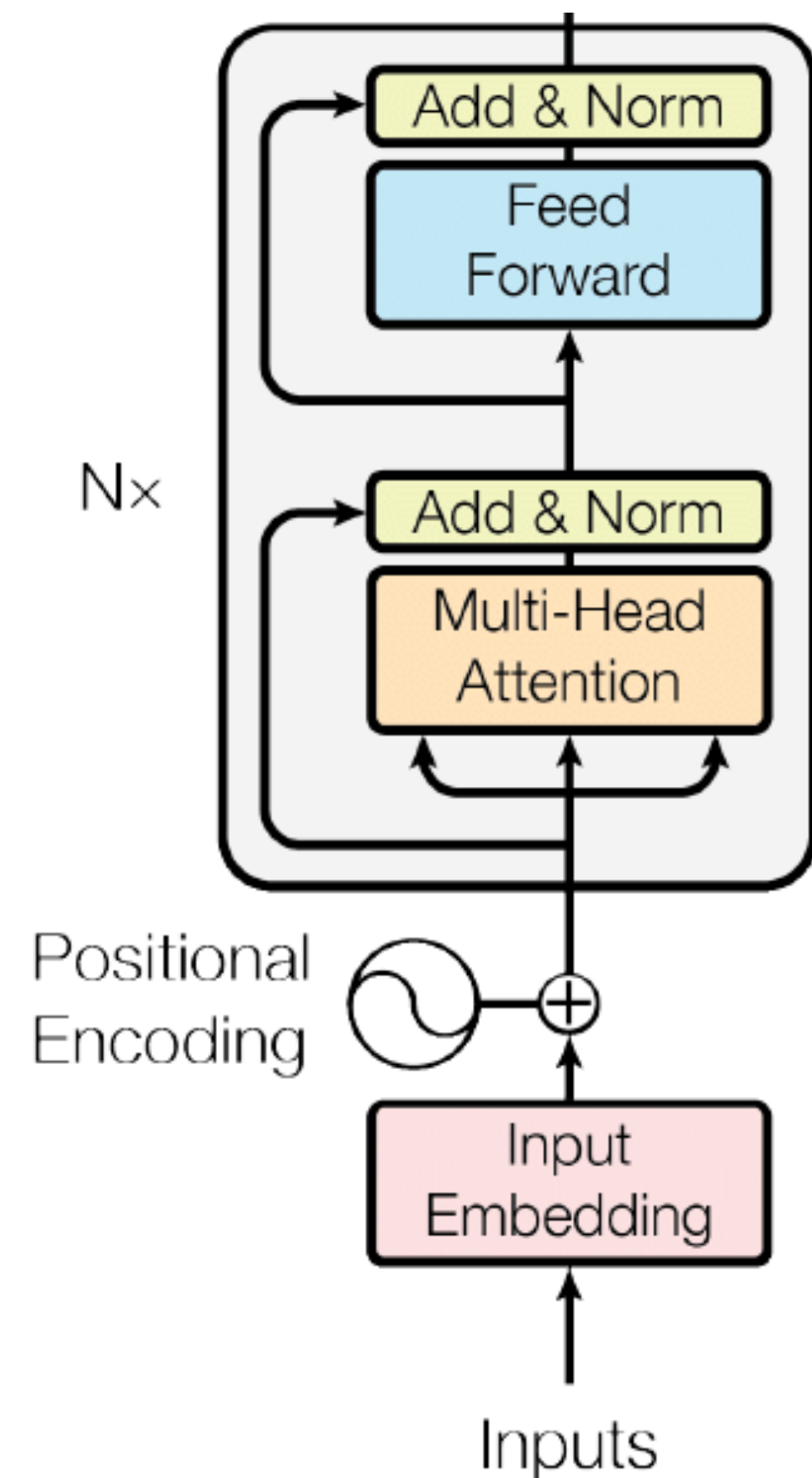
Architecture	Pretraining objective	Examples
Transformer Encoder	Masked language models	BERT, RoBERTa, ELECTRA
Transformers Encoder-Decoder	Span Corruption	T5, BART
Transformers Decoder	Autoregressive language models (Causal language models)	GPT-2, GPT-3, Llama

Recap: Masked Language Modeling (MLM)

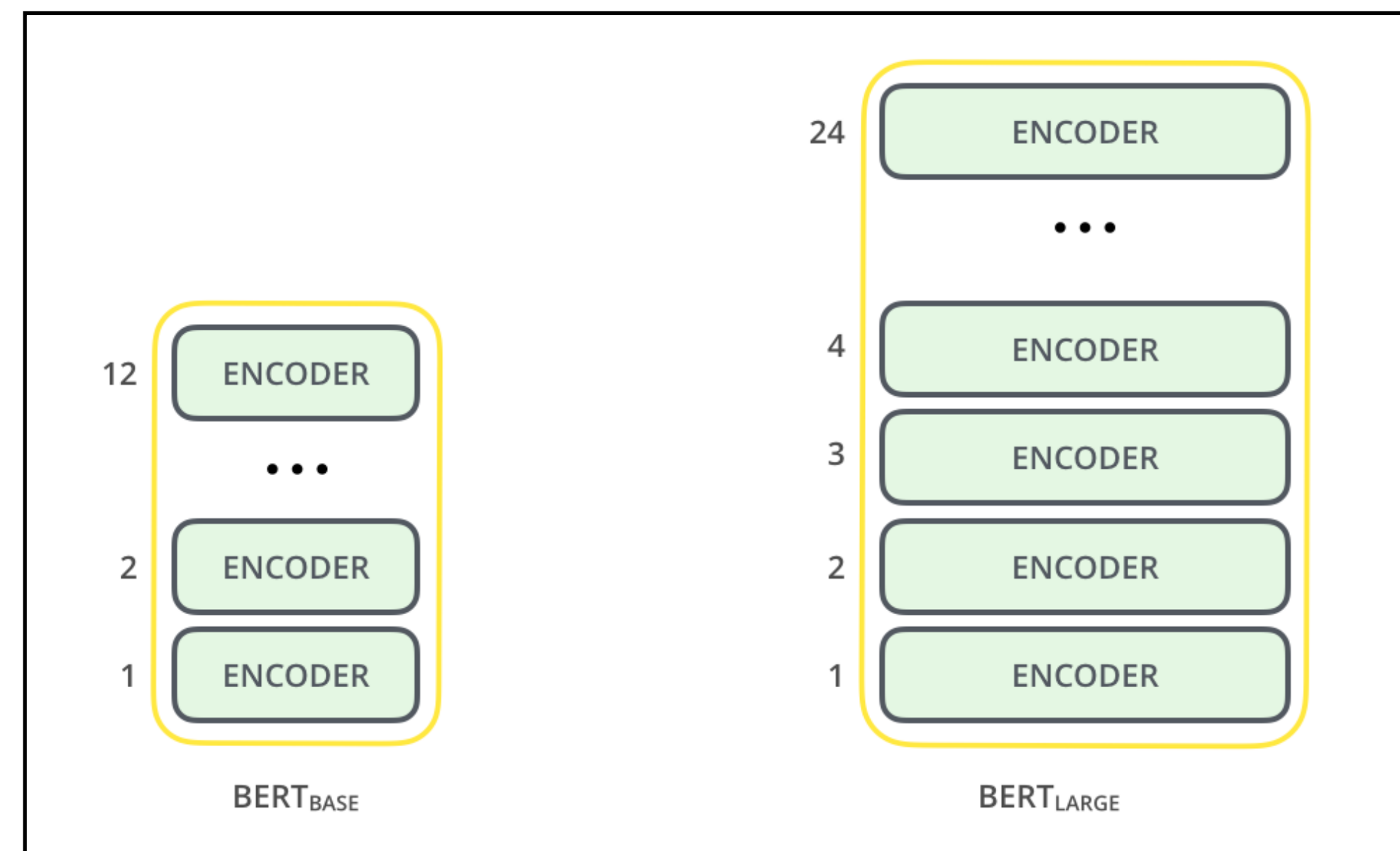
1. Prepare raw text: "The capital city of California is Sacramento."
2. Mask 15% of the tokens: "The [MASK] city of California is [MASK]."
3. Feed into the Transformers encoder and train with a masked language modeling objective.



Recap: BERT (Devlin et al. 2018)



- BERT-base: 12 layers, 768 hidden size, 12 attention heads, 110M parameters
- BERT-large: 24 layers, 1024 hidden size, 16 attention heads, 340M parameters



Recap: Pretraining / finetuning

Experiments on GLUE (Wang et al., 2019)

of examples range between 2.5K and 392K

System	MNLI-(m/mm)	QQP	QNLI	SST-2	CoLA	STS-B	MRPC	RTE	Average
	392k	363k	108k	67k	8.5k	5.7k	3.5k	2.5k	-
Pre-OpenAI SOTA	80.6/80.1	66.1	82.3	93.2	35.0	81.0	86.0	61.7	74.0
BiLSTM+ELMo+Attn	76.4/76.1	64.8	79.8	90.4	36.0	73.3	84.9	56.8	71.0
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BERT _{LARGE}	86.7/85.9	72.1	92.7	94.9	60.5	86.5	89.3	70.1	82.1

Today we are going to see other uses of pre-trained models:

- 1) few-shot examples (e.g., 32)
- 2) No fine-tuning (= no gradient updates)

Recap: RoBERTa (Liu et al. 2019)

- BERT is still under-trained!
- Removed the next sentence prediction training - it adds more noise than benefits!
- Trained longer with 10x data & bigger batch sizes
- Pre-trained on 1,024 V100 GPUs for one day in 2019

Model	data	bsz	steps	SQuAD (v1.1/2.0)	MNLI-m	SST-2
RoBERTa						
with BOOKS + WIKI	16GB	8K	100K	93.6/87.3	89.0	95.3
+ additional data (§3.2)	160GB	8K	100K	94.0/87.7	89.3	95.6
+ pretrain longer	160GB	8K	300K	94.4/88.7	90.0	96.1
+ pretrain even longer	160GB	8K	500K	94.6/89.4	90.2	96.4
BERT _{LARGE}						
with BOOKS + WIKI	13GB	256	1M	90.9/81.8	86.6	93.7

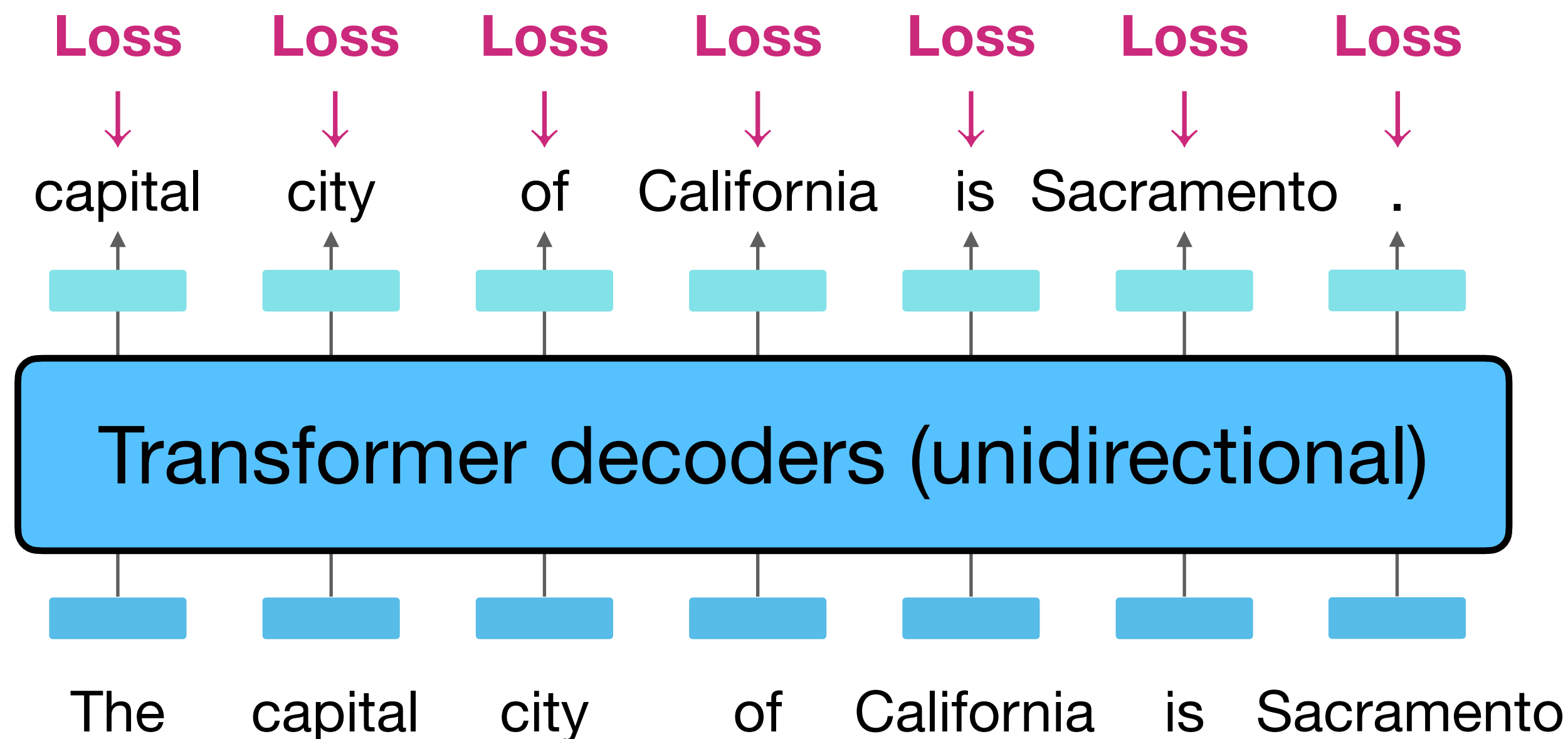


Pre-training for three types of architectures

Architecture	Pretraining objective	Examples
Transformer Encoder	Masked language models	BERT, RoBERTa, ELECTRA
Transformers Encoder-Decoder	Span Corruption	T5, BART
Transformers Decoder	Autoregressive language models (Causal language models)	GPT-2, GPT-3, Llama

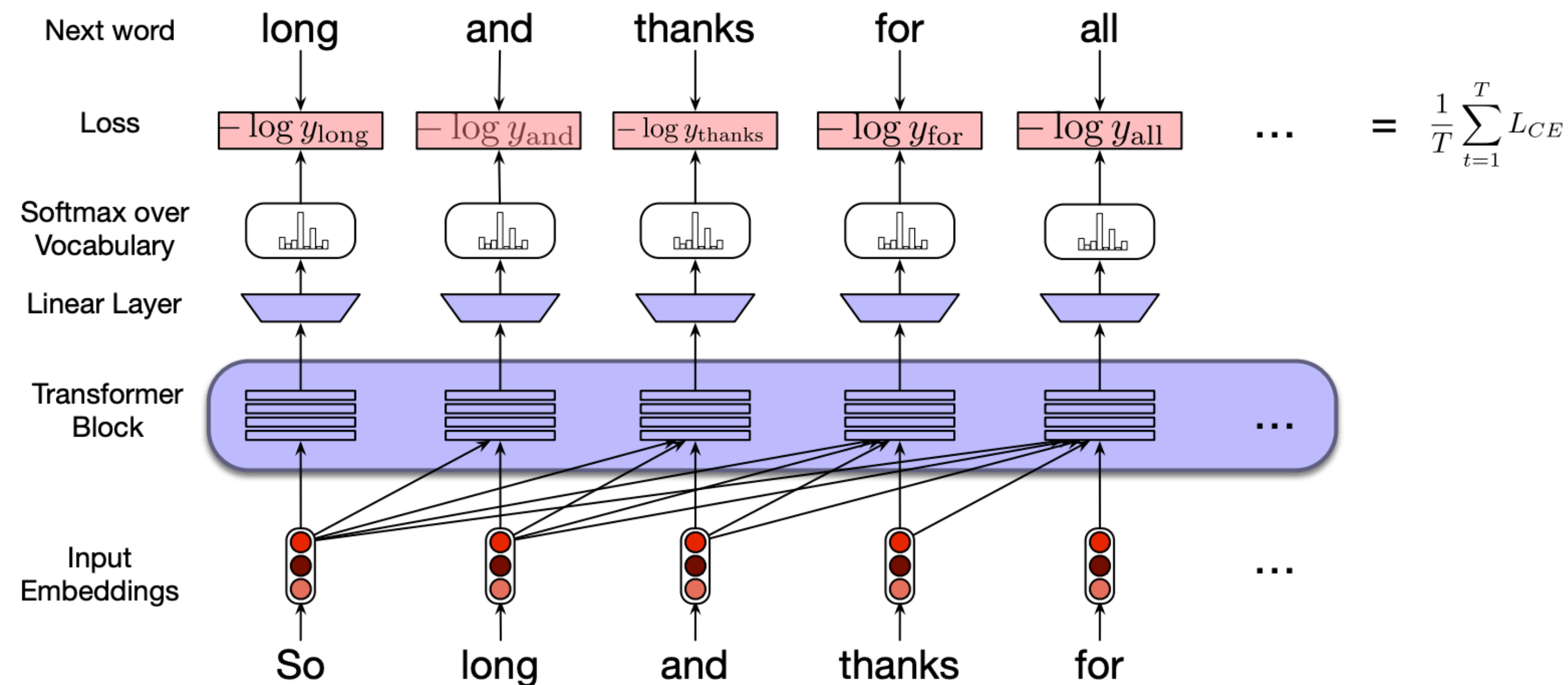
Recap: Causal (Generative) Language Modeling

1. Prepare raw text: “The capital city of California is Sacramento.”
2. Feed into the Transformers encoder and train with a (causal) language modeling objective.



Recap: GPT

- Use a **Transformer decoder** (unidirectional; left-to-right) instead of LSTMs
- Use **language modeling** as a pre-training objective
- Trained on longer segments of text (**512 BPE tokens**), not just single sentences



Recap: Experimental results: GLUE

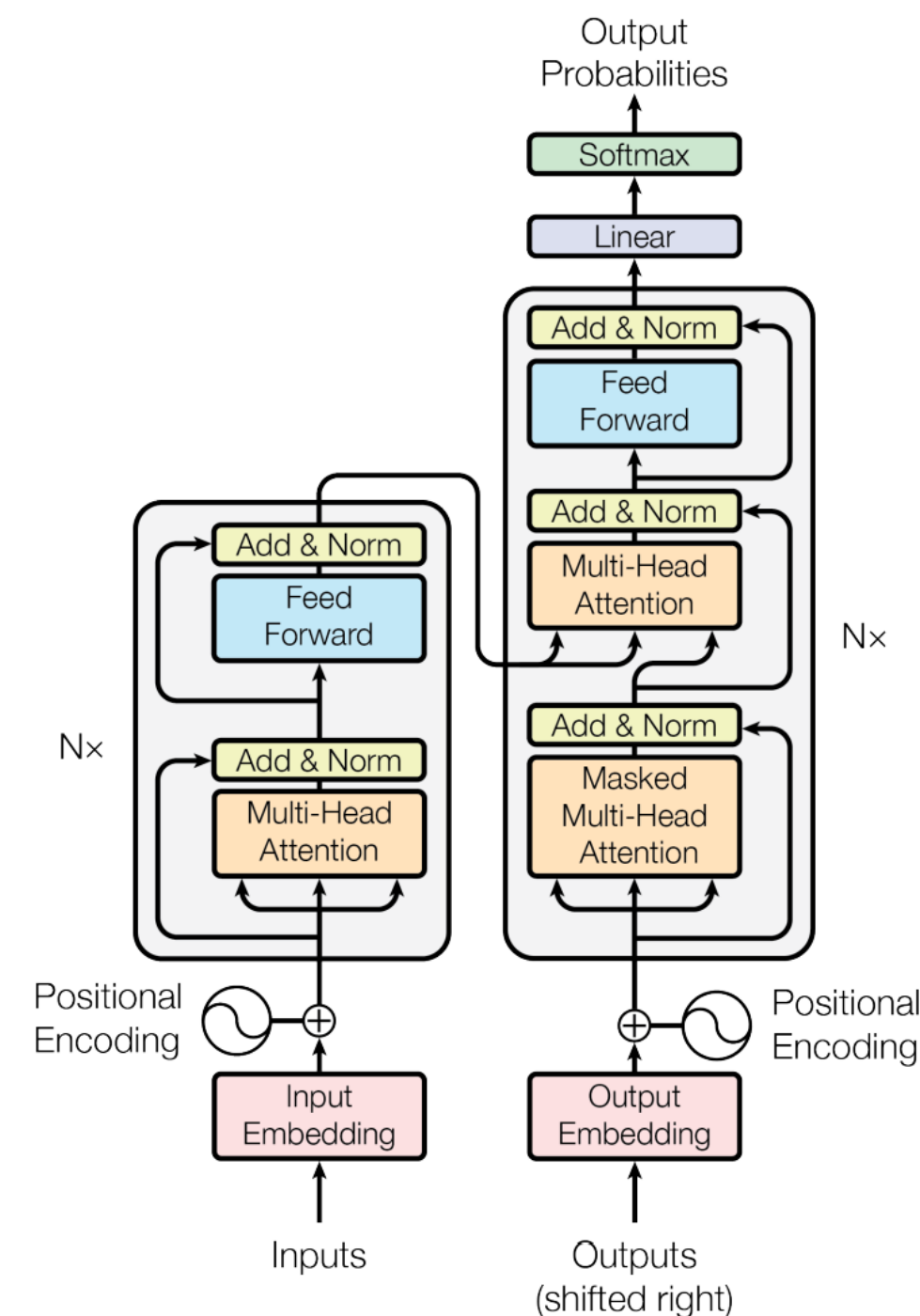
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Pre-training for three types of architectures

Architecture	Pretraining objective	Examples
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Transformers Decoder	Autoregressive language models (Causal language models)	GPT-2, GPT-3, Llama

T5

- Encoder-only models (e.g., BERT) enjoy the benefits of bidirectionally but they can't generate text.
- Decoder-only models (e.g., GPT) can do generation but they don't benefit from bidirectionally for reading inputs.
- Encoder-decoder combines the best of both worlds!



Original text

Thank you ~~for inviting~~ me to your party ~~last~~ week.

Inputs

Thank you <X> me to your party <Y> week.

← encoder

Targets

<X> for inviting <Y> last <Z>

← decoder

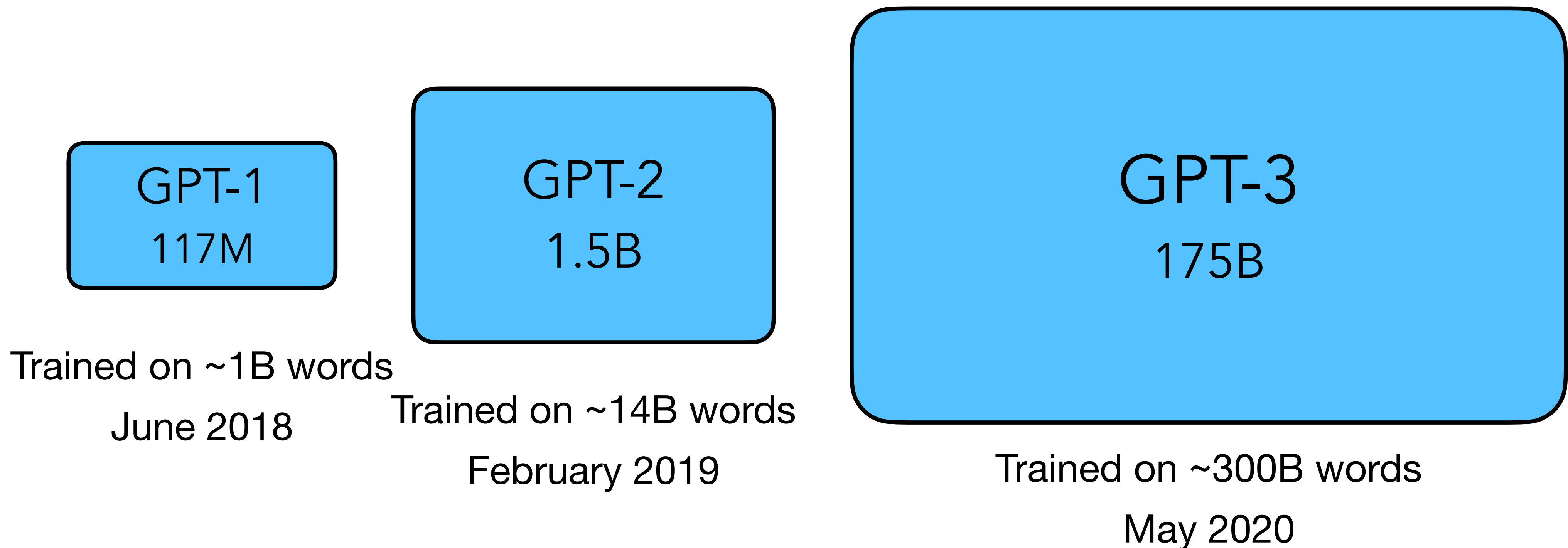
Pre-training for three types of architectures

Architecture	Pretraining objective	Examples
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From GPT to GPT-2 to GPT-3

From GPT to GPT-2 to GPT-3

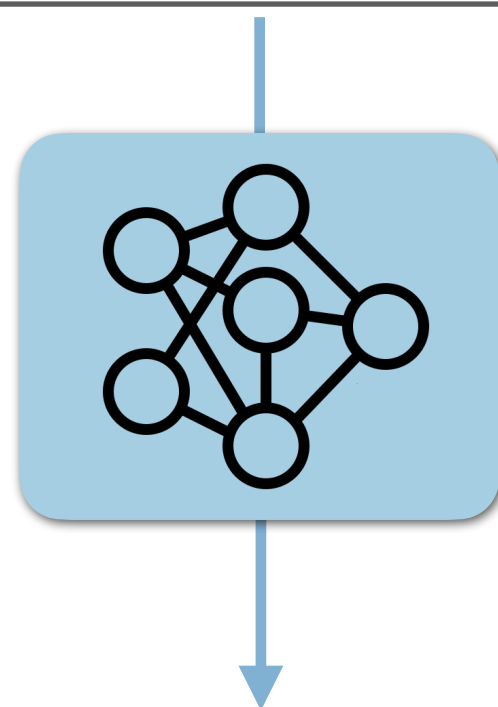
- All **decoder-only** Transformer-based language models
- **More** parameters, **larger** training corpora



GPT-2: language models are *zero-shot* learners

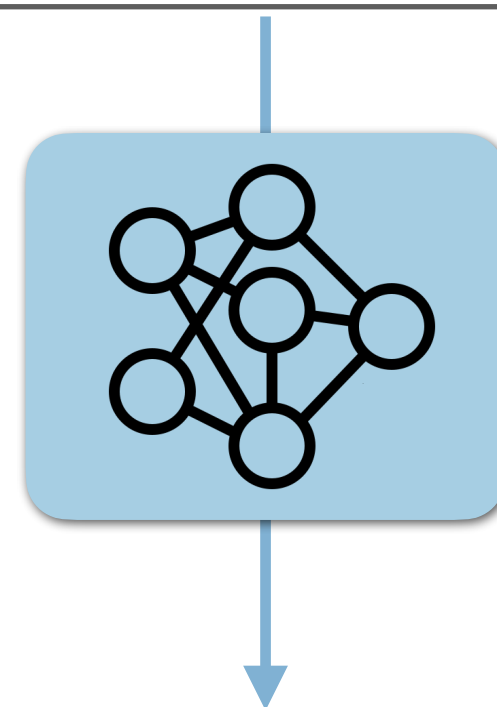
Notation: Given text | Generation

Question: What is the capital city of Washington?
Answer: _____



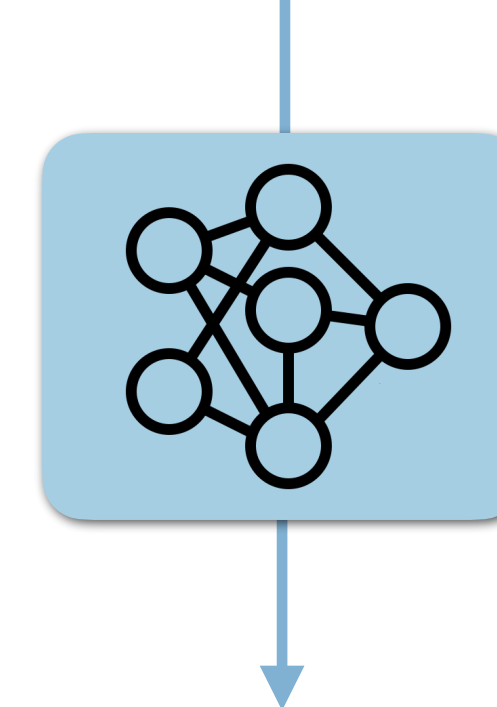
Olympia ██████████
Seattle ████████
Redmond █
...

Review: The company anticipated its operating profit to improve.
Sentiment: _____



Positive ██████████
Negative █

English: Hello
French: _____



Bonjour ██████████
Hola █
Ciao █
...

What can you do with a Language Model?

Notation: Given text | Generation

Answer the following question.
Q: What is the capital city of California?
A: Sacramento

Translate this text.
English: The weather is nice today.
Korean: 오늘 날씨가 좋네요.

Write a short story about a girl living in California.

Title: The Orange Tree

In a quiet neighborhood on the edge of Sacramento, twelve-year-old Maya lived in a small yellow house with her grandmother. The backyard was wild and beautiful, with tall sunflowers, patches of mint, and an old orange tree that her grandfather had planted decades ago. (...)

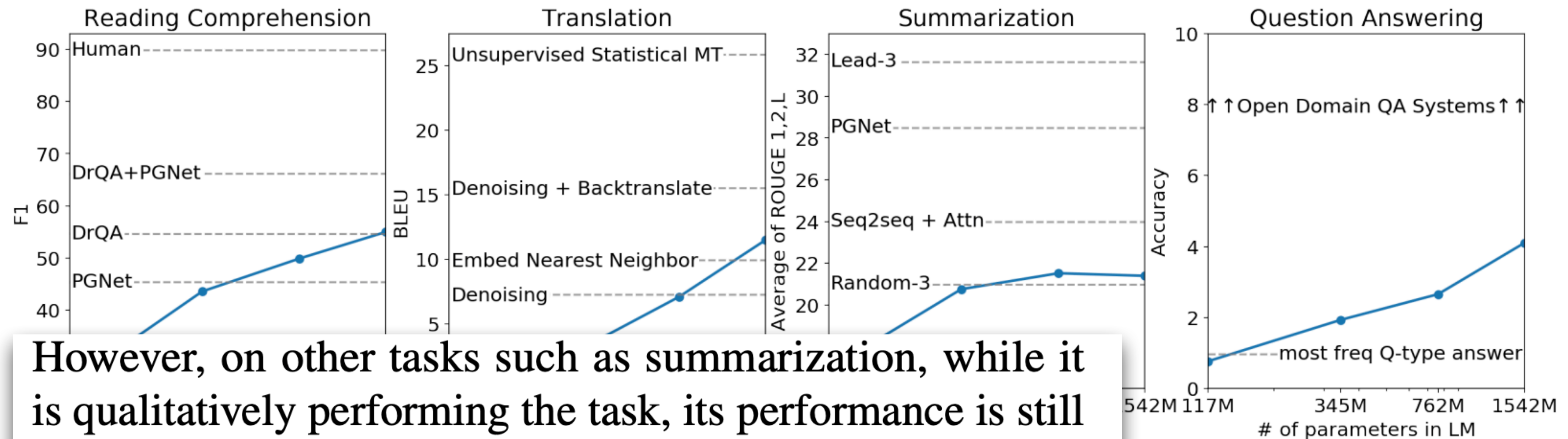
How do I get to Sacramento City Hall from UC Berkeley by car?

Instructions: Here are driving directions (approximately 1hr 30min, ~85 miles)

1. Get on I-80 E from Berkeley.
2. Follow I-80 E toward Sacramento.
3. Take exit 519B toward J Street.
4. Turn left on 9th Street.
5. Destination: 915 I Street, Sacramento, CA 95814

A.K.A. “Prompting”

GPT-2: language models are *zero-shot* learners



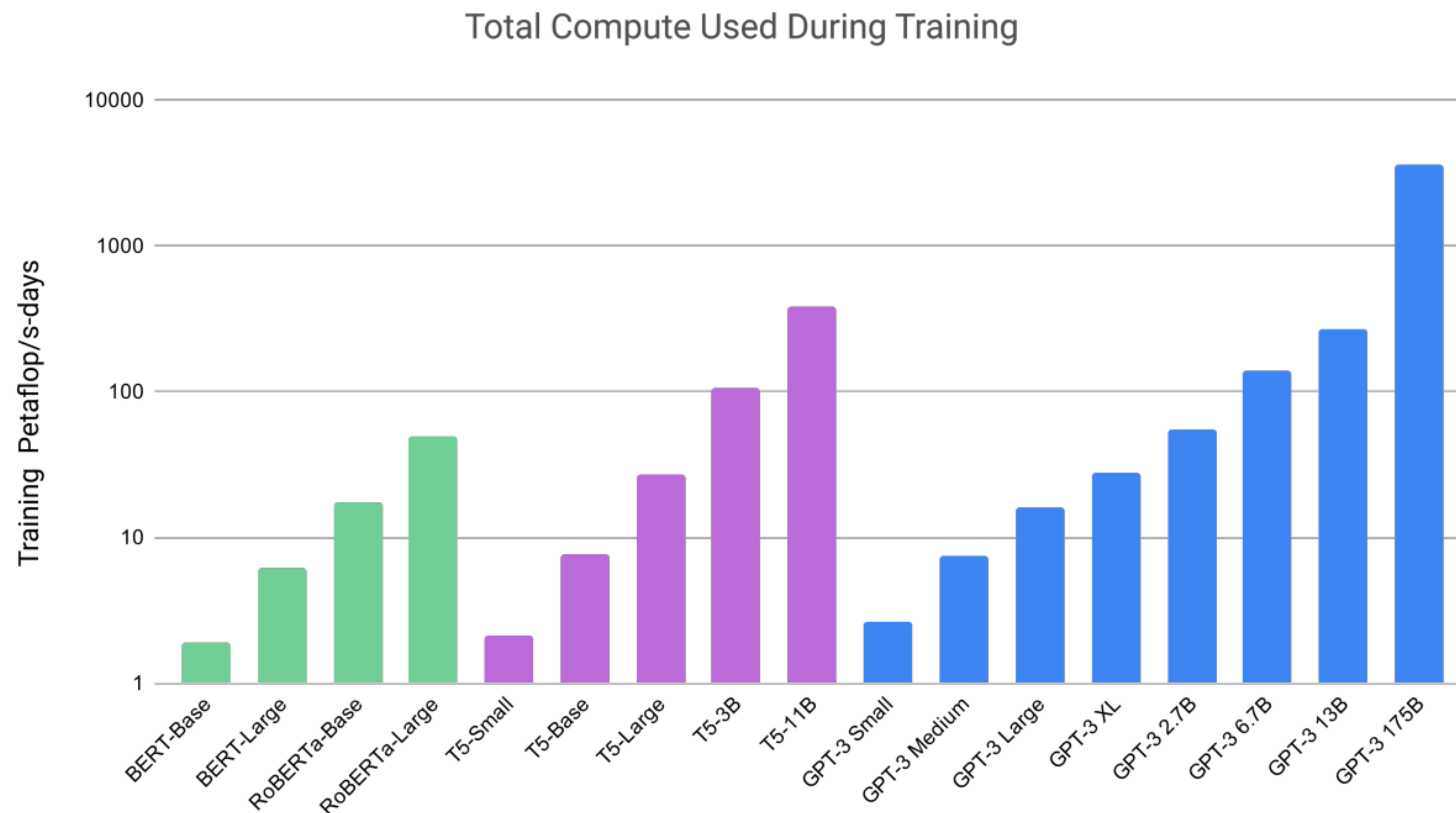
However, on other tasks such as summarization, while it is qualitatively performing the task, its performance is still only rudimentary according to quantitative metrics. While suggestive as a research result, in terms of practical applications, the zero-shot performance of GPT-2 is still far from use-able.

tasks. Reading Comprehension results on on CNN and Daily Mail (See et al., ns detailed descriptions of each result.

(Although, not to all...)

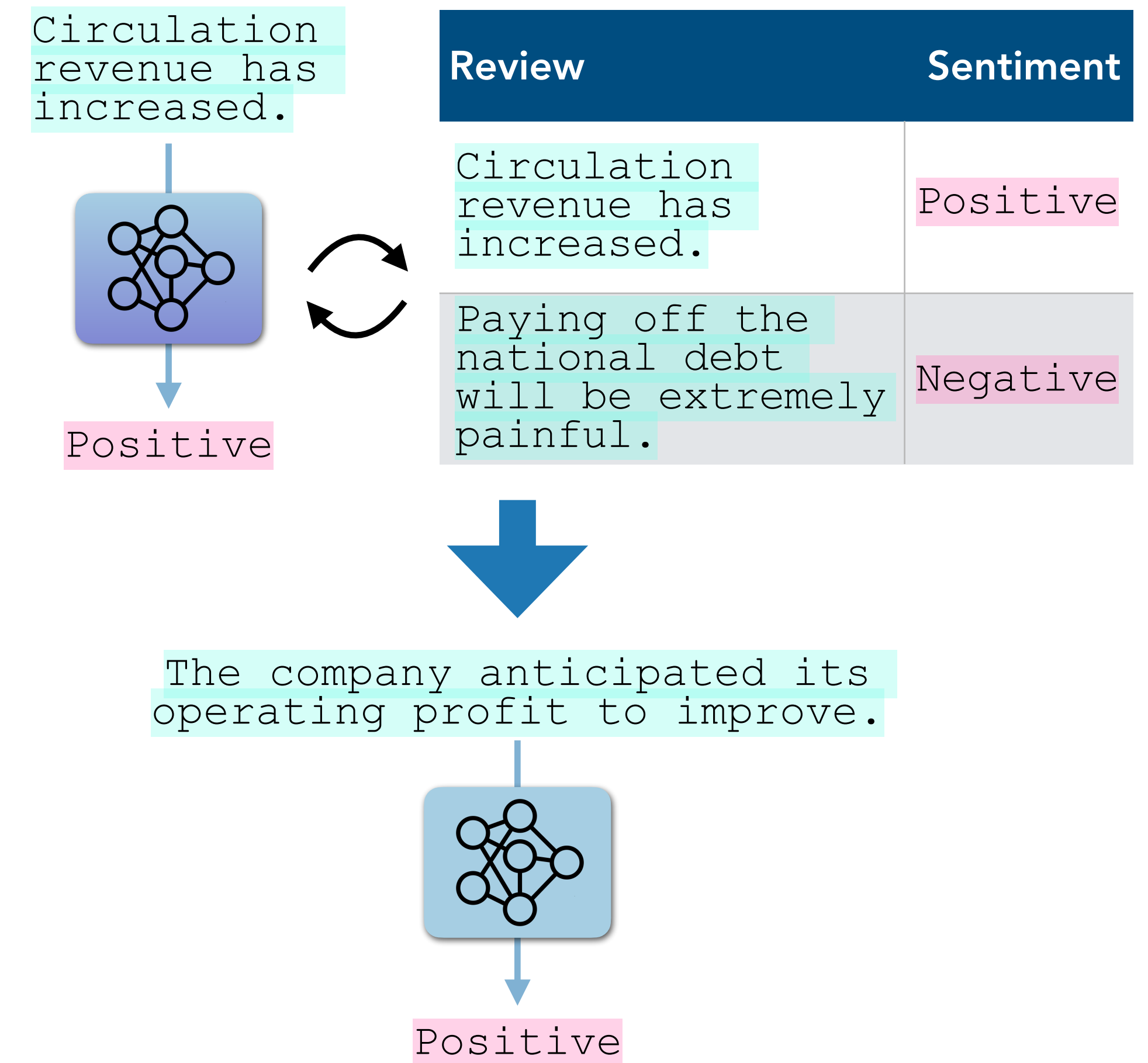
GPT-3: language models are *few-shot* learners

GPT-2 → GPT-3: 1.5B → **175B** (# of parameters), ~14B → **300B** (# of tokens)

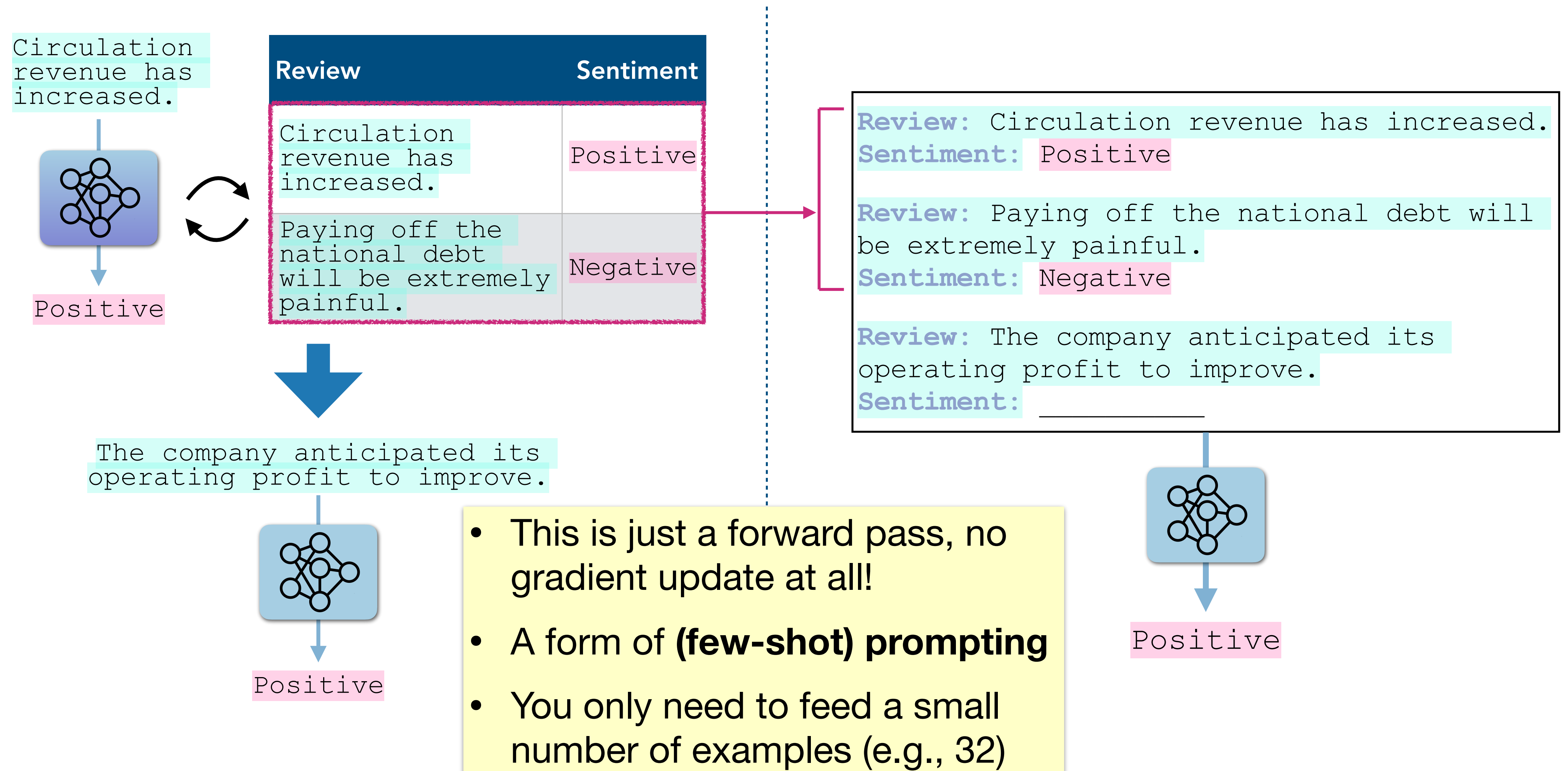


GPT-3: language models are *few-shot* learners

- Before GPT-3, fine-tuning is the default way of doing learning in models like BERT/T5/GPT-2
 - SST-2 has 67k examples, SQuAD has 88k (passage, answer, question) triples
- Fine-tuning requires computing the gradient and applying a parameter update on every example (or every K examples in a mini-batch)
- However, this is very expensive for the 175B GPT-3 model



Finetuning vs. In-context learning



GPT-3's in-context learning

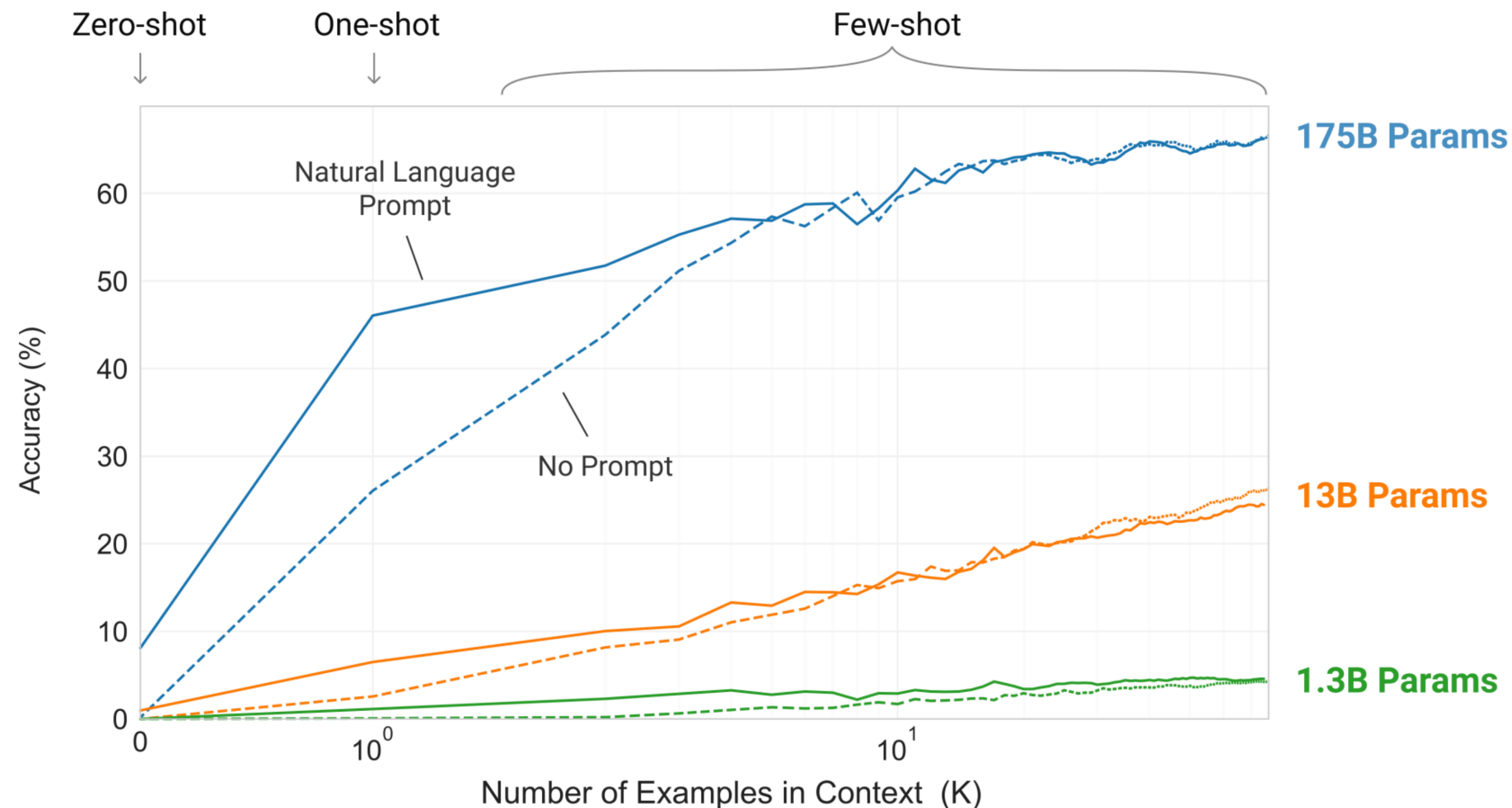
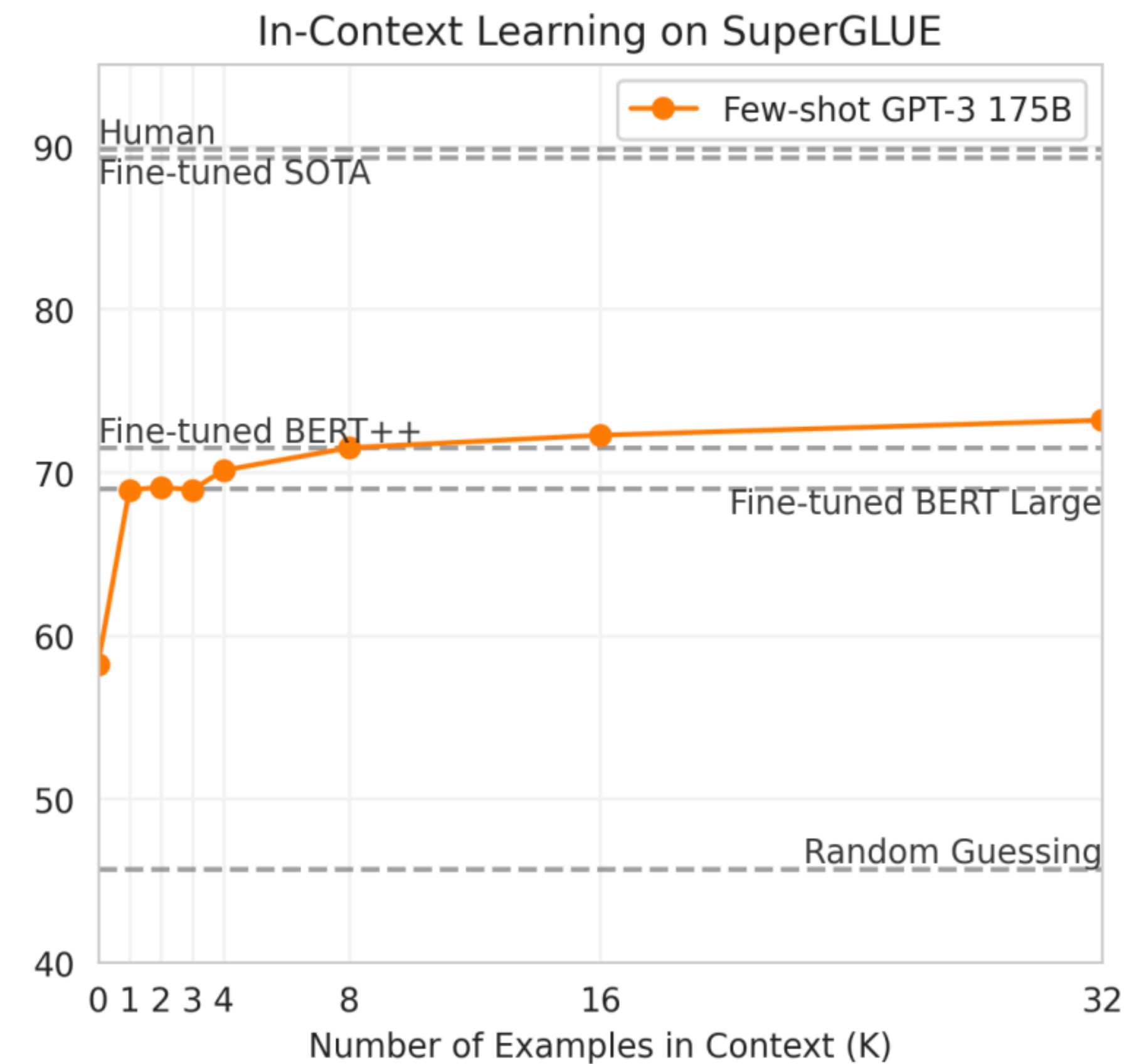
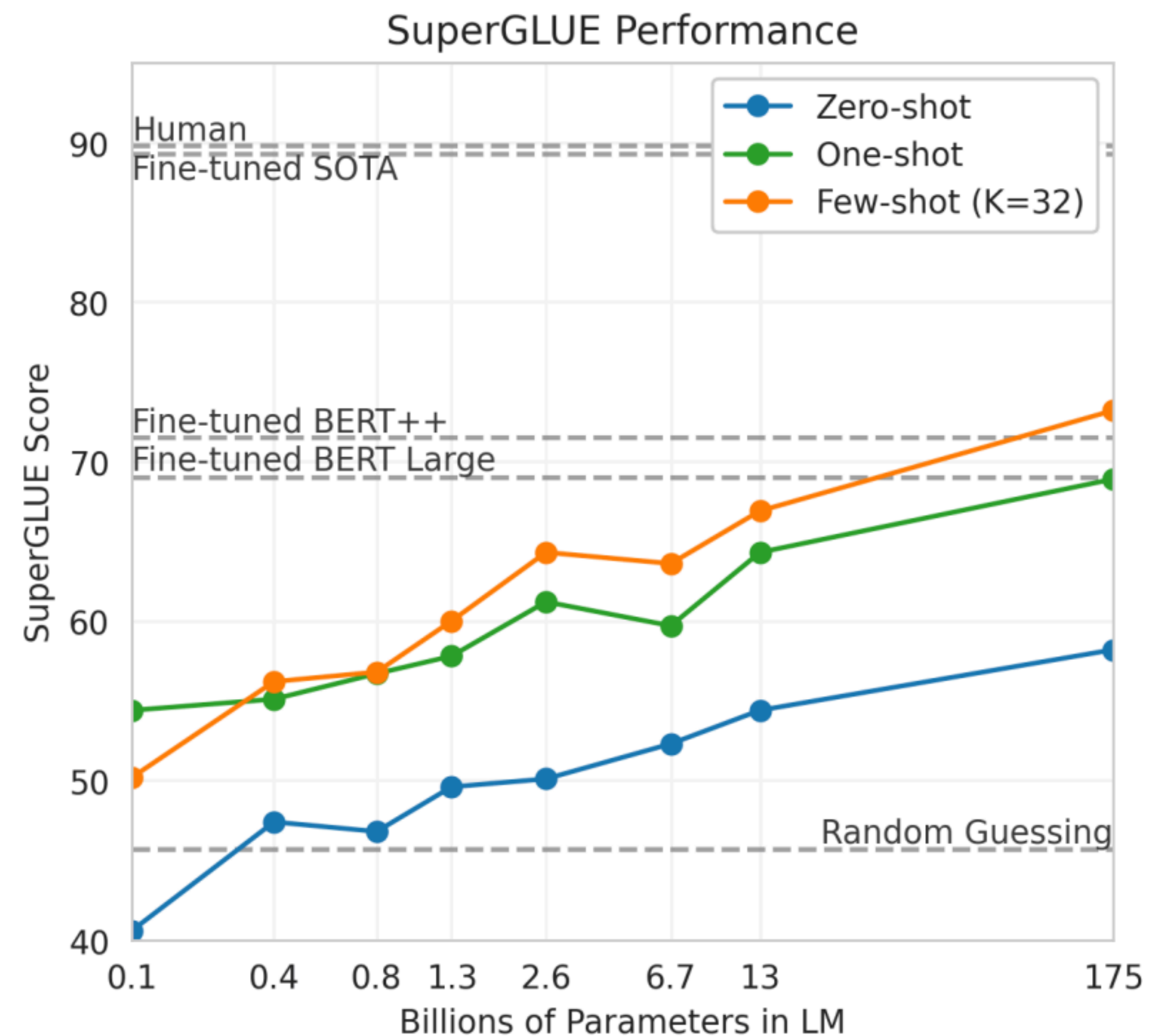
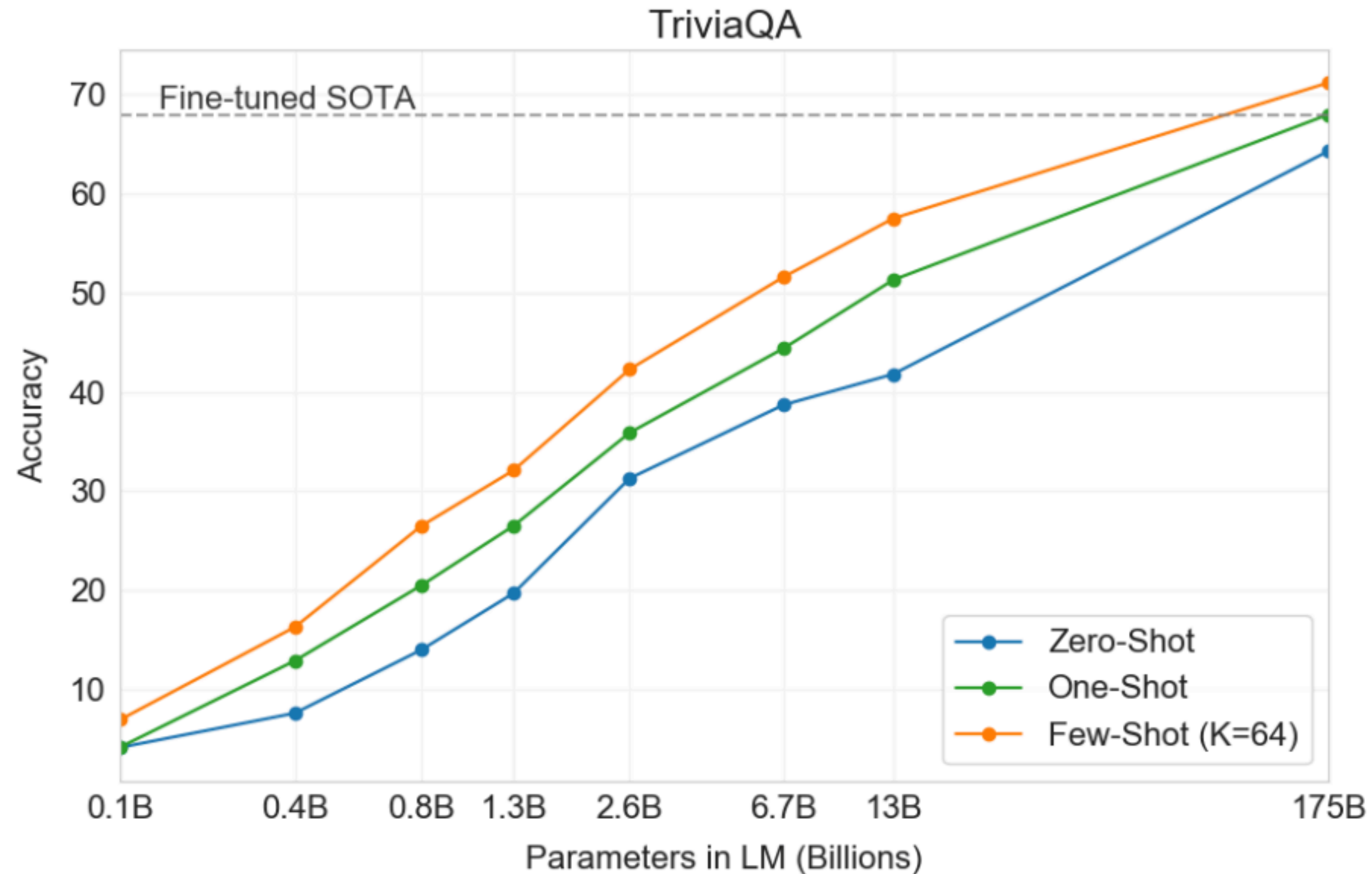


Figure 1.2: Larger models make increasingly efficient use of in-context information. We show in-context learning performance on a simple task requiring the model to remove random symbols from a word, both with and without a natural language task description (see Sec. 3.9.2). The steeper “in-context learning curves” for large models demonstrate improved ability to learn a task from contextual information. We see qualitatively similar behavior across a wide range of tasks.

GPT-3 performance on SuperGLUE



GPT-3 performance on TriviaQA



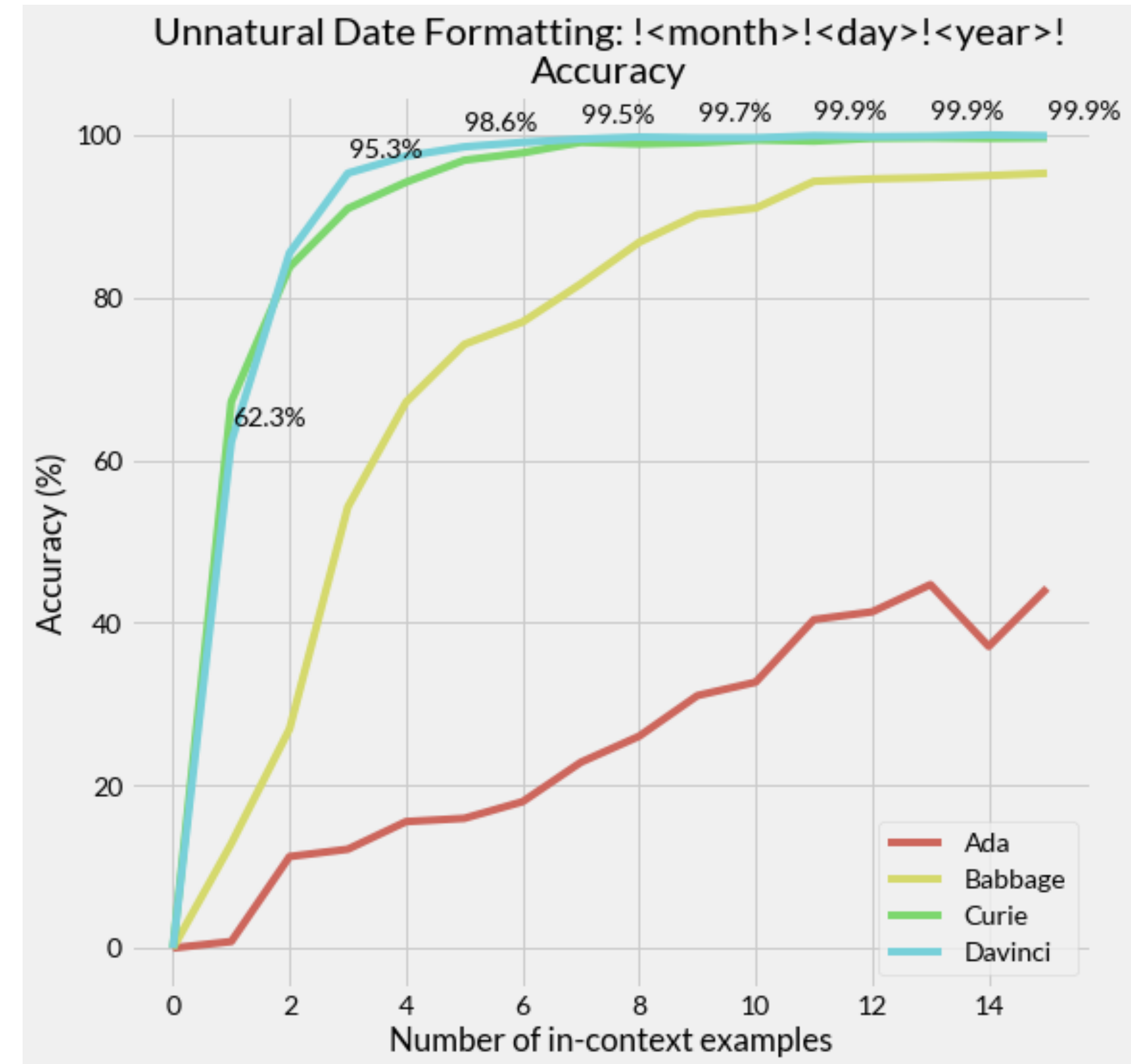
GPT-3's in-context learning

```
Input: 2014-06-01
Output: !06!01!2014!
Input: 2007-12-13
Output: !12!13!2007!
Input: 2010-09-23
Output: !09!23!2010!
Input: 2005-07-23
Output: !07!23!2005!
```

*in-context
examples*

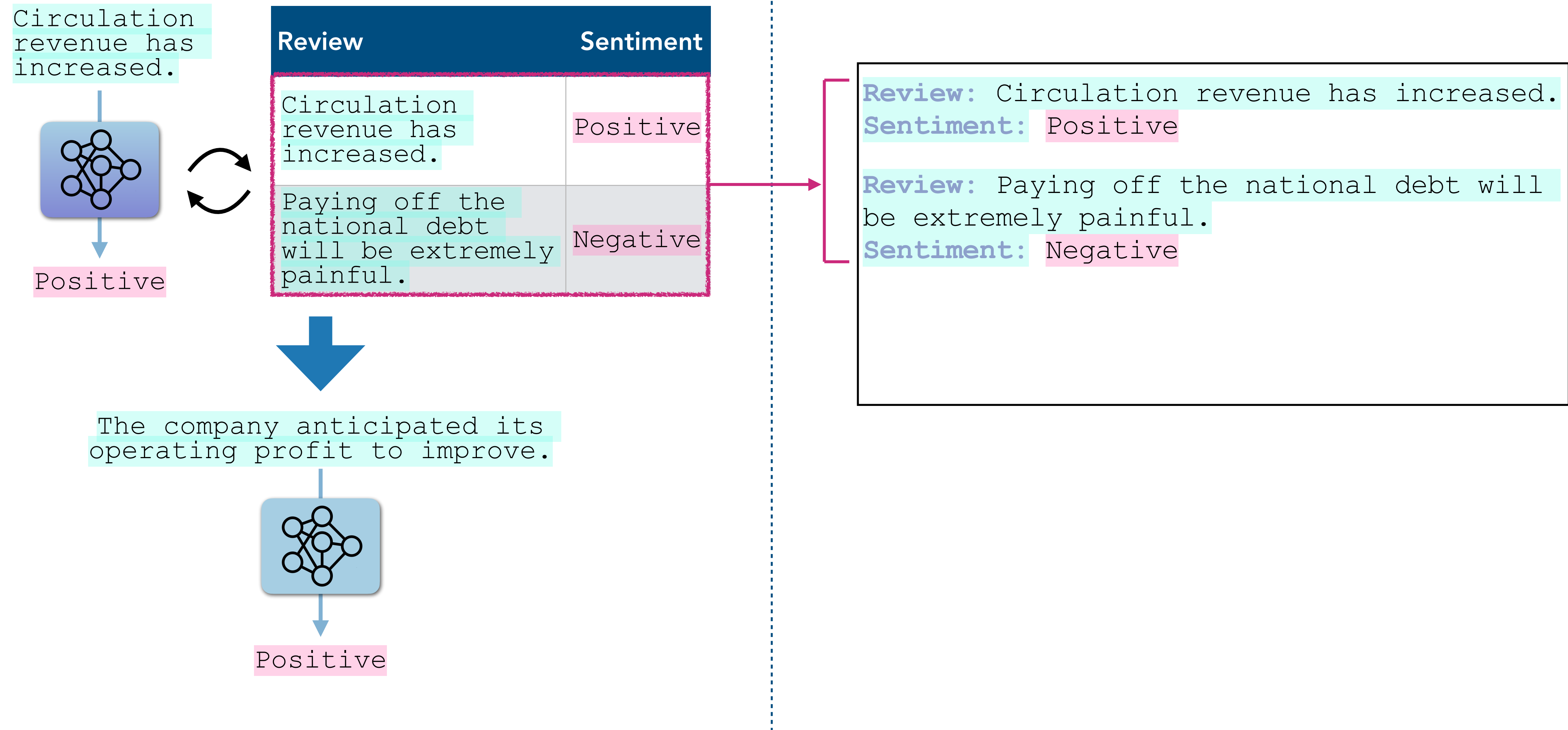
test example

└── *model completion*

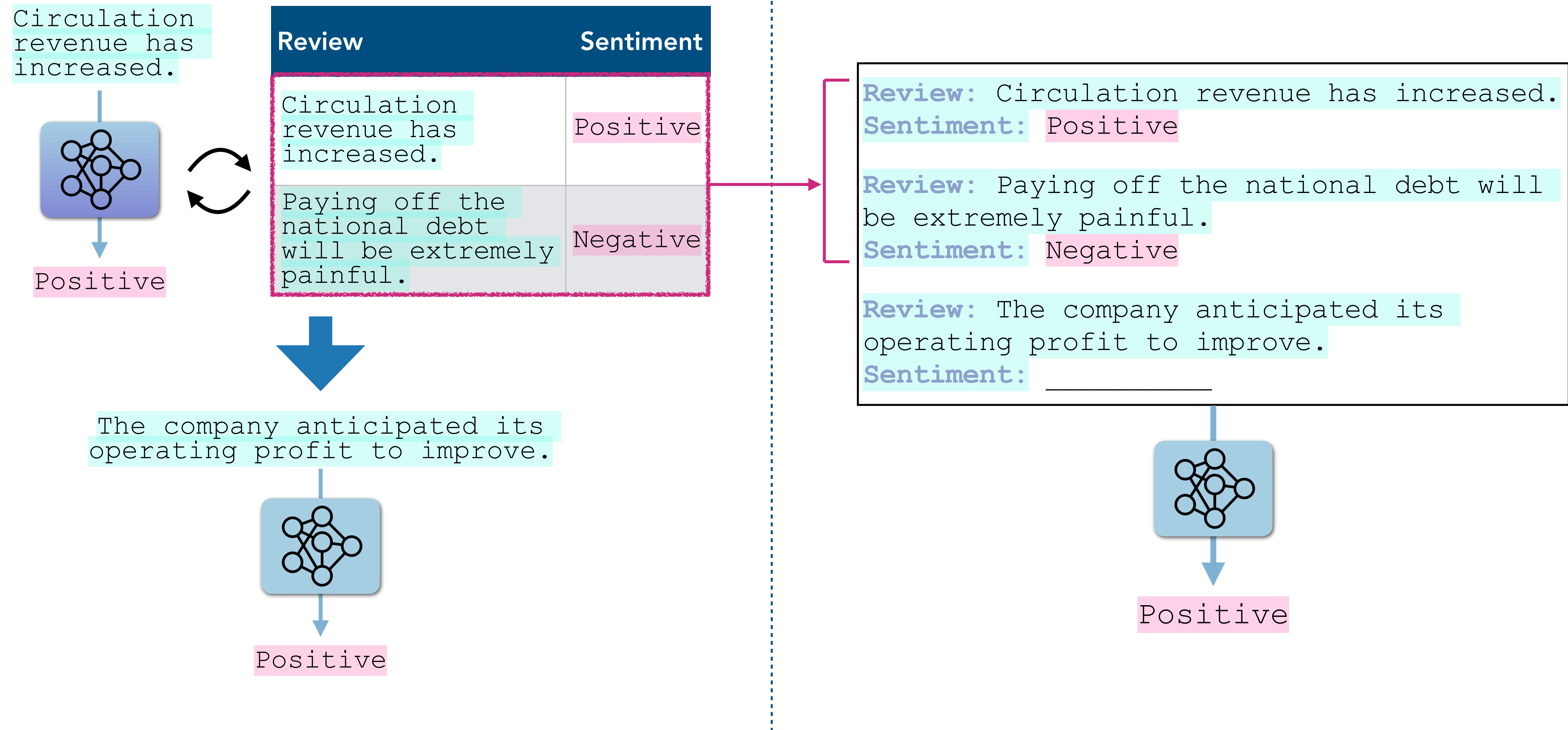


More about in-context learning (ICL)

Finetuning vs. In-context learning



Finetuning vs. In-context learning



Chain-of-thought

Labeled
data

Question: Leah had 32 chocolates and her sister had 42.
If they ate 35, how many pieces do they have left?
Answer: 39

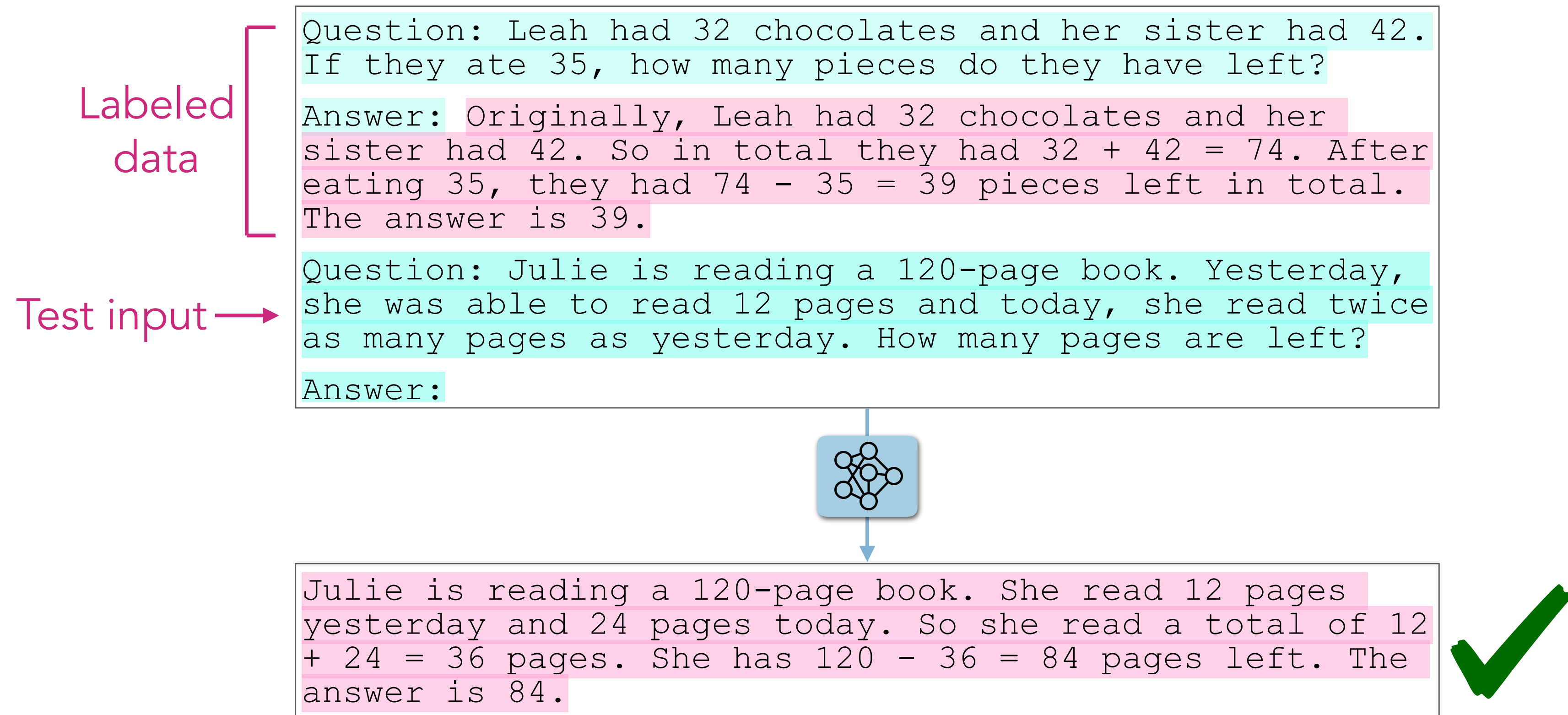
Chain-of-thought

Labeled
data

Question: Leah had 32 chocolates and her sister had 42.
If they ate 35, how many pieces do they have left?

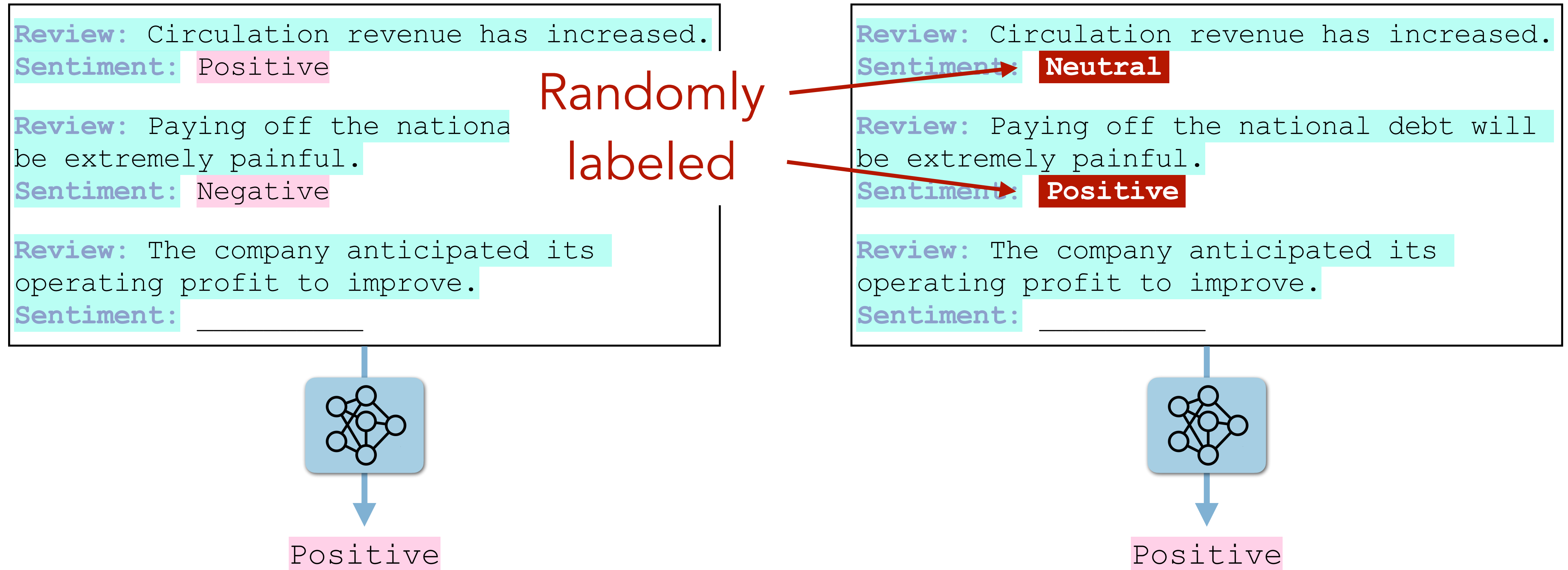
Answer: Originally, Leah had 32 chocolates and her
sister had 42. So in total they had $32 + 42 = 74$. After
eating 35, they had $74 - 35 = 39$ pieces left in total.
The answer is 39.

Chain-of-thought

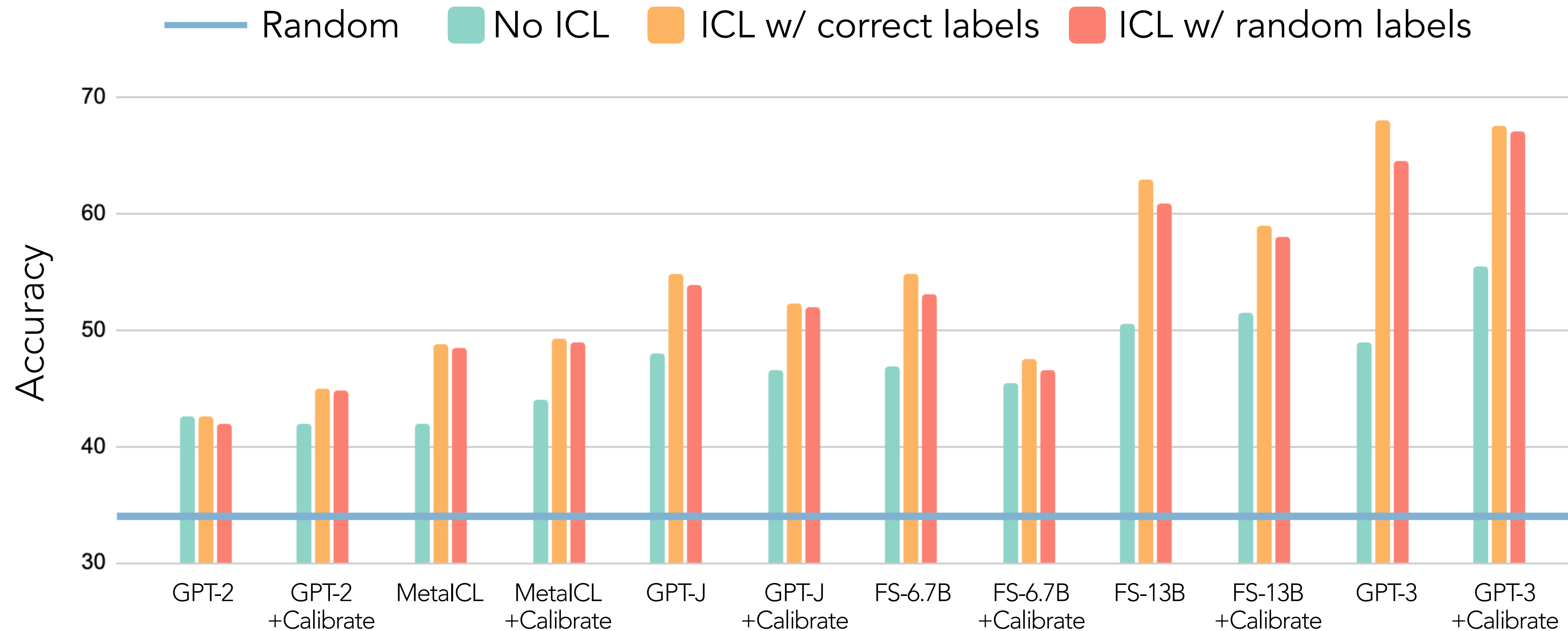


- You need to use CoT with in-context learning
- A little more about it will be covered in a later lecture (“Reasoning models”)

Does in-context learning *learn*?



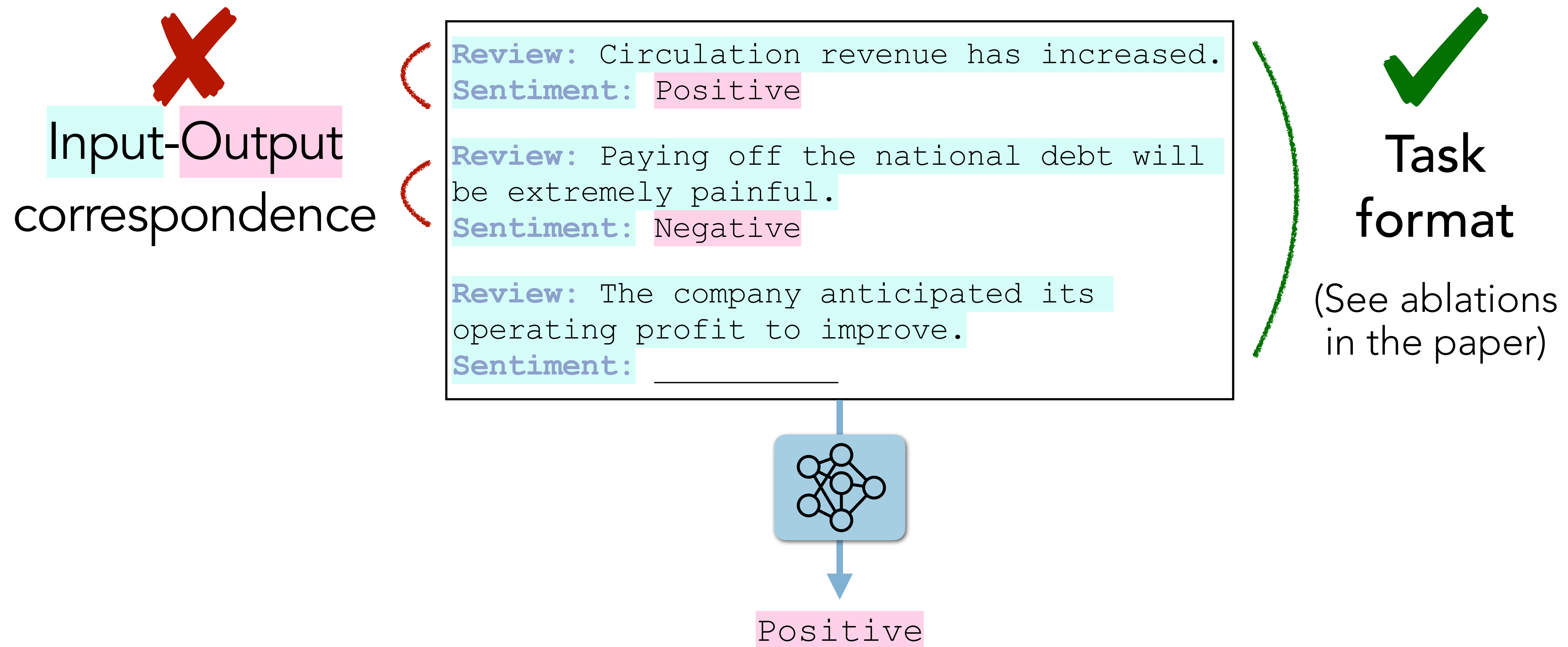
Does in-context learning *learn*?



Recovers **95%** of ICL accuracy

In-context learning: Hypothesis

I. Labeled data teaches format, not correspondence



In-context learning: Hypothesis

2. Correspondence is learned from LM training data

Mr. and Mrs. Dursley, of number four, Privet Drive, were proud to say that they were perfectly normal, thank you very much. They were the last people you'd expect to be involved in anything strange or mysterious, because they just didn't hold with such nonsense. Mr. Dursley was the director of a firm called Grunnings, which made



It works on every level which is, perhaps, fitting for a film about levels and whether you are at the top or bottom in life.

[Full Review](#) | Feb 7, 2020

★★★★★ **Works well**

Reviewed in the United States on February 6, 2010

Verified Purchase

For such an inexpensive little thing, it works great. The dogs react immediately to it as though it means something wonderful.

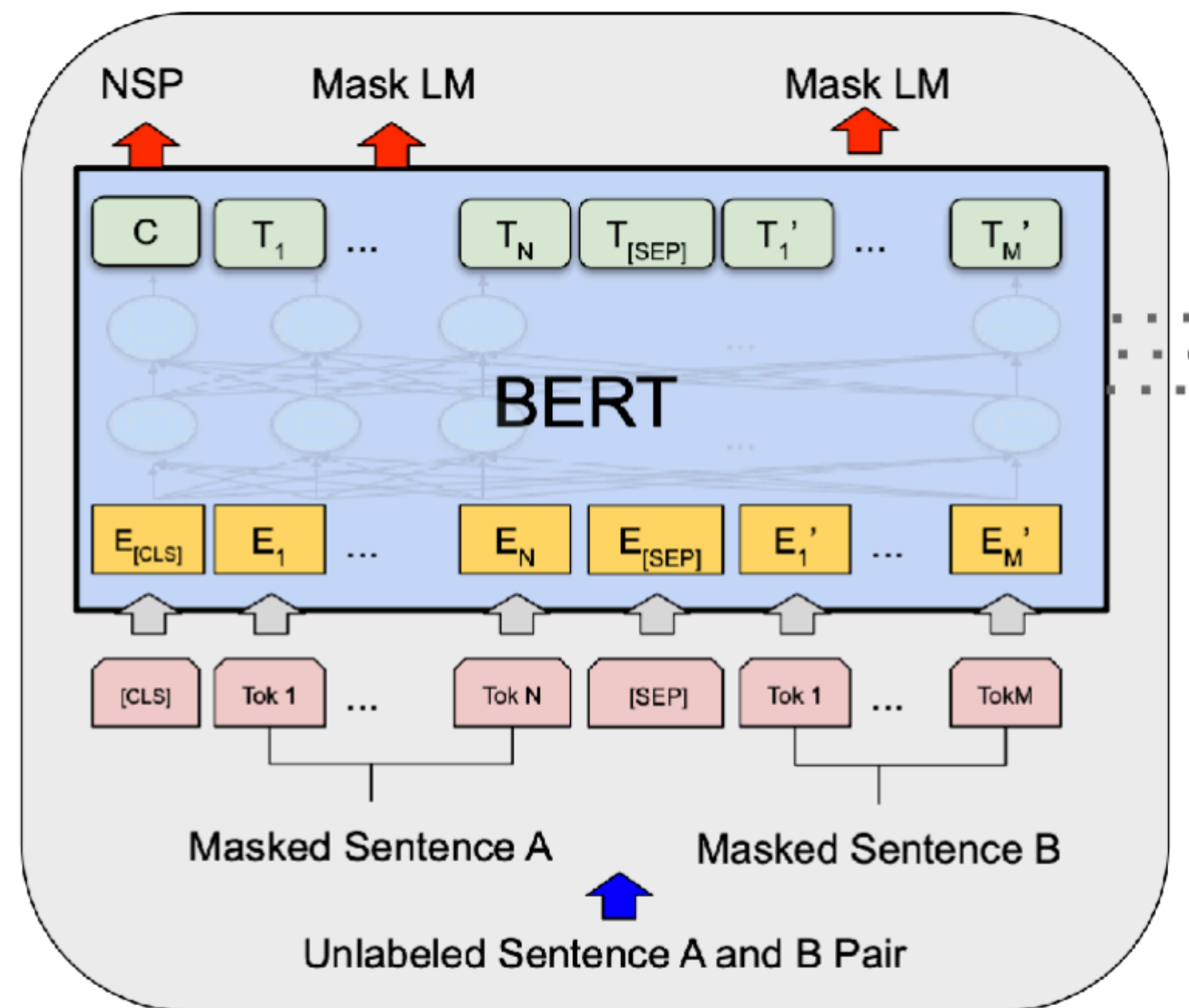
★★★★★ Reviewed April 12, 2017

Really good coffee

In the land of great coffee shops, this one stands out. Not super fancy inside - its right near the college - it carries a fun ambience with great baked goods and really good coffee. Great service and a large area allowing a lot of... **More**

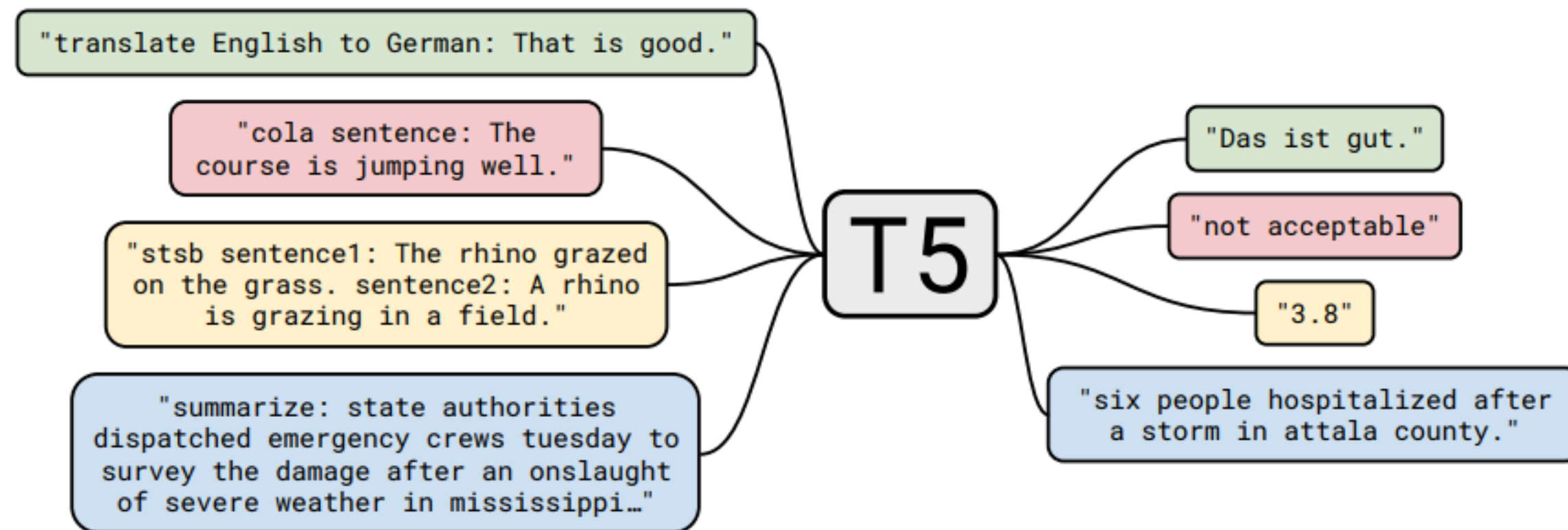
What happened to the NLP field after GPT-3?

Recap: Three major forms of pre-training



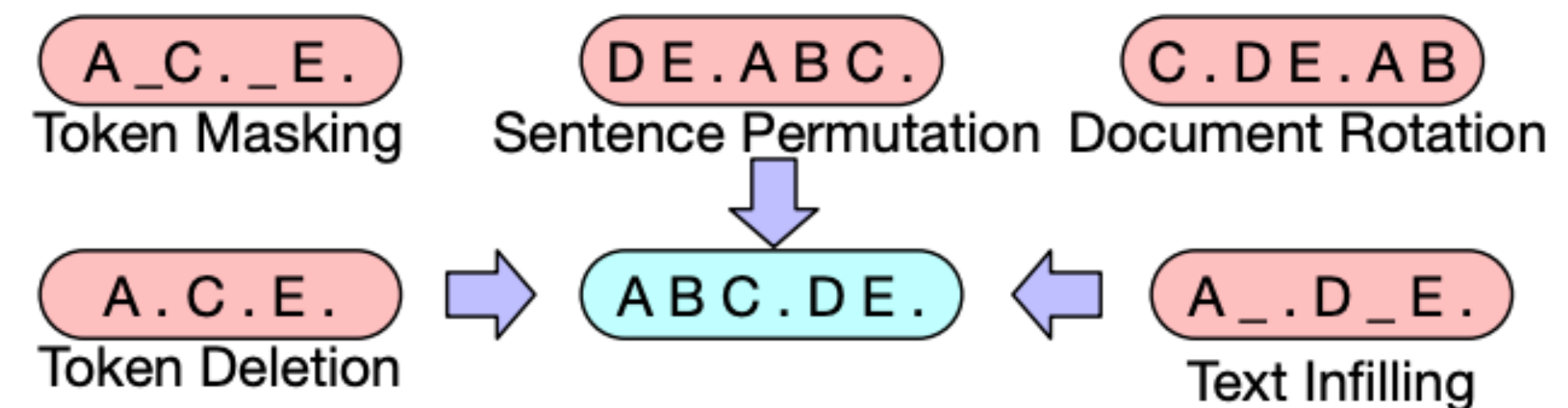
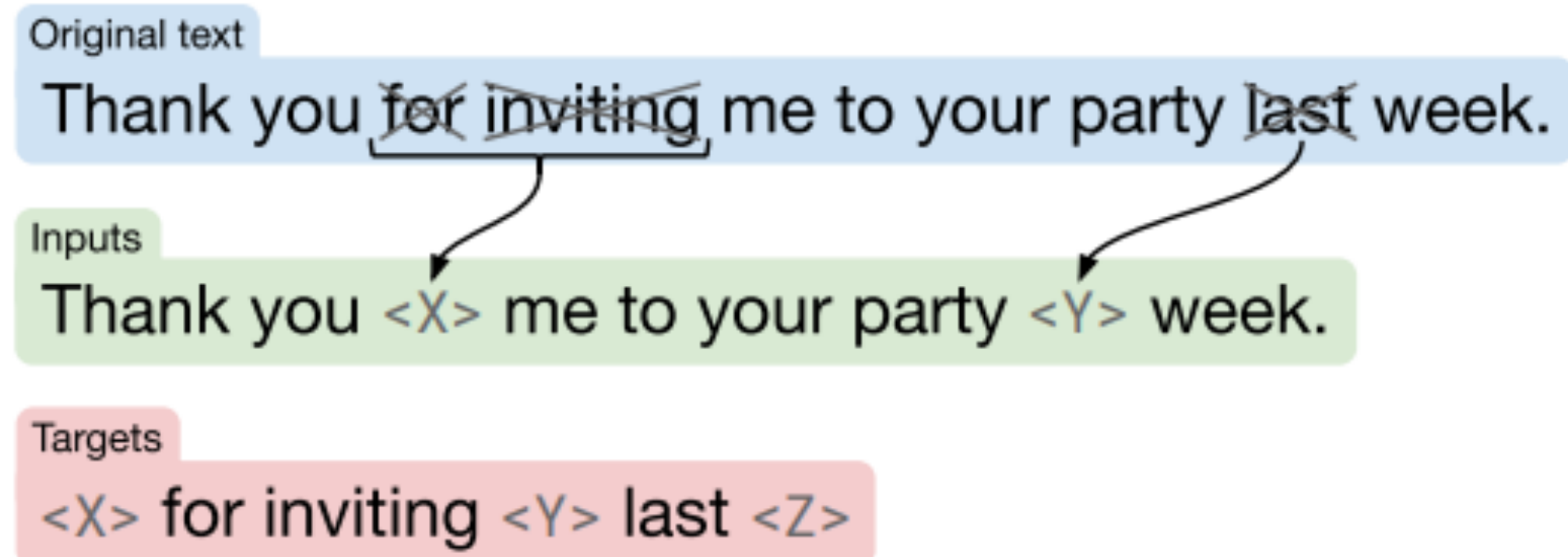
- **Pre-training objectives:** masked language modeling (+ optional: next sentence prediction)
- **Model architecture:** Transformer encoder
- **Examples:** BERT, RoBERTa, ALBERT, ELECTRA
(2018) (2019) (2019) (2020)
(BERT: 110M or 330M parameters)

Recap: Three major forms of pre-training

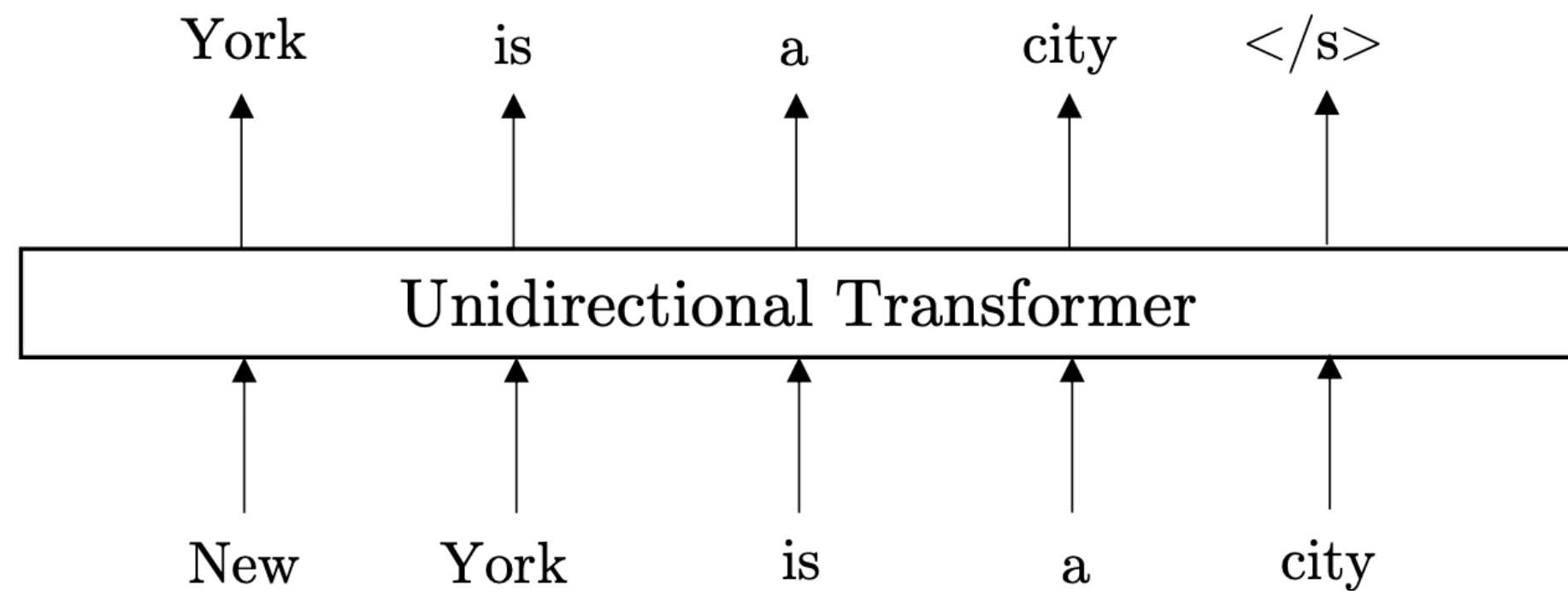


- **Pre-training objectives:** random span masking and many other variants
- **Model architecture:** Transformer encoder-decoder
- **Examples:** T5, BART (2019)

(T5: 60M-11B parameters)



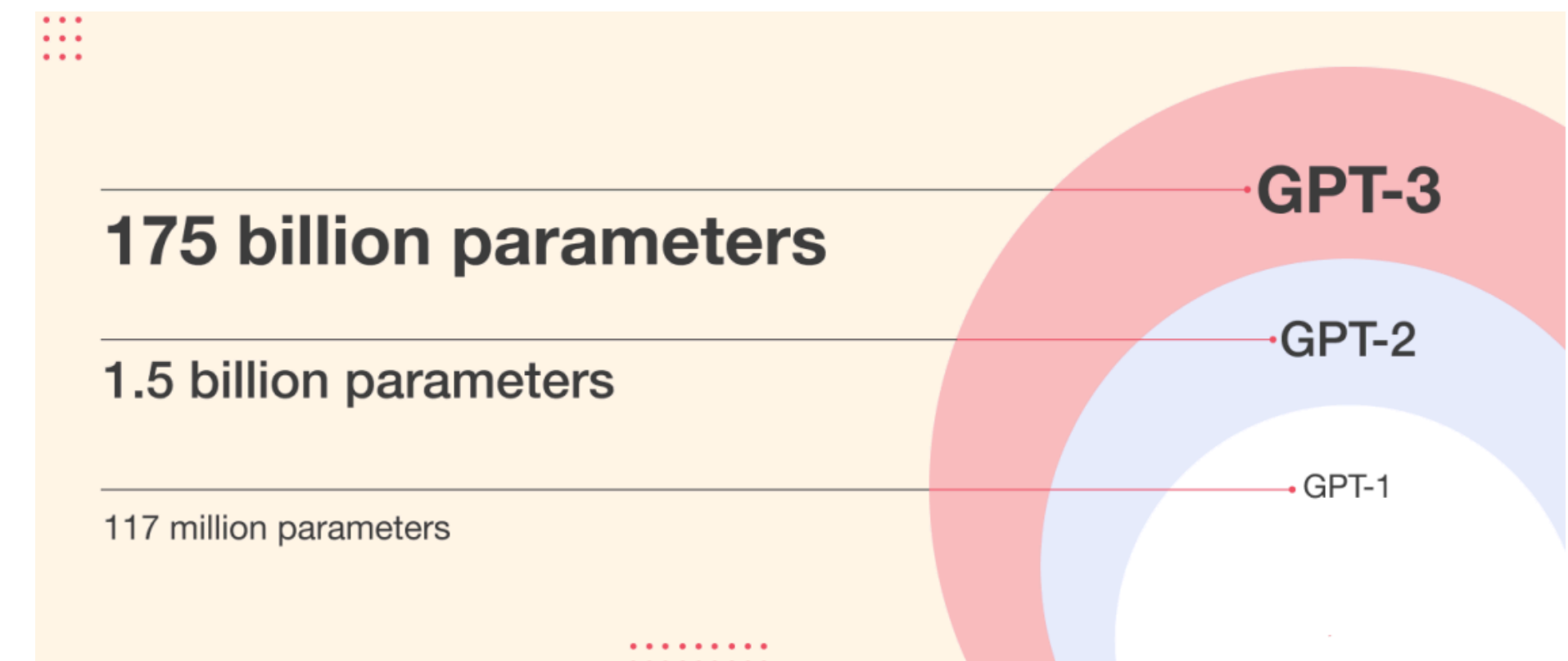
Recap: Three major forms of pre-training



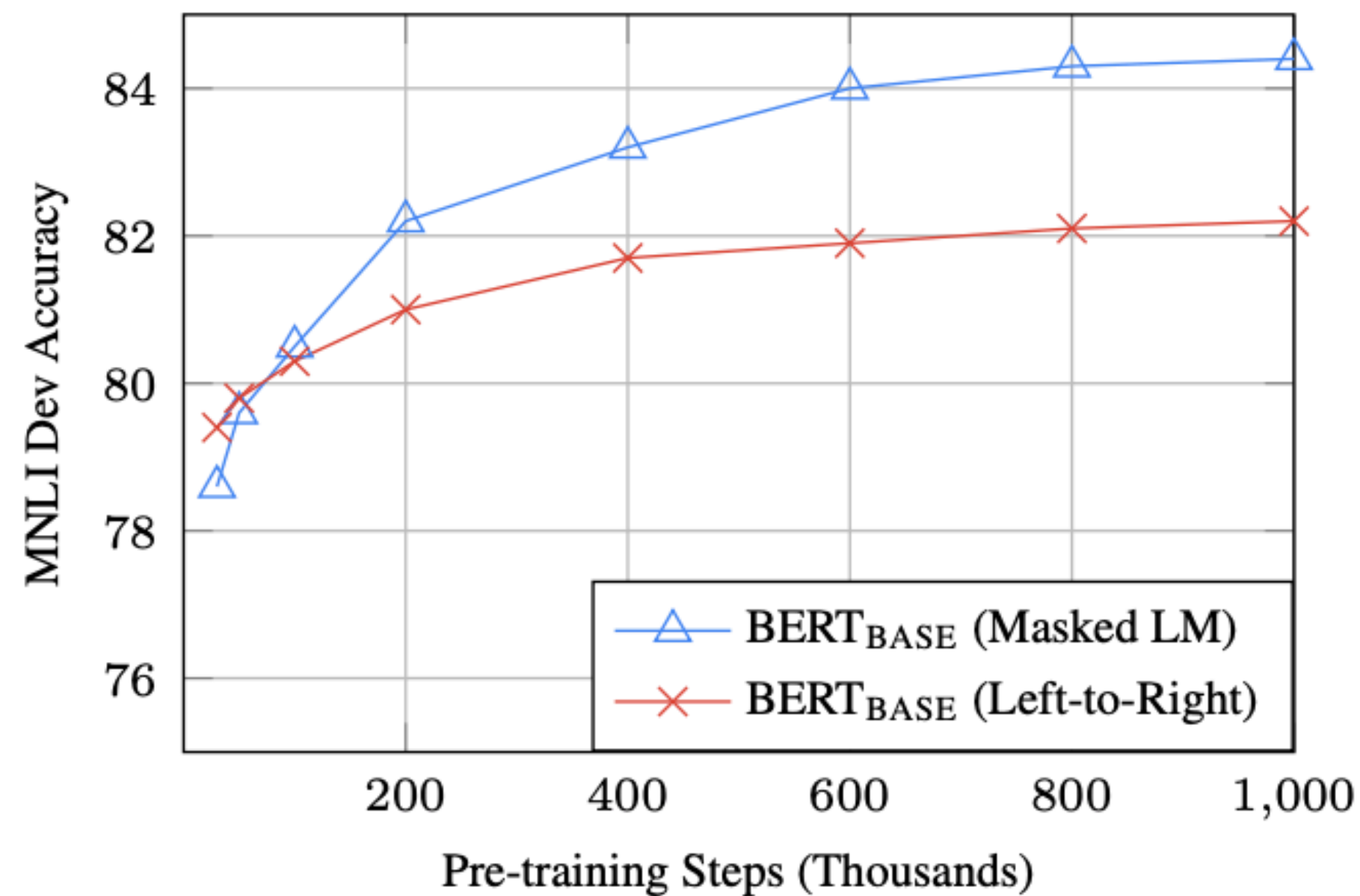
$$\log p(\mathbf{x}) = \sum_{t=1}^T \log p(x_t | \mathbf{x}_{<t})$$

- **Pre-training objectives:** next-token prediction
- **Model architecture:** Transformer decoder
- **Examples:** almost all modern LMs you see today!
 - GPT-1, GPT-2, GPT-3, ChatGPT, GPT-4, ...
 - LLaMA models
 - PaLM, Gemini, Gemma, ..
 - Claude
 - Mistral

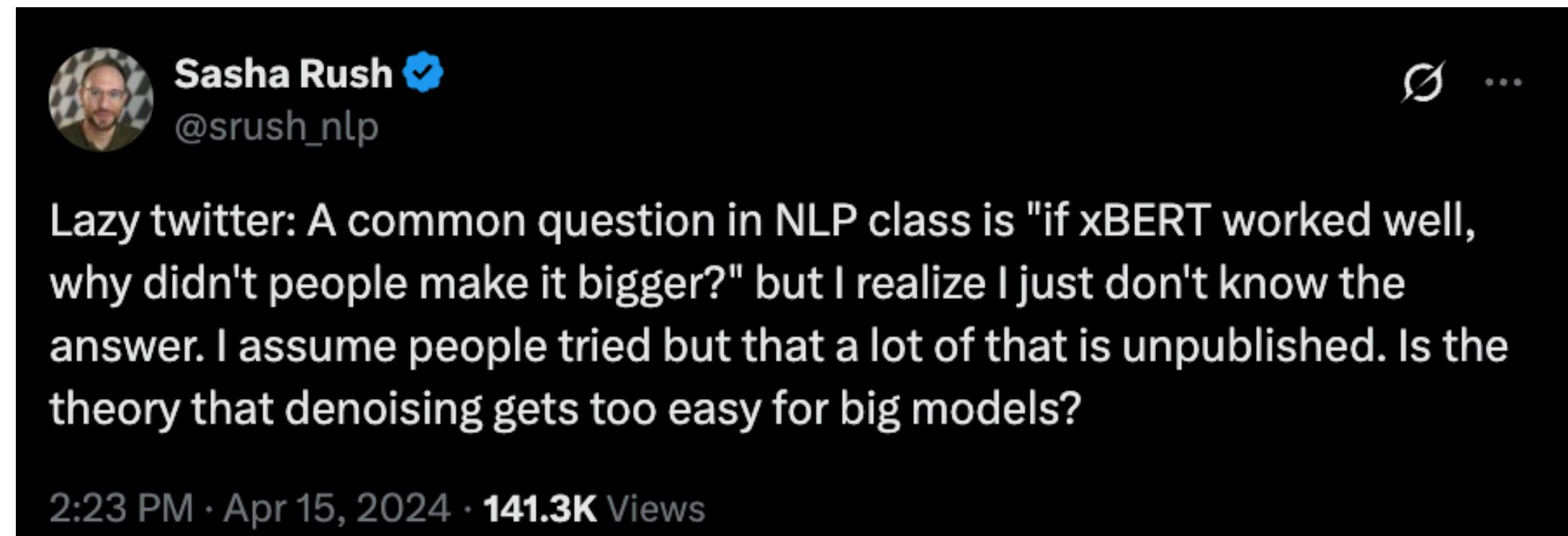
(2018-today)



Why do autoregressive LMs win out?



(Devlin et al. , 2018)



What happened to BERT & T5? On Transformer Encoders, PrefixLM and Denoising Objectives



<https://www.yitay.net/blog/model-architecture-blogpost-encoders-prefixlm-denoising>

Why do autoregressive LMs win out?

- Bidirectional attention is only important at smaller scale?
- “Masking objectives” can be still combined with autoregressive LMs

Training

Original Document

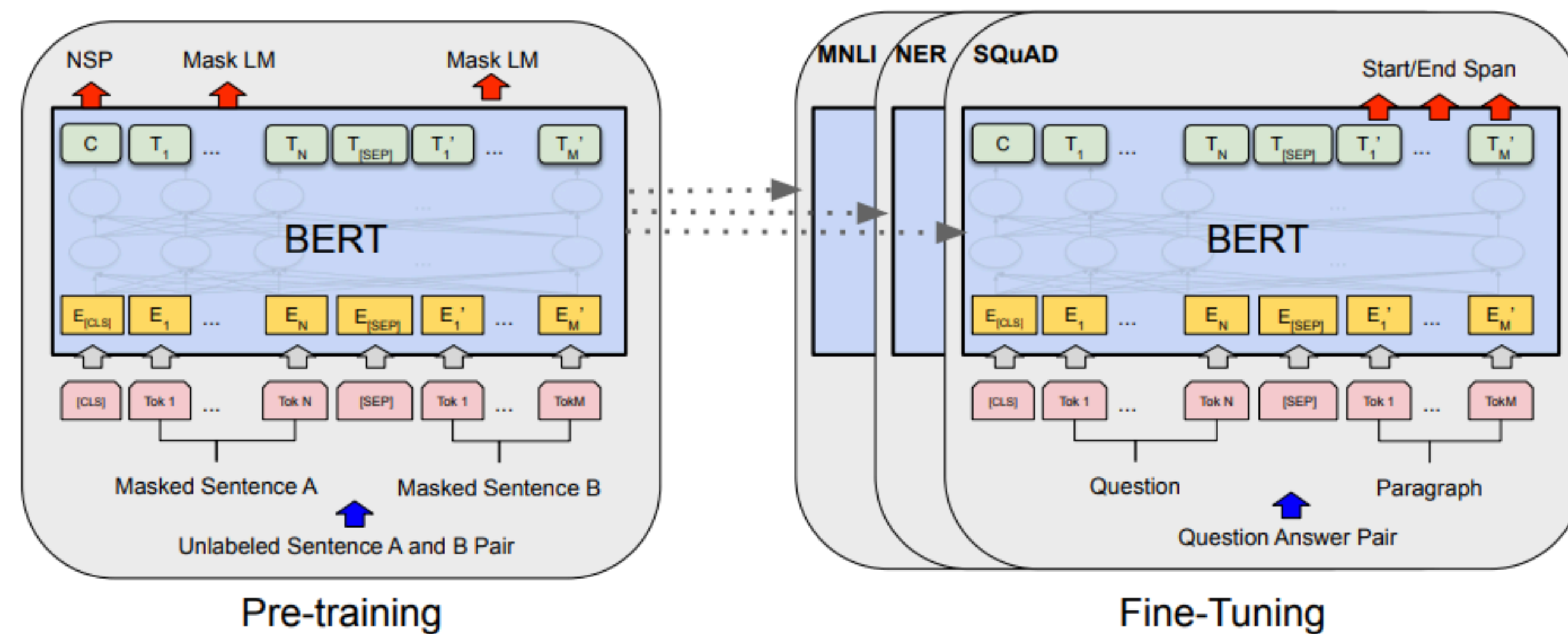
```
def count_words(filename: str) -> Dict[str, int]:
    """Count the number of occurrences of each word in the file."""
    with open(filename, 'r') as f:
        word_counts = {}
        for line in f:
            for word in line.split():
                if word in word_counts:
                    word_counts[word] += 1
                else:
                    word_counts[word] = 1
    return word_counts
```

Masked Document

```
def count_words(filename: str) -> Dict[str, int]:
    """Count the number of occurrences of each word in the file."""
    with open(filename, 'r') as f:
        <MASK:0> in word_counts:
            word_counts[word] += 1
        else:
            word_counts[word] = 1
    return word_counts
<MASK:0> word_counts = {}
for line in f:
    for word in line.split():
        if word <EOM>
```

(Fried et al. , 2022) InCoder: A Generative Model for Code Infilling and Synthesis

Fine-tuning vs prompting



You still need many annotated examples for each single task!

SST-2 (sentiment analysis): 67k

SQuAD (question answering): 100k

Fine-tuning vs prompting

In-context learning is a special form of prompting
(with a few annotated examples provided)

Circulation revenue has increased by 5%
in Finland. // Positive

Panostaja did not disclose the purchase
price. // Neutral

Paying off the national debt will be
extremely painful. // Negative

The company anticipated its operating
profit to improve. // _____

LM

Circulation revenue has increased by
5% in Finland. // Finance

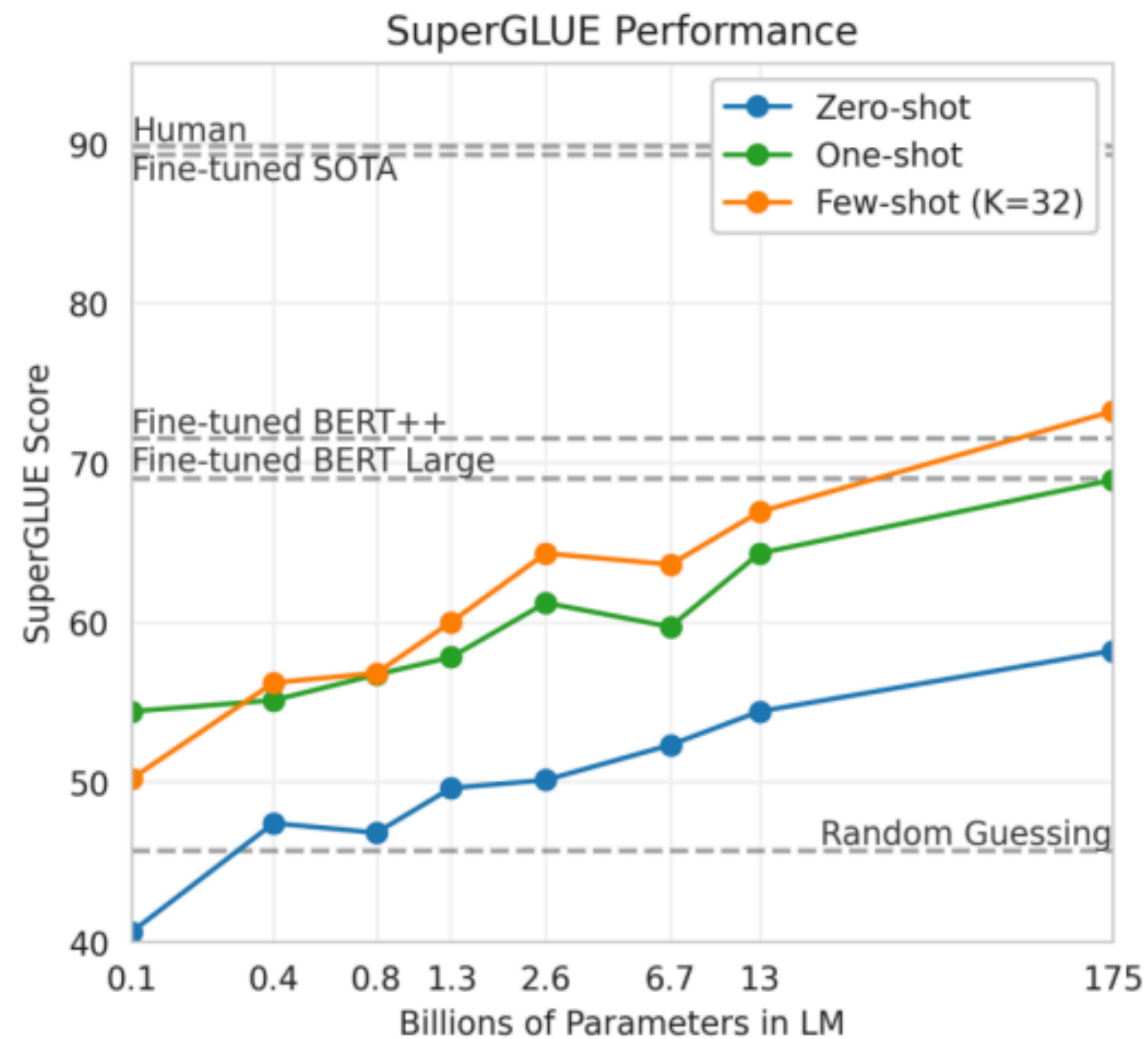
They defeated ... in the NFC
Championship Game. // Sports

Apple ... development of in-house
chips. // Tech

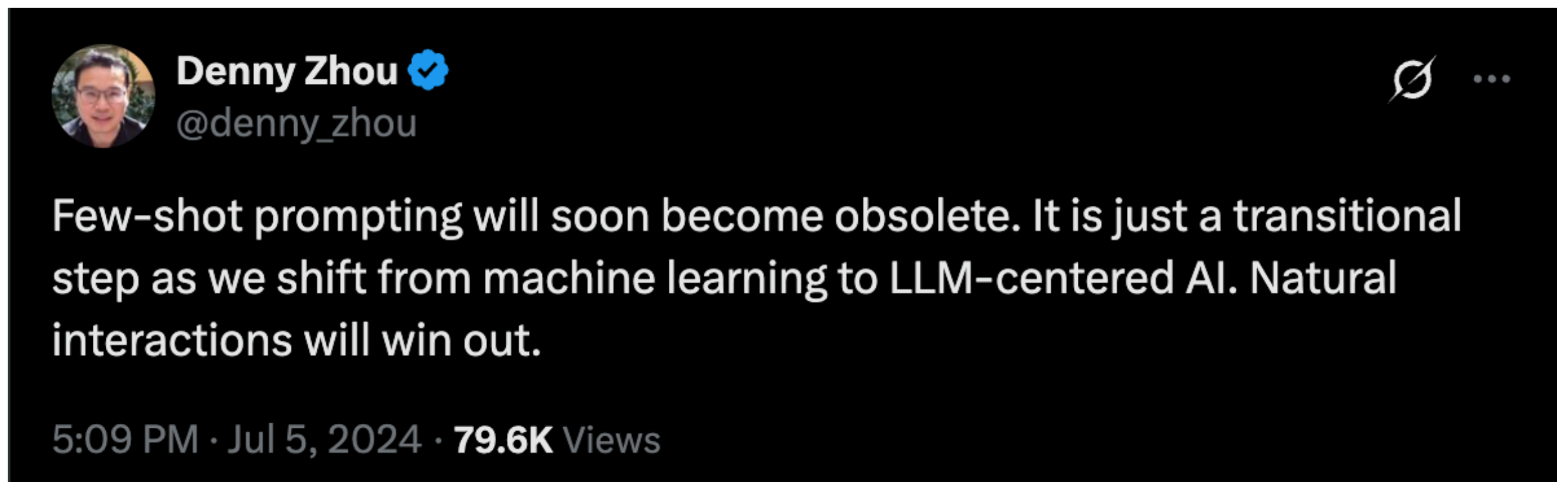
The company anticipated its operating
profit to improve. // _____

LM

Fine-tuning vs prompting



From GPT-3



Paradigm shift: one model does it all

NLP before 2020:

Sentiment
analysis

Question
answering

Machine
translation

Text
summarization

Prompt: Translate the following sentence from
English to German: “The dinner was great”



Machine
translation

Large language
models



Completion: Das Abendessen war großartig

Paradigm shift: one model does it all

NLP before 2020:

Sentiment
analysis

Question
answering

Machine
translation

Text
summarization

Prompt: Given the following paragraph [...], **how**
would you phrase it in a few words?

Text
summarization

Large language
models

Completion: Graffiti artist Banksy is
believed to be behind [...]

- “**Foundation Model**” (Bommasani et al., 2021)
- Zero or very few human-annotated examples required

What is an NLP task at all?



(Wang et al. , 2022)

HellaSwag (Commonsense)



A woman is outside with a bucket and a dog. The dog is running around trying to avoid a bath. She...

- A. rinses the bucket off with soap and blow dry the dog's head.
- B. uses a hose to keep it from getting soapy.
- C. gets the dog wet, then it runs away again.**
- D. gets into a bath tub with the dog.



PIQA (Physical reasoning)



To separate egg whites from the yolk using a water bottle, you should...

a. **Squeeze** the water bottle and press it against the yolk. **Release**, which creates suction and lifts the yolk.

b. **Place** the water bottle and press it against the yolk. **Keep pushing**, which creates suction and lifts the yolk.



What is an NLP task at all?



(Wang et al. , 2022)

MMLU (Massive multitask language understanding)

Three contrasting tactics that CSO's can engage in to meet their aims are _____ which typically involves research and communication, _____, which may involve physically attacking a company's operations or _____, often involving some form of _____.

(A) Non-violent direct action, Violent direct action, Indirect action, Boycott
(B) Indirect action, Instrumental action, Non-violent direct action, Information campaign
(C) **Indirect action, Violent direct action, Non-violent direct-action Boycott.**
(D) Non-violent direct action, Instrumental action, Indirect action, Information campaign

Figure 17: A Business Ethics example.

How many attempts should you make to cannulate a patient before passing the job on to a senior colleague?

(A) 4 (B) 3 (C) **2** (D) 1

Figure 18: A Clinical Knowledge example.

In a given population, 1 out of every 400 people has a cancer caused by a completely recessive allele, b. Assuming the population is in Hardy-Weinberg equilibrium, which of the following is the expected proportion of individuals who carry the b allele but are not expected to develop the cancer?

(A) 1/400 (B) 19/400 (C) 20/400 (D) **38/400**

Figure 19: A College Biology example.

What is an NLP task at all?



(Wang et al. , 2022)

GSM8K (Grade school math)

Problem

The battery charge in Mary's cordless vacuum cleaner lasts ten minutes. It takes her four minutes to vacuum each room in her house. Mary has three bedrooms, a kitchen, and a living room. How many times does Mary need to charge her vacuum cleaner to vacuum her whole house?

Solution

Mary has $3 + 1 + 1 = 5$ rooms in her house.

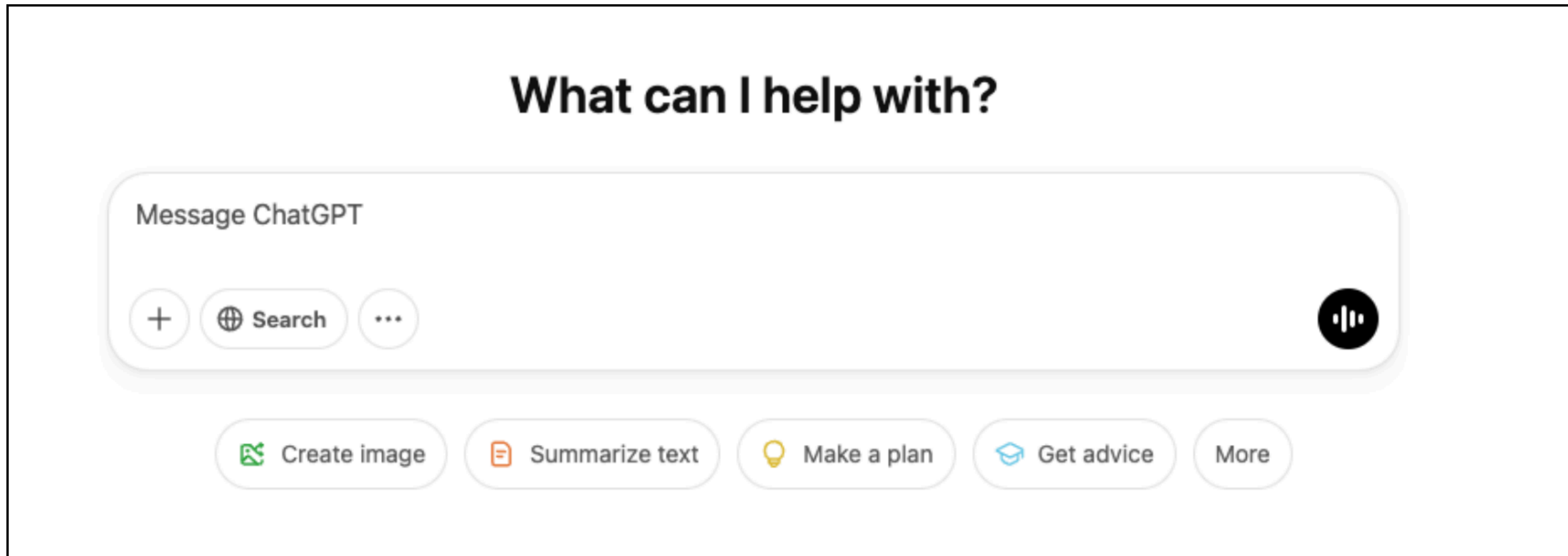
At 4 minutes a room, it will take her $4 * 5 = 20$ minutes to vacuum her whole house.

At 10 minutes a charge, she will need to charge her vacuum cleaner $20 / 10 = 2$ times to vacuum her whole house.

Final Answer

2

This is how we use ChatGPT today



Use-case	Prompt
Brainstorming	List five ideas for how to regain enthusiasm for my career
Generation	Write a short story where a bear goes to the beach, makes friends with a seal, and then returns home.

generation	Write an outline for an essay about John von Neumann and his contributions to computing: I. Introduction, his life and background A: His early life B:
rewrite	Covert my resume into a profile overview. {resume} Profile overview:
rewrite	Rephrase this for me: "I can't seem to find out how to work this darn thing." Alternate phrasing: "

A conceptual shift in NLP tasks

- There isn't a reliable way to evaluate

Rank* (UB)	Rank (StyleCtrl)	Model	Arena Score	95% CI	Votes	Organization	License
1	1	Gemini-2.5-Pro-Exp-03-25	1439	+7/-10	5858	Google	Proprietary
2	5	Llama-4-Maverick-03-26-Experimental	1417	+13/-12	2520	Meta	Llama
2	1	ChatGPT-4o-latest (2025-03-26)	1410	+8/-10	4899	OpenAI	Proprietary
2	4	Grok-3-Preview-02-24	1403	+6/-6	12391	xAI	Proprietary
3	2	GPT-4.5-Preview	1398	+5/-7	12312	OpenAI	Proprietary
6	7	Gemini-2.0-Flash-Thinking-Exp-01-21	1380	+4/-4	24298	Google	Proprietary
6	4	Gemini-2.0-Pro-Exp-02-05	1380	+4/-4	20289	Google	Proprietary
6	4	DeepSeek-V3-0324	1369	+10/-10	3526	DeepSeek	MIT
8	5	DeepSeek-R1	1358	+5/-5	14259	DeepSeek	MIT
9	14	Gemini-2.0-Flash-001	1354	+5/-5	20028	Google	Proprietary
9	4	o1-2024-12-17	1351	+5/-4	26722	OpenAI	Proprietary
12	14	Gemma-3-27B-it	1341	+5/-5	8420	Google	Gemma
12	14	Qwen2.5-Max	1340	+6/-3	18906	Alibaba	Proprietary
12	10	o1-preview	1335	+3/-4	33182	OpenAI	Proprietary
15	14	o3-mini-high	1325	+4/-4	15927	OpenAI	Proprietary

©b1bor91 (Floor Blano)

A conceptual shift in NLP tasks

- We still use ICL to evaluate a base model!

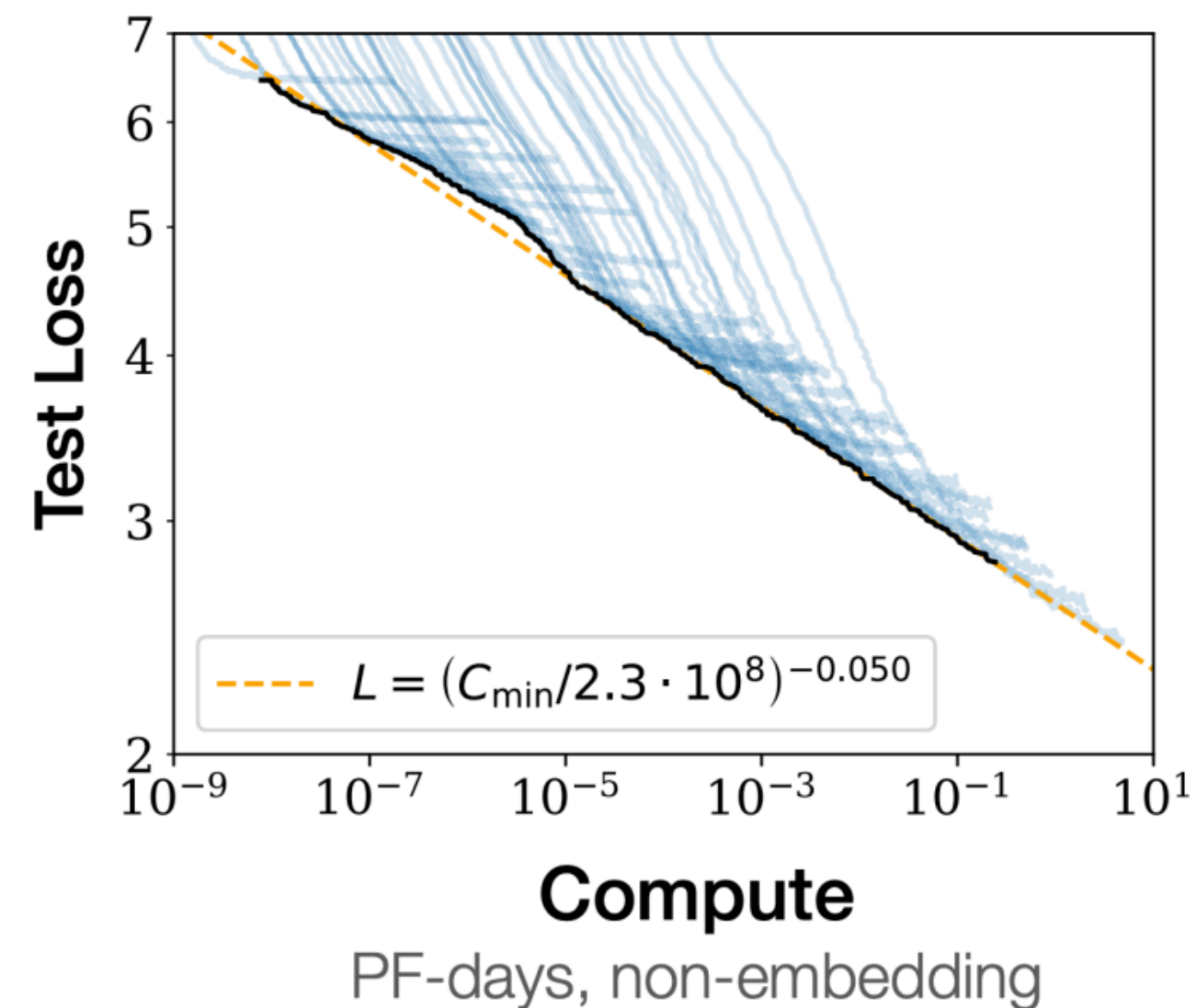
Data Curation	MMLU	HSwag	PIQA	WinoG	CSQA	SIQA	ARC-e	ARC-c	OBQA	Avg
Baseline corpus	30.3	57.5	71.3	56.1	59.0	49.9	62.2	34.0	44.0	51.6
+ Clusters	31.8	59.4	73.4	58.2	58.7	50.7	66.1	35.2	44.8	53.2
+ Topic	31.4	56.2	72.1	54.8	61.3	47.8	70.3	40.6	49.0	53.7
+ Format	31.7	60.9	74.1	56.9	60.1	47.4	65.8	35.9	47.6	53.4
+ Topic × Format	32.7	60.1	73.4	56.5	62.3	49.3	69.7	38.8	49.0	54.6
	↑2.4	↑2.6	↑2.1	↑0.4	↑3.3	↓0.6	↑7.5	↑4.8	↑5.0	↑3.0

(Wettig et al., 2025)

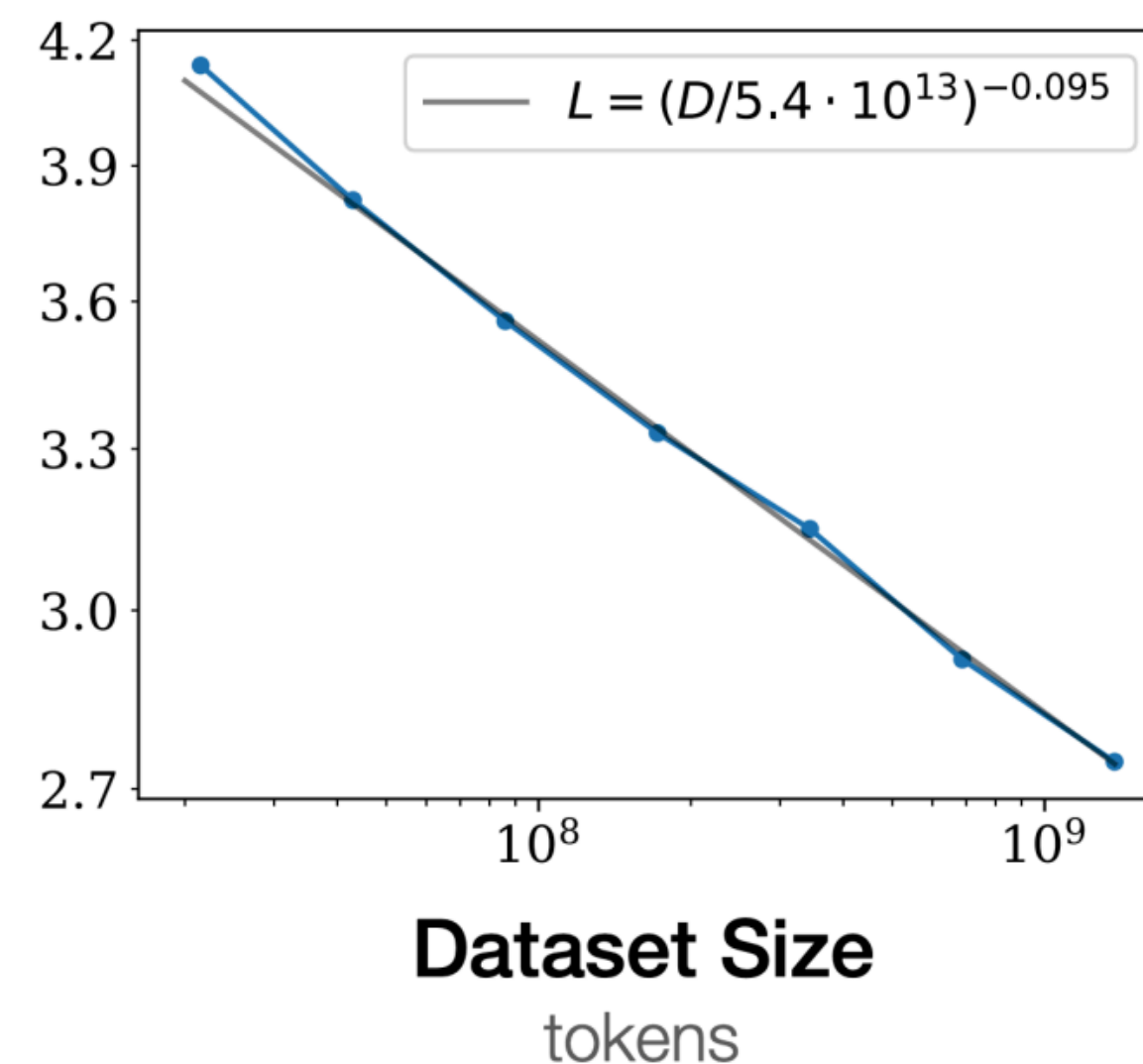
- What is a base model?
 - A model that has only gone through pre-training (next-token prediction on massive corpora)
 - vs models that are “post-trained” (**next week**)

LLMs after GPT-3

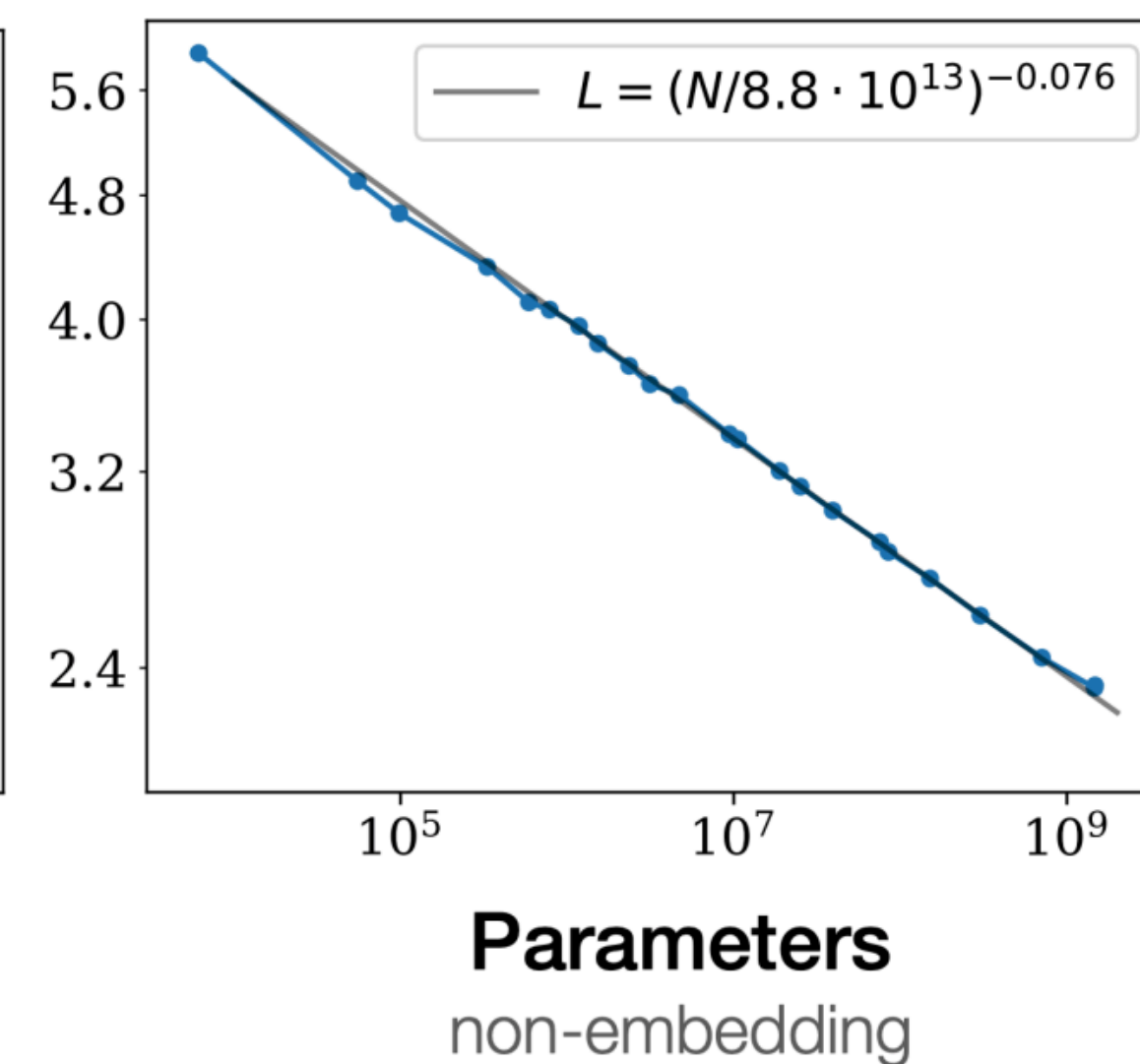
Scaling Laws: How to Optimally Allocate Compute: Model Params vs Dataset Size (Next Lecture)



$$\text{Loss} \propto (\text{Compute})^{-\alpha}$$



$$\text{Loss} \propto (\text{Data})^{-\beta}$$



$$\text{Loss} \propto (\text{Model params})^{-\gamma}$$

Loss goes down predictably wrt compute, data, model size!

(Kaplan et al., 2020) Scaling Laws for Neural Language Models

Scaling Laws: How to Optimally Allocate Compute: Model Params vs Dataset Size (Next Lecture)

$$\hat{L}(N, D) \triangleq E + \frac{A}{N^\alpha} + \frac{B}{D^\beta}.$$

L: loss

N: number of params

D: dataset size

E, A, B, α , β : fit based on data

Model	Size (# Parameters)	Training Tokens
LaMDA (Thoppilan et al., 2022)	137 Billion	168 Billion
GPT-3 (Brown et al., 2020)	175 Billion	300 Billion
Jurassic (Lieber et al., 2021)	178 Billion	300 Billion
Gopher (Rae et al., 2021)	280 Billion	300 Billion
MT-NLG 530B (Smith et al., 2022)	530 Billion	270 Billion
<i>Chinchilla</i>	70 Billion	1.4 Trillion

Rule of thumb: Increase dataset size proportional to model size (e.g. 20 token per param)

Many LLMs became proprietary

The creators of a revolutionary AI system that can write news stories and works of fiction - dubbed “deepfakes for text” - have taken the unusual step of not releasing their research publicly, for fear of potential misuse.

OpenAI, an nonprofit research company backed by Elon Musk, Reid Hoffman, Sam Altman, and others, says its new AI model, called GPT2 is so good and the risk of malicious use so high that **it is breaking from its normal practice of releasing the full research to the public** in order to allow more time to discuss the ramifications of the technological breakthrough.

The Guardian. February 2019.

Key LLM artifacts

Year	Model	Size	Status	Developer
2020	GPT-3	175B	Proprietary	OpenAI
2021	MT-NLG	530B	Proprietary	Microsoft/Nvidia
2022	PaLM	540B	Proprietary	Google
2022	Chinchilla	70B	Proprietary	Google DeepMind
2022	BLOOM	176B	Open-weight	BigScience
2023	Llama 1	7B - 65B	Open-weight	Meta
2023	Pythia	70M - 12B	Open-source	EleutherAI
2023	GPT-4	~1.8T (MoE)	Proprietary	OpenAI
2023	Llama 2	7B - 70B	Open-weight	Meta
2023	Mixtral	7B	Open-weight	Mistral AI
2024	OLMo	1B - 7B	Open-source	Ai2
2024	Llama 3	8B - 405B	Open-weight	Meta
2024	Wen 2.5	0.5B - 72B	Open-weight	Alibaba
2025	DeepSeek-V3	671B	Open-weight	DeepSeek

Open-Weight Models

RESEARCH

Introducing LLaMA: A foundational, 65-billion-parameter large language model

February 24, 2023



- Smaller models trained on 1.4T, high-quality & publicly available data
- “LLaMA-13B outperforms GPT-3 (175B) on most benchmarks, and LLaMA-65B is competitive with the best models, Chinchilla-70B and PaLM-540B”

Quick quiz

Which statement is incorrect?

- (A) Llama-3 8B is roughly comparable performance to GPT-3 175B.
- (B) Llama-3 [2023] is trained with 50x larger data than GPT-3.
- (C) One of the best language model trained primarily by university in the US as of 2025 is trained with 8x larger data than GPT-3.
- (D) None

(D) is correct.

A, B: Llama-3 8B trained for 15-trillion tokens $\approx$ GPT-3 175B trained for 0.3-trillion tokens

C: DCLM 7B trained for 3-trillion tokens (and is comparable to Llama-3 8B/GPT-3 175B, at least on the benchmarks they evaluated on).

Recent models are trained for much longer

- Llama-3: 8B, 70B, 405B trained on 15T tokens
- Qwen-2.5: 0.5B, 1.5B, 3B, 7B, 14B, 32B, 72B trained on 18T tokens
- DeepSeek V3: Mixture-of-Experts; 671B (37B active) trained on 14.8T tokens

Open source (pretraining) data (Next lecture)

Dataset	Example LMs	Tokens	Sources	Dataset	Example LMs	Tokens	Sources
C4 (Oct 2019)	T5, FLAN-T5	175B	Common Crawl	OpenWebMath (Oct 2023)	Llema	15B	Common Crawl
Pile (Dec 2020)	GPT-J, GPT-NeoX, Pythia	387B	Common Crawl, arXiv, PubMed, Books3, Gutenberg, Wikipedia, etc...	RedPajama v2 (Oct 2023)	-	30T	Common Crawl
The Stack v1 (Nov 2022)	StarCoder	200B	Software Heritage	Amber (Dec 2023)	Amber	1.3T	C4, RefinedWeb, the Stack, RedPajama v1
RedPajama v1 (Apr 2023)	INCITE	1.2T	Common Crawl, C4, Github, arXiv, Gutenberg, Books3, Wikipedia, Internet Archive (Stack Exchange)	Dolma 1.7 (Apr 2024)	OLMo 0424	2.3T	Dolma, RefinedWeb, RP's StackExchange, Flan, OpenWebMath, ...
RefinedWeb (Jun 2023)	Falcon	580B*	Common Crawl	FineWeb (May 2024)	-	15T	Common Crawl
Dolma (Aug 2023)	OLMo	3.1T	Common Crawl, C4, Semantic Scholar, Pushshift Reddit, Gutenberg, the Stack, Wikipedia, Wikibooks	Matrix (May 2024)	MAP-Neo	4.7T	RedPajama v2, Dolma, CulturaX, Amber, SlimPajama, Falcon, crawled Chinese web
				DCLM (Jun 2024)	DCLM-Baseline	4T	Common Crawl

Timeline summary

- 2017: Transformers introduced, mainly for supervised learning (MT).
- 2018: Early pre-training ideas (e.g., ELMO); BERT introduced: first widely-used pre-trained Transformer (key: masked LM objective); GPT-1 appeared slightly earlier (generative LM) but saw less adoption
- 2019: GPT-2: more data, more parameters, introduced zero-shot prompting, with mixed results
- 2020: GPT-3: strong zero-shot and few-shot prompting; emergence of in-context learning; began attracting serious attention
- Late 2022: ChatGPT — clear inflection point for the field and the public; success driven largely by post-training ← Covered in a later lecture!

Questions?

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Princeton COS 484 by Danqi Chen, Tri Dao, Vikram Ramaswamy